Statistics

Chapter 7 — Instructor Solutions Manual

October 10, 2026 • Total Exercises: 76

PROBLEM STATEMENT

Chapter 7, Page 226

Step 1

A uniform distribution is defined over the interval from 6 to 10.

a. In this case $a = 6$ and $b = 10$

b. Mean of the distribution μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{6+10}{2} \\ &= \frac{16}{2} \\ &= 8\end{aligned}$$

c. Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(10-6)^2}{12}} \\ &= \sqrt{\frac{16}{12}} \\ &= \sqrt{1.33} \\ &= 1.155\end{aligned}$$

d. Area of the distribution A

For the uniform distribution, the height of the rectangle is $\frac{1}{b-a}$ and the base of the distribution is $b-a$. Hence

$$\begin{aligned}A &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times (b-a) \\ &= 1\end{aligned}$$

Step 2

e. Probability of a value more than 7

If the value is more than 7, it means that the value is between 7 and 10. So, the base is $10 - 7 = 3$.

$$\begin{aligned} P(7 \leq \text{value} \leq 10) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 3 \\ &= \frac{1}{10-6} \times 3 \\ &= \frac{3}{4} \\ &= 0.75 \end{aligned}$$

f. Probability of a value is between 7 and 9

The value is between 7 and 9. So, the base is $9 - 7 = 2$.

$$\begin{aligned} P(7 \leq \text{value} \leq 9) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 2 \\ &= \frac{1}{10-6} \times 2 \\ &= \frac{2}{4} \\ &= 0.5 \end{aligned}$$

🎯 Result

a. $a = 6$ and $b = 10$

b. 8

c. 1.155

d. area is 1

e. 0.75

f. 0.5

PROBLEM STATEMENT

Chapter 7, Page 226

Step 1

A uniform distribution is defined over the interval from 2 to 5.

a. In this case $a = 2$ and $b = 5$

b. Mean of the distribution μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{2+5}{2} \\ &= \frac{7}{2} \\ &= 3.5\end{aligned}$$

c. Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(5-2)^2}{12}} \\ &= \sqrt{\frac{9}{12}} \\ &= \sqrt{0.75} \\ &= 0.866\end{aligned}$$

d. Area of the distribution A

For the uniform distribution, the height of the rectangle is $\frac{1}{b-a}$ and the base of the distribution is $b-a$. Hence

$$\begin{aligned}A &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times (b-a) \\ &= 1\end{aligned}$$

Step 2

e. Probability of a value more than 2.6

If the value is more than 2.6, it means that the value is between 2.6 and 5. So, the base is $5 - 2.6 = 2.4$.

$$\begin{aligned} P(2.6 \leq \text{value} \leq 5) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 2.4 \\ &= \frac{1}{5-2} \times 2.4 \\ &= \frac{2.4}{3} \\ &= 0.8 \end{aligned}$$

f. Probability of a value is between 2.9 and 3.7

The value is between 2.9 and 3.7. So, the base is $3.7 - 2.9 = 0.8$.

$$\begin{aligned} P(2.9 \leq \text{value} \leq 3.7) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 0.8 \\ &= \frac{1}{5-2} \times 0.8 \\ &= \frac{0.8}{3} \\ &= 0.267 \end{aligned}$$

🎯 Result

a. $a = 2$ and $b = 5$

b. 3.5

c. 0.866

d. Area is 1

e. 0.8

f. 0.267

PROBLEM STATEMENT

Chapter 7, Page 226

Step 1

Common stock is uniformly distributed between 20 and 30 per share

$\Rightarrow a = 20$ and $b = 30$

For the uniform distribution, the height of the rectangle is $\frac{1}{b-a}$

a. Probability that the stock price will be more than $\$27$

Stock price more than $\$27$ means that the stock price lies between 27 and 30. So the base is $30 - 27 = 3$. Hence

$$\begin{aligned} P(27 < \text{stock price} < 30) &= \text{Height} \times \text{base} \\ &= \frac{1}{b-a} \times 3 \\ &= \frac{1}{30-20} \times 3 \\ &= \frac{3}{10} \\ &= 0.3 \end{aligned}$$

b. Probability that the stock price will be less than or equal to $\$24$

Stock price less than or equal to $\$24$ means that the stock price lies between 20 and 24. So the base is $24 - 20 = 4$. Hence

$$\begin{aligned} P(20 < \text{stock price} \leq 24) &= \text{Height} \times \text{base} \\ &= \frac{1}{b-a} \times 4 \\ &= \frac{1}{30-20} \times 4 \\ &= \frac{4}{10} \\ &= 0.4 \end{aligned}$$

🎯 Result

a. 0.3

b. 0.4

PROBLEM STATEMENT

Chapter 7, Page 227

Step 1

Money spent on insurance is uniformly distributed between 400 and 3800 per year

$\Rightarrow a = 400$ and $b = 3800$

a. Mean amount spent on insurance μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{400+3800}{2} \\ &= \frac{4200}{2} \\ &= 2100\end{aligned}$$

b. Standard deviation of the amount spent on insurance σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(3800-400)^2}{12}} \\ &= \sqrt{\frac{(3400)^2}{12}} \\ &= \sqrt{963333} \\ &= 981.5\end{aligned}$$

c. Probability of insurance amount less than 2000

If the insurance amount is less than 2000, it means that the insurance amount is between 400 and 2000. So, the base is $2000 - 400 = 1600$.

$$\begin{aligned}P(400 < \text{insurance amount} < 2000) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 1600 \\ &= \frac{1}{3800-400} \times 1600 \\ &= \frac{1600}{3400} \\ &= 0.471\end{aligned}$$

Step 2

d. Probability of insurance amount more than 3000

If the insurance amount is more than 3000, it means that the insurance amount is between 3000 and 3800. So, the base is $3800 - 3000 = 800$.

$$\begin{aligned} P(3000 < \text{insurance amount} < 3800) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 800 \\ &= \frac{1}{3800 - 400} \times 800 \\ &= \frac{800}{3400} \\ &= 0.235 \end{aligned}$$

Result

a. 2100

b. 981.5

c. 0.471

d. 0.235

PROBLEM STATEMENT

Chapter 7, Page 227

Step 1

a. The rainfall follows the uniform distribution between 0.5 and 3 inches,

$\Rightarrow a = 0.5$ and $b = 3$

b. Mean amount of rainfall μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{0.5+3}{2} \\ &= \frac{3.5}{2} \\ &= 1.75\end{aligned}$$

Standard deviation of the rainfall σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(3-0.5)^2}{12}} \\ &= \sqrt{\frac{6.25}{12}} \\ &= \sqrt{0.521} \\ &= 0.723\end{aligned}$$

c. Probability of less than an inch of the rain

If the rainfall is less than an inch, it means that the rainfall is between 0.5 and 1. So, the base is $1 - 0.5 = 0.5$.

$$\begin{aligned}P(0.5 < \text{rainfall} < 1) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 0.5 \\ &= \frac{1}{3-0.5} \times 0.5 \\ &= \frac{0.5}{2.5} \\ &= 0.2\end{aligned}$$

Step 2

d. Probability of exactly 1 inch of rainfall

If the rainfall is exactly 1, then the base is 0

$$\begin{aligned} P(\text{rainfall} = 1) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 0 \\ &= \frac{1}{3. - 0.5} \times 0 \\ &= 0 \end{aligned}$$

e. Probability of rainfall more than 1.50 inches

If the rainfall is more than 1.50, it means that the rainfall is between 1.50 and 3. So, the base is $3 - 1.5 = 1.5$.

$$\begin{aligned} P(1.50 < \text{rainfall} < 3) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 1.5 \\ &= \frac{1}{3 - 0.5} \times 1.5 \\ &= \frac{1.5}{2.5} \\ &= 0.6 \end{aligned}$$

🎯 Result

a. $a = 0.5$ and $b = 3$

b. $\mu = 1.75$ and $\sigma = 0.723$

c. 0.2

d. 0

e. 0.6

PROBLEM STATEMENT

Chapter 7, Page 227

Solution A

Step 1

The problem involves uniform distribution. We need to apply the features of the uniform distribution to answer the corresponding questions. With these, we are going to use the following formulas:

$$\mu = \frac{a+b}{2}$$

$$\sigma = \sqrt{\frac{(b-a)^2}{12}}$$

$$A = (\text{height})(\text{base})$$

Where a and b are the minimum and maximum points, respectively, on the distribution.

Step 2

(a) Values of a and b:

The distribution for the data follows the uniform distribution. This indicates that a minimum a and maximum b occurs for the shape of the distribution is rectangular and the height of the distribution is uniform between a and b. We are talking about technical support time and a technician can resolve a technical difficulty between 30 seconds and 10 minutes.

Thus, the minimum of the distribution a is 30 seconds and the maximum of the distribution b is 10 minutes. Converting seconds to minutes we have the values for a and b:

$$\text{minimum} = a = 30 \text{ sec} \times \frac{1 \text{ min}}{60 \text{ sec}} = 0.5 \text{ min}$$

$$\text{maximum} = b = 10 \text{ min}$$

Therefore, $\boxed{a = 0.5}$ and $\boxed{b = 10}$.

Step 3

(b) Mean and Standard Deviation:

The **mean** of a uniform distribution can be calculated by:

$$\mu = \frac{a+b}{2}$$

Substituting $a = 0.5$ and $b = 10$ into the mean formula then evaluating we obtain:

$$\begin{aligned}
\mu &= \frac{a+b}{2} && (\text{Write the formula.}) \\
&= \frac{0.5+10}{2} && (\text{Substitute the values.}) \\
&= \frac{10.5}{2} && (\text{Add.}) \\
&= 5.25 && (\text{Divide.})
\end{aligned}$$

Thus, the mean of the uniform distribution is $\boxed{5.25 \text{ min}}$. This indicates that the average time a technician can fix the problem is 5.25 minutes.

Step 4

The **standard deviation** of a uniform distribution can be calculated by:

$$\sigma = \sqrt{\frac{(b-a)^2}{12}}$$

Substituting $a = 0.5$ and $b = 10$ then evaluating we obtain:

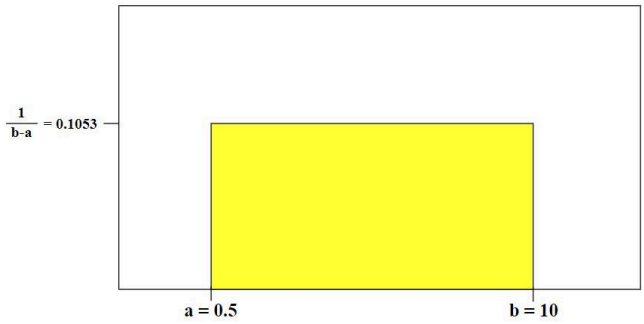
$$\begin{aligned}
\sigma &= \sqrt{\frac{(b-a)^2}{12}} && (\text{Write the formila.}) \\
&= \sqrt{\frac{(10-0.5)^2}{12}} && (\text{Substitute the values.}) \\
&= \sqrt{\frac{(9.5)^2}{12}} && (\text{Subtract.}) \\
&= \sqrt{\frac{90.25}{12}} && (\text{Evaluate the exponent.}) \\
&= \sqrt{7.5208} && (\text{Divide.}) \\
&= 2.7424 && (\text{Evaluate the square root.})
\end{aligned}$$

Thus, the standard distribution is 2.7424 minutes. This is the measure of variation in technician's support time.

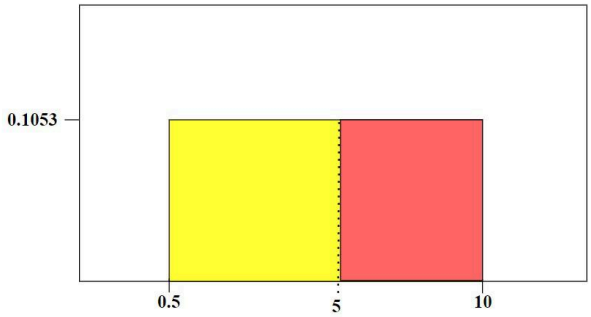
Step 5

(c) Percent for More Than 5 minutes of Resolving the Problem:

With the uniform distribution illustration for the support time:



let us locate the region for 5 minutes to 10 minutes of support time:

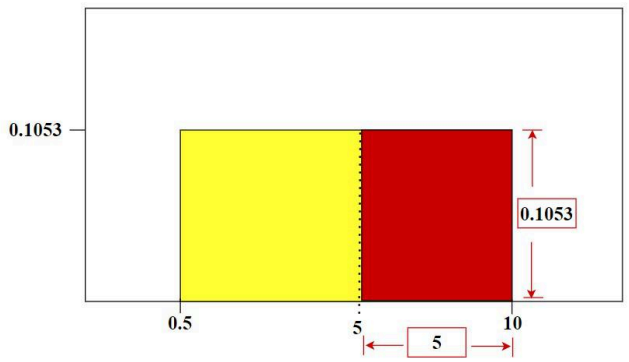


Step 6

The total area for the uniform distribution area is 1 which corresponds to a probability of 1. Thus, we need to find the area of the red region to find its probability. Let the probability of resolving the problem for more than 5 minutes be:

$$P(5 < \text{ support time } < 10)$$

Based on the values given, the height and the base of the shaded region are:



The probability is equal to the area of the shaded region. Using the area formula: $A = (\text{height})(\text{base})$ to find the probability we obtain:

$$\begin{aligned} P(5 < \text{ support time } < 10) &= (\text{height})(\text{base}) \\ &= (0.1053)(5) \\ &= 0.5265 \end{aligned}$$

Thus, the probability of resolving the problem in more than 5 minutes is 0.5265. In percent, $0.5265 = \boxed{52.65\%}$

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Step 7

(d) End points for the 50%:

Let x be the minimum point, and y be the maximum point on the probability distribution. We want to know the endpoints from which the probability area yields 50%. Thus, we will be working on the probability equality:

🎯 Result

$$P(x < \text{ support time } < y) = 0.50$$

From above, we know that the height is 0.1053 while the base with 0.50 probability is the difference of y and x that is $y - x$. We then equate the area to the probability that we want to show then simplify to find the base that is:

$$P(x < \text{ support time } < y) = A$$

$$P(x < \text{ support time } < y) = (\text{height})(\text{base})$$

$$0.50 = (0.1053)(y - x)$$

$$\frac{0.50}{0.1053} = \frac{(0.1053)(y - x)}{0.1053}$$

$$4.75 = y - x$$

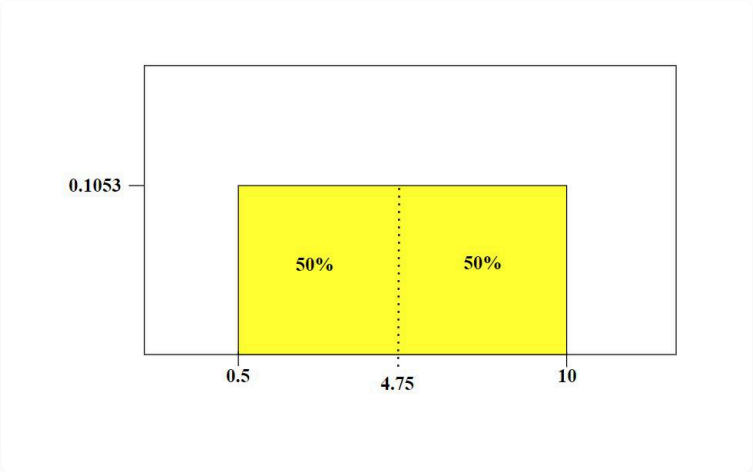
Thus, the endpoints for a 50% probability are found using the equation:

$$y - x = 4.75$$

Which means that any enpoints with the difference of 4.75 gives 50% probability.

Two pairs of endpoints that are evident in the probability illustration are:

$$\frac{b - a}{2} = \frac{10 - 0.5}{2} = \frac{9.5}{2} = 4.75$$



By dividing the uniform distribution into two equal parts, we can find the endpoints for the 50% problem-solving times. The two regions will have the endpoints:

$$a \text{ to } 4.75 \longrightarrow 0.5 \text{ to } 4.75$$

$$4.75 \text{ to } b \longrightarrow 4.75 \text{ to } 10$$

(a) $a = 0.5, b = 10$

(b) $\mu = 5.25, \sigma = 2.7424$

(c) 52.65%

(d) $y - x = 4.75$

Solution B

Step 1

a. Technical support time follows uniform distribution between 0.5 minutes ($\frac{30}{60}$ minutes) and 10 minutes.

$\Rightarrow a = 0.5$ and $b = 10$

b. Mean time to resolve the problem μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{0.5+10}{2} \\ &= \frac{10.5}{2} \\ &= 5.25\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(10-0.5)^2}{12}} \\ &= \sqrt{\frac{90.25}{12}} \\ &= \sqrt{7.521} \\ &= 1.155\end{aligned}$$

c. Probability that the problem takes more than 5 minutes

If the problem takes more than 5 minutes, it means that the time taken by problems is between 5 and 10. So, the base is $10 - 5 = 5$.

$$\begin{aligned}P(5 < \text{support time} < 10) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 5 \\ &= \frac{1}{10-0.5} \times 5 \\ &= \frac{5}{9.5} \\ &= 0.53\end{aligned}$$

Step 2

d. Suppose that the probability that problem solving time is between x and y is 50% or 0.5.

So, the base is $y - x$

$$\begin{aligned} P(x < \text{support time} < y) &= \text{height} \times \text{base} \\ \Rightarrow 0.5 &= \frac{1}{b-a} \times (y-x) \\ \Rightarrow 0.5 &= \frac{1}{10-0.5} \times (y-x) \\ \Rightarrow 0.5 &= \frac{y-x}{9.5} \\ \Rightarrow y - x &= 4.75 \end{aligned}$$

As long as the difference between the two end points of the times is 4.75 minutes, the problem solving time will be 0.5.

🎯 Result

a. $a = 0.5$ and $b = 10$

b. $\mu = 5.25$ and $\sigma = 1.155$

c. 0.53

d. $y - x = 4.75$

PROBLEM STATEMENT

Chapter 7, Page 233

Step 1

The **normal probability distribution** is dependent on the value of the mean and the standard deviation.

However, this then implies that a different mean and/or a different standard deviation leads to a different normal distribution and thus there are many possible normal probability distributions.

Since there are many possible normal probability distributions, we refer to the normal probability distributions as a family.

🎯 Result

There are many possible normal probability distributions.

PROBLEM STATEMENT

Chapter 7, Page 233

Step 1

We need to list the major characteristics of a normal probability distribution:

- dependent on the mean μ and standard deviation σ (that is, different means and/or standard deviations lead to different normal distributions).
- Symmetric about the mean μ
- Peak of the probability distribution lies as the mean μ
- Standard deviation determines the width of the probability distribution (larger standard deviation implies wider probability distribution)
- By the empirical rule, about 68% of the data values are within 1 standard deviation of the mean, about 95% of the data values are within 2 standard deviations of the mean, and about 99.7% (nearly all) of the observations are within 3 standard deviations of the mean.

Result

Symmetric about the mean μ

PROBLEM STATEMENT

Chapter 7, Page 233

Solution A

Step 1

Given:

$$\mu = \text{Mean} = 500$$
$$\sigma = \text{Standard deviation} = 10$$

The **Empirical Rule** tells us:

Approximately 68% of the observations are within 1 standard deviation of the mean.

Approximately 95% of the observations are within 2 standard deviations of the mean.

Approximately 99.7% of the observations are within 3 standard deviations of the mean.

Step 2

(a) By the empirical rule, we know that 68% of the observations are within 1 standard deviation of the mean:

$$\mu - \sigma = 500 - 10 = 490$$
$$\mu + \sigma = 500 + 10 = 510$$

Thus 68% of all observations are between 490 and 510.

Step 3

(b) By the empirical rule, we know that 95% of the observations are within 2 standard deviations of the mean:

$$\mu - 2\sigma = 500 - 2(10) = 500 - 20 = 480$$
$$\mu + 2\sigma = 500 + 2(10) = 500 + 20 = 520$$

Thus 95% of all observations are between 480 and 520.

Step 4

(c) By the empirical rule, we know that 99.7% of the observations (which are thus nearly all observations) are within 3 standard deviations of the mean:

$$\mu - 3\sigma = 500 - 3(10) = 500 - 30 = 470$$
$$\mu + 3\sigma = 500 + 3(10) = 500 + 30 = 530$$

Thus 99.7% of all observations are between 470 and 530.

🎯 Result

(a) Between 490 and 510

(b) Between 480 and 520

(c) Between 470 and 530

Solution B

Step 1

The mean of the probability distribution is $\mu = 500$, and the standard deviation is $\sigma = 10$.

$\textbf{a. 68% of the observations}$

About 68% of the area under the normal curve is within one standard deviation of the mean i.e $\mu \pm 1\sigma$

$$\mu + 1\sigma = 500 + 1(10) = 510$$

and

$$\mu - 1\sigma = 500 - 1(10) = 490$$

About 68% of the observations lie between 490 and 510.

$\textbf{b. 95% of the observations}$

About 95% of the area under the normal curve is within one standard deviation of the mean i.e $\mu \pm 2\sigma$

$$\mu + 2\sigma = 500 + 2(10) = 520$$

and

$$\mu - 2\sigma = 500 - 2(10) = 480$$

About 95% of the observations lie between 480 and 520.

c. All of the obsevations

Practically all of the area under the normal curve is within one standard deviation of the mean i.e $\mu \pm 3\sigma$

$$\mu + 3\sigma = 500 + 3(10) = 530$$

and

$$\mu - 3\sigma = 500 - 3(10) = 470$$

Practically all of the observations lie between 470 and 530.

Result

a. 490 and 510

b. 480 and 520

c. 470 and 530

PROBLEM STATEMENT

Chapter 7, Page 233

Step 1

Mean of the normal distribution is $\mu = 60$ and the standard deviation is $\sigma = 5$

$\textbf{a. Observations lie between } 55 \text{ and } 65\}$

Suppose that area under the normal curve with these observations is within z standard deviation of the mean.

$$\begin{aligned} &\Rightarrow \mu - z\sigma = 55 \text{ and } \mu + z\sigma = 65 \\ &\Rightarrow 60 - 5z = 55 \text{ and } 60 + 5z = 65 \\ &\Rightarrow 5z = 60 - 55 \text{ and } 5z = 65 - 60 \\ &\Rightarrow 5z = 5 \text{ and } 5z = 5 \\ &\Rightarrow z = 1 \text{ and } z = 1 \end{aligned}$$

As per the empirical rule, about 68% of the area under the normal curve is within one standard deviation of the mean.

$\textbf{b. Observations lie between } 50 \text{ and } 70\}$

Suppose that area under the normal curve with these observations is within z standard deviation of the mean.

$$\begin{aligned} &\Rightarrow \mu - z\sigma = 50 \text{ and } \mu + z\sigma = 70 \\ &\Rightarrow 60 - 5z = 50 \text{ and } 60 + 5z = 70 \\ &\Rightarrow 5z = 60 - 50 \text{ and } 5z = 70 - 60 \\ &\Rightarrow 5z = 10 \text{ and } 5z = 10 \\ &\Rightarrow z = 2 \text{ and } z = 2 \end{aligned}$$

As per the empirical rule, about 95% of the area under the normal curve is within two standard deviation of the mean.

Step 2

$\textbf{c. Observations lie between 45 and 75}$

Suppose that area under the normal curve with these observations is within z standard deviation of the mean.

$$\begin{aligned}
&\Rightarrow \mu - z\sigma = 45 \text{ and } \mu + z\sigma = 75 \\
&\Rightarrow 60 - 5z = 45 \text{ and } 60 + 5z = 75 \\
&\Rightarrow 5z = 60 - 45 \text{ and } 5z = 75 - 60 \\
&\Rightarrow 5z = 15 \text{ and } 5z = 15 \\
&\Rightarrow z = 3 \text{ and } z = 3
\end{aligned}$$

As per the empirical rule, Practically all of the area under the normal curve is within three standard deviation of the mean.

Result

a. 68%

b. 95%

c. Practically all area

PROBLEM STATEMENT

Chapter 7, Page 233

Step 1

Let X denote the salary of Rob and Rachel. So, $X = \$50000$.

Mean salary in the Rachel's industry is $\mu_1 = \$35000$ and standard deviation is $\sigma_1 = \$8000$.

Mean salary in theRob's industry is $\mu_2 = \$60000$ and standard deviation is $\sigma_2 = \$5000$.

So, z **value for Rob**

$$\begin{aligned} z_1 &= \frac{X - \mu_1}{\sigma_1} \\ &= \frac{50000 - 35000}{8000} \\ &= \frac{15000}{8000} \\ &= 1.875 \end{aligned}$$

z **value for Rachel**

$$\begin{aligned} z_2 &= \frac{X - \mu_2}{\sigma_2} \\ &= \frac{50000 - 60000}{5000} \\ &= \frac{-10000}{5000} \\ &= -2 \end{aligned}$$

Rob is earning below average and Rachel is earning above average.

Result

$z_1 = 1.875$ and $z_2 = -2$

PROBLEM STATEMENT

Chapter 7, Page 233

Step 1

Mean labor cost to repair the heat pump is $\mu = \$90$ and the standard deviation is $\sigma = \$22$.

Labor cost for the first heat pump is $X_1 = \$75$ and the labor cost for the second heat pump is $X_2 = \$100$.

z value for the first heat pump

$$\begin{aligned} z_1 &= \frac{X_1 - \mu}{\sigma} \\ &= \frac{75 - 90}{22} \\ &= \frac{-15}{22} \\ &= -0.6818 \end{aligned}$$

z value for the second heat pump

$$\begin{aligned} z_2 &= \frac{X_2 - \mu}{\sigma} \\ &= \frac{100 - 90}{22} \\ &= \frac{10}{22} \\ &= 0.45 \end{aligned}$$

Result

$z_1 = -0.6818$ and $z_2 = 0.45$

PROBLEM STATEMENT

Chapter 7, Page 236

Step 1

Normal population has mean $\mu = 20$ and standard deviation $\sigma = 4$.

a. z value with $X = 25$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{25 - 20}{4} \\ &= \frac{5}{4} \\ &= 1.25 \end{aligned}$$

b. X is between 20 and 25

$$\begin{aligned} P(20 < X < 25) &= P(20 - \mu < X - \mu < 25 - \mu) \\ &= P\left(\frac{20 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{25 - \mu}{\sigma}\right) \\ &= P\left(\frac{20 - 20}{4} < z < \frac{25 - 20}{4}\right) \\ &= P(0 < z < 1.25) \\ &= 0.3944 \text{ from the appendix B1} \end{aligned}$$

c. X is less than 18

$$\begin{aligned} P(X < 18) &= P(X - \mu < 18 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{18 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{18 - 20}{4}\right) \\ &= P(z < \frac{-2}{4}) \\ &= P(z < -0.5) \\ &= P(z > 0.5) \\ &= 0.5 - P(0 < z < 0.5) \\ &= 0.5 - 0.1915 \text{ from the appendix B1} \\ &= 0.3085 \end{aligned}$$

🎯 **Result**

a. $z = 1.25$

b. 0.3994

c. 0.3085

PROBLEM STATEMENT

Chapter 7, Page 236

Step 1

Normal population has mean $\mu = 12.2$ and standard deviation $\sigma = 2.5$.

a. z value with $X = 14.3$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{14.3 - 12.2}{2.5} \\ &= \frac{2.1}{2.5} \\ &= 0.84 \end{aligned}$$

b. X is between 12.2 and 14.3

$$\begin{aligned} P(12.2 < X < 14.3) &= P(12.2 - \mu < X - \mu < 14.3 - \mu) \\ &= P\left(\frac{12.2 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{14.3 - \mu}{\sigma}\right) \\ &= P\left(\frac{12.2 - 12.2}{2.5} < z < \frac{14.3 - 12.2}{2.5}\right) \\ &= P(0 < z < 0.84) \\ &= 0.2995 \text{ from the appendix B1} \end{aligned}$$

c. X is less than 10

$$\begin{aligned} P(X < 10) &= P(X - \mu < 10 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{10 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{10 - 12.2}{2.5}\right) \\ &= P(z < \frac{-2.2}{2.5}) \\ &= P(z < -0.88) \\ &= P(z > 0.88) \\ &= 0.5 - P(0 < z < 0.88) \\ &= 0.5 - 0.3106 \text{ from the appendix B1} \\ &= 0.1894 \end{aligned}$$

🎯 **Result**

a. $z = 0.84$

b. 0.2995

c. 0.1894

PROBLEM STATEMENT

Chapter 7, Page 237

Step 1

Mean hourly salary was $\mu = 20.50$ and standard deviation $\sigma = 3.5$.

Let X be the hourly wages.

a. X is between 20.50 and 24

$$\begin{aligned} P(20.5 < X < 24) &= P(20.5 - \mu < X - \mu < 24 - \mu) \\ &= P\left(\frac{20.5 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{24 - \mu}{\sigma}\right) \\ &= P\left(\frac{20.5 - 20.5}{3.5} < z < \frac{24 - 20.5}{3.5}\right) \\ &= P(0 < z < 1) \\ &= 0.3413 \ \text{ from the appendix B1} \end{aligned}$$

b. X is more than 24

$$\begin{aligned} P(X > 24) &= P(X - \mu > 24 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{24 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{24 - 20.5}{3.5}\right) \\ &= P(z > 1) \\ &= 0.5 - P(0 < z < 1) \\ &= 0.5 - 0.3413 \ \text{ from the appendix B1} \\ &= 0.1587 \end{aligned}$$

Step 2

c. X is less than 19

$$
\begin{aligned}
P(X < 19) &= P(X - \mu < 19 - \mu) \\
&= P(\frac{X - \mu}{\sigma} < \frac{19 - \mu}{\sigma}) \\
&= P(z < \frac{19 - 20.5}{3.5}) \\
&= P(z < \frac{-0.5}{3.5}) \\
&= P(z < -0.143) \\
&= P(z > 0.143) \\
&= 0.5 - P(0 < z < 0.143) \\
&= 0.5 - 0.0557 \text{ from the appendix B1} \\
&= 0.4443
\end{aligned}
$$

🎯 Result

a. 0.3413

b. 0.1587

c. 0.4443

PROBLEM STATEMENT

Chapter 7, Page 237

Step 1

Mean of normal probability distribution is $\mu = 400$ pounds and standard deviation $\sigma = 10$ pounds.

Let X be the variable in pounds.

a. X is between 400 and 415 pounds

$$\begin{aligned} P(400 < X < 415) &= P(400 - \mu < X - \mu < 415 - \mu) \\ &= P\left(\frac{400 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{415 - \mu}{\sigma}\right) \\ &= P\left(\frac{400 - 400}{10} < z < \frac{415 - 400}{10}\right) \\ &= P(0 < z < 1.5) \\ &= 0.4332 \text{ from the appendix B1} \end{aligned}$$

b. X is between mean 400 and 395 pounds

$$\begin{aligned} P(395 < X < 400) &= P(395 - \mu < X - \mu < 400 - \mu) \\ &= P\left(\frac{395 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{400 - \mu}{\sigma}\right) \\ &= P\left(\frac{395 - 400}{10} < \frac{X - \mu}{\sigma} < \frac{400 - 400}{10}\right) \\ &= P(-0.5 < z < 0) \\ &= P(0 < z < 0.5) \\ &= 0.1915 \text{ from the appendix B1} \end{aligned}$$

Step 2

c. X is less than 395 pounds

$$
\begin{aligned}
P(X < 395) &= P(X - \mu < 395 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} < \frac{395 - \mu}{\sigma}\right) \\
&= P\left(< z < \frac{395 - 400}{10}\right) \\
&= P(z < -0.5) \\
&= P(z > 0.5) \\
&= 0.5 - P(0 < z < 0.5) \\
&= 0.5 - 0.1915 \ \text{ from the appendix B1} \\
&= 0.3085
\end{aligned}
$$

🎯 Result

a. 0.4332

b. 0.1915

c. 0.3085

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean of the normal distribution is $\mu = 50$ and the standard deviation is $\sigma = 4$

Let X be the random variable.

$\textbf{a. } X \text{ is between } 44 \text{ and } 55 \}$

$$\begin{aligned} P(44 < X < 55) &= P(44 - \mu < X - \mu < 55 - \mu) \\ &= P\left(\frac{44 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{55 - \mu}{\sigma}\right) \\ &= P\left(\frac{44 - 50}{4} < z < \frac{55 - 50}{4}\right) \\ &= P\left(\frac{-6}{4} < z < \frac{5}{4}\right) \\ &= P(-1.5 < z < 1.25) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.5 < z < 0)$, $P(0 < z < 1.25)$ and add the two probabilities.

$$\begin{aligned} P(44 < X < 55) &= P(-1.5 < z < 0) + P(0 < z < 1.25) \\ &= P(0 < z < 1.5) + P(0 < z < 1.25) \\ &= 0.4332 + 0.3944 \ \ (\text{from Appendix B1.}) \\ &= 0.8276 \end{aligned}$$

Step 2

$\textbf{b. X is greater than 55 }$

$$\begin{aligned} P(X > 55) &= P(X - \mu > 55 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{55 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{55 - 50}{4}\right) \\ &= P\left(z > \frac{5}{4}\right) \\ &= P(z > 1.25) \\ &= 0.5 - P(0 < z < 1.25) \\ &= 0.5 - 0.3944 \ \ (\text{from Appendix B1.}) \\ &= 0.1056 \end{aligned}$$

$\textbf{c. X is between 52 and 55 }$

$$\begin{aligned} P(52 < X < 55) &= P(52 - \mu < X - \mu < 55 - \mu) \\ &= P\left(\frac{52 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{55 - \mu}{\sigma}\right) \\ &= P\left(\frac{52 - 50}{4} < z < \frac{55 - 50}{4}\right) \\ &= P\left(\frac{2}{4} < z < \frac{5}{4}\right) \\ &= P(0.5 < z < 1.25) \end{aligned}$$

The two points are on the same sides of the mean, so calculate $P(0 < z < 0.5)$, $P(0 < z < 1.25)$ and subtract the two probabilities.

$$\begin{aligned} P(52 < X < 55) &= P(0 < z < 1.25) - P(0 < z < 0.5) \\ &= 0.3944 - 0.1915 \ \ (\text{from Appendix B1.}) \\ &= 0.2029 \end{aligned}$$

Result

a. 0.8276

b. 0.1056

c. 0.2029

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean of the normal distribution is $\mu = 80$ and the standard deviation is $\sigma = 14$

Let X be the random variable.

$\textbf{a. } X \text{ is between } 75 \text{ and } 90 \}$

$$\begin{aligned} P(75 < X < 90) &= P(75 - \mu < X - \mu < 90 - \mu) \\ &= P\left(\frac{75 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{90 - \mu}{\sigma}\right) \\ &= P\left(\frac{75 - 80}{14} < z < \frac{90 - 80}{14}\right) \\ &= P\left(\frac{-5}{14} < z < \frac{10}{14}\right) \\ &= P(-0.357 < z < 0.714) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.357 < z < 0), P(0 < z < 0.714)$ and add the two probabilities.

$$\begin{aligned} P(75 < X < 90) &= P(-0.357 < z < 0) + P(0 < z < 0.714) \\ &= P(0 < z < 0.357) + P(0 < z < 0.714) \\ &= 0.1406 + 0.2611 \ \ (\text{from Appendix B1.}) \\ &= 0.4017 \end{aligned}$$

Step 2

$\textbf{b. } X \text{ is less than } 75 \}$

$$\begin{aligned} P(X<75) &= P(X-\mu<75-\mu) \\ &= P\left(\frac{X-\mu}{\sigma}<\frac{75-\mu}{\sigma}\right) \\ &= P\left(z<\frac{75-80}{14}\right) \\ &= P\left(z<\frac{-5}{14}\right) \\ &= P(z<-0.357) \\ &= P(z>0.357) \\ &= 0.5-P(0<z<0.357) \\ &= 0.5-0.1406 \ \ (\text{from Appendix B1.}) \\ &= 0.3594 \end{aligned}$$

$\textbf{c. } X \text{ is between } 52 \text{ and } 55 \}$

$$\begin{aligned} P(55<X<70) &= P(55-\mu<X-\mu<70-\mu) \\ &= P\left(\frac{55-\mu}{\sigma}<\frac{X-\mu}{\sigma}<\frac{70-\mu}{\sigma}\right) \\ &= P\left(\frac{55-80}{14}<z<\frac{70-80}{14}\right) \\ &= P\left(\frac{-25}{14}<z<\frac{-10}{14}\right) \\ &= P(-1.78<z<-0.714) \end{aligned}$$

The two points are on the same sides of the mean, so calculate $P(-1.78<z<0)$, $P(-0.714<z<0)$ and subtract the two probabilities.

$$\begin{aligned} P(55<X<70) &= P(-1.78<z<0)-P(-0.714<z<0) \\ &= P(0<z<1.78)-P(0<z<0.714) \\ &= 0.4625-0.2611 \ \ (\text{from Appendix B1.}) \\ &= 0.2014 \end{aligned}$$

Result

a. 0.4017

b. 0.3594

c. 0.2014

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean tax refund is $\mu = \$2708$ and the standard deviation is $\sigma = \$650$

Let X be the tax refunds.

$\textbf{a. } X \text{ is more than } \$3000 \}$

$$\begin{aligned} P(X > 3000) &= P(X - \mu > 3000 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{3000 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{3000 - 2708}{650}\right) \\ &= P\left(z > \frac{292}{650}\right) \\ &= P(z > 0.45) \\ &= 0.5 - P(0 < z < 0.45) \\ &= 0.5 - 0.1736 \ \ (\text{from Appendix B1.}) \\ &= 0.3264 \end{aligned}$$

32.64% of the refunds are more than $\$3000$.

Step 2

$\textbf{b. } X \text{ is more than } \$3000 \text{ but less than } \$3500 \text{ }$

$$\begin{aligned} P(3000 < X < 3500) &= P(3000 - \mu < X - \mu < 3500 - \mu) \\ &= P\left(\frac{3000 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{3500 - \mu}{\sigma}\right) \\ &= P\left(\frac{3000 - 2708}{650} < z < \frac{3500 - 2708}{650}\right) \\ &= P\left(\frac{292}{650} < z < \frac{792}{650}\right) \\ &= P(0.45 < z < 1.22) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0 < z < 0.45)$, $P(0 < z < 1.22)$ and subtract the two probabilities.

$$\begin{aligned} P(3000 < X < 3500) &= P(0 < z < 1.22) - P(0 < z < 0.45) \\ &= 0.3888 - 0.1736 \ \ (\text{from Appendix B1.}) \\ &= 0.2152 \end{aligned}$$

21.52% of the refunds are more than $\$3000$ but less than $\$3500$.

Step 3

$\textbf{c. } X \text{ is more than } \$2500 \text{ but less than } \$3500 \text{ }$

$$\begin{aligned} P(2500 < X < 3500) &= P(2500 - \mu < X - \mu < 3500 - \mu) \\ &= P\left(\frac{2500 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{3500 - \mu}{\sigma}\right) \\ &= P\left(\frac{2500 - 2708}{650} < z < \frac{3500 - 2708}{650}\right) \\ &= P\left(\frac{-208}{650} < z < \frac{792}{650}\right) \\ &= P(-0.32 < z < 1.22) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.32 < z < 0)$, $P(0 < z < 1.22)$ and add the two probabilities.

$$\begin{aligned} P(2500 < X < 3500) &= P(-0.32 < z < 0) + P(0 < z < 1.22) \\ &= P(0 < z < 0.32) + P(0 < z < 1.22) \\ &= 0.1255 + 0.3888 \ \ (\text{from Appendix B1.}) \\ &= 0.5143 \end{aligned}$$

51.43% of the refunds are more than $\$2500$ but less than $\$3500$.

🎯 Result

a. 32.64%

b. 21.52%

c. 51.43%

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean of the number of viewers in American Idol is $\mu = 29$ million and the standard deviation is $\sigma = 5$ million.

Let X be the number of viewers.

$\textbf{a. } X \text{ is between } 30 \text{ and } 34 \}$

$$\begin{aligned} P(30 < X < 34) &= P(30 - \mu < X - \mu < 34 - \mu) \\ &= P\left(\frac{30 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{34 - \mu}{\sigma}\right) \\ &= P\left(\frac{30 - 29}{5} < z < \frac{34 - 29}{5}\right) \\ &= P\left(\frac{1}{4} < z < \frac{5}{5}\right) \\ &= P(0.25 < z < 1) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0 < z < 0.25)$, $P(0 < z < 1)$ and subtract the two probabilities.

$$\begin{aligned} P(30 < X < 34) &= P(0 < z < 1) - P(0 < z < 0.25) \\ &= 0.3413 - 0.0987 \ \ (\text{from Appendix B1.}) \\ &= 0.2426 \end{aligned}$$

Step 2

$\textbf{b. } X \text{ is at least } 23 \text{ million }}$

$$
\begin{aligned}
P(X > 23) &= P(X - \mu > 23 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} > \frac{23 - \mu}{\sigma}\right) \\
&= P\left(z > \frac{23 - 29}{5}\right) \\
&= P\left(z > \frac{-6}{5}\right) \\
&= P(z > -1.2) \\
&= P(z < 1.2) \\
&= 0.5 + P(0 < z < 1.2) \\
&= 0.5 + 0.3849 \ \ (\text{from Appendix B1.}) \\
&= 0.8849
\end{aligned}
$$

Step 3

$\textbf{c. } X \text{ exceeds } 40 \text{ million }}$

$$
\begin{aligned}
P(X > 40) &= P(X - \mu > 40 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} > \frac{40 - \mu}{\sigma}\right) \\
&= P\left(z > \frac{40 - 29}{5}\right) \\
&= P\left(z > \frac{11}{5}\right) \\
&= P(z > 2.2) \\
&= 0.5 - P(0 < z < 2.2) \\
&= 0.5 - 0.4861 \ \ (\text{from Appendix B1.}) \\
&= 0.0139
\end{aligned}
$$

Result

a. 0.2426

b. 0.8849

c. 0.0139

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean of the distribution of the lengths of time listeners are tuned is $\mu = 15$ minutes and the standard deviation is $\sigma = 3.5$

Let X be the length of time a listener is tuned.

$\textbf{a. } X \text{ is more than } 20 \text{ minutes }$

$$\begin{aligned} P(X > 20) &= P(X - \mu > 20 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{20 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{20 - 15}{3.5}\right) \\ &= P\left(z > \frac{5}{3.5}\right) \\ &= P(z > 1.43) \\ &= 0.5 - P(0 < z < 1.43) \\ &= 0.5 - 0.4236 \ \ (\text{from Appendix B1.}) \\ &= 0.0764 \end{aligned}$$

Step 2

$\textbf{b. } X \text{ is } 20 \text{ minutes or less }$

$$\begin{aligned} P(X \geq 20) &= P(X - \mu < 20 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{20 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{20 - 15}{3.5}\right) \\ &= P\left(z < \frac{5}{3.5}\right) \\ &= P(z < 1.43) \\ &= 0.5 + P(0 < z < 1.43) \\ &= 0.5 + 0.4236 \ \ (\text{from Appendix B1.}) \\ &= 0.9236 \end{aligned}$$

Step 3

$\textbf{c. } X \text{ is between } 10 \text{ and } 12 \text{ minutes }$

$$\begin{aligned} P(10 < X < 12) &= P(10 - \mu < X - \mu < 12 - \mu) \\ &= P\left(\frac{10 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{12 - \mu}{\sigma}\right) \\ &= P\left(\frac{10 - 15}{3.5} < z < \frac{12 - 15}{3.5}\right) \\ &= P\left(\frac{-5}{3.5} < z < \frac{-3}{3.5}\right) \\ &= P(-1.43 < z < -0.86) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(-1.43 < z < 0)$, $P(-0.86 < z < 0)$ and subtract the two probabilities.

$$\begin{aligned} P(10 < X < 12) &= P(-1.43 < z < -0.86) \\ &= P(0.86 < z < 1.43) \\ &= P(0 < z < 1.43) - P(0 < z < 0.86) \\ &= 0.4236 - 0.3051 \ \ (\text{from Appendix B1.}) \\ &= 0.1185 \end{aligned}$$

Result

a. 0.0764

b. 0.9236

c. 0.1185

PROBLEM STATEMENT

Chapter 7, Page 239

Step 1

Mean of the commuting time in New York is $\mu = 38.3$ minutes and the standard deviation is $\sigma = 7.5$ minutes.

Let X be the commuting time.

$\textbf{a. } X \text{ is less than } 30 \text{ minutes }$

$$\begin{aligned} P(X < 30) &= P(X - \mu < 30 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{30 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{30 - 38.3}{7.5}\right) \\ &= P\left(z < \frac{-8.3}{7.5}\right) \\ &= P(z < -1.11) \\ &= P(z > 1.11) \\ &= 0.5 + P(0 < z < 1.11) \\ &= 0.5 + 0.3665 \ \ (\text{from Appendix B1.}) \\ &= 0.8665 \end{aligned}$$

Step 2

$\textbf{b. X is between 30 and 35 minutes }$

$$\begin{aligned} P(30 < X < 35) &= P(30 - \mu < X - \mu < 35 - \mu) \\ &= P\left(\frac{30 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{35 - \mu}{\sigma}\right) \\ &= P\left(\frac{30 - 38.3}{7.5} < z < \frac{35 - 38.3}{7.5}\right) \\ &= P\left(\frac{-8.3}{7.5} < z < \frac{-3.3}{7.5}\right) \\ &= P(-1.11 < z < -0.44) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(-1.11 < z < 0)$, $P(-0.44 < z < 0)$ and subtract the two probabilities.

$$\begin{aligned} P(30 < X < 35) &= P(-1.11 < z < 0) - P(-0.44 < z < 0) \\ &= P(0 < z < 1.11) - P(0 < z < 0.44) \\ &= 0.3665 - 0.1700 \ \ (\text{from Appendix B1.}) \\ &= 0.1965 \end{aligned}$$

Step 3

$\textbf{c. X is between 30 and 40 minutes }$

$$\begin{aligned} P(30 < X < 40) &= P(30 - \mu < X - \mu < 40 - \mu) \\ &= P\left(\frac{30 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{40 - \mu}{\sigma}\right) \\ &= P\left(\frac{30 - 38.3}{7.5} < z < \frac{40 - 38.3}{7.5}\right) \\ &= P\left(\frac{-8.3}{7.5} < z < \frac{1.7}{7.5}\right) \\ &= P(-1.11 < z < 0.23) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.11 < z < 0)$, $P(0 < z < 0.23)$ and add the two probabilities.

$$\begin{aligned} P(30 < X < 40) &= P(-1.11 < z < 0) - P(0 < z < 0.23) \\ &= P(0 < z < 1.11) + P(0 < z < 0.23) \\ &= 0.3665 + 0.0910 \ \ (\text{from Appendix B1.}) \\ &= 0.4575 \end{aligned}$$

🎯 **Result**

a. 0.8665

b. 0.1965

c. 0.4575

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Mean of the normal distribution is $\mu = 50$ and the standard deviation is $\sigma = 4$.

Let X be the random variable.

Now,

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ \Rightarrow z &= \frac{X - 50}{4} \\ \text{and } z_1 &= \frac{X_1 - 50}{4} \end{aligned}$$

95 percent of the observations lie below X_1.

$$\begin{aligned} \Rightarrow P(z < z_1) &= 0.95 \\ \Rightarrow 0.5 + P(0 < z < z_1) &= 0.95 \\ \Rightarrow P(0 < z < z_1) &= 0.95 - 0.5 \\ \Rightarrow P(0 < z < z_1) &= 0.45 \\ \Rightarrow z_1 &= 1.64 \ \text{(from Appendix B1.)} \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 50}{4} \\ \Rightarrow 1.64 &= \frac{X_1 - 50}{4} \\ \Rightarrow X_1 - 50 &= 6.56 \\ \Rightarrow X_1 &= 50 + 6.65 \\ \Rightarrow X_1 &= 56.65 \end{aligned}$$

95 percent of the observations will lie below 56.65.

Result

$$56.65$$

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Mean of the normal distribution is $\mu = 80$ and the standard deviation is $\sigma = 14$.

Let X be the random variable.

Now,

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ \Rightarrow z &= \frac{X - 80}{14} \\ \text{and } z_1 &= \frac{X_1 - 80}{14} \end{aligned}$$

80 percent of the observations lie above X_1.

$$\begin{aligned} \Rightarrow P(z > z_1) &= 0.80 \\ \Rightarrow 0.5 + P(-z_1 < z < 0) &= 0.80 \\ \Rightarrow P(-z_1 < z < 0) &= 0.80 - 0.5 \\ \Rightarrow P(-z_1 < z < 0) &= 0.30 \\ \Rightarrow -z_1 &= 0.84 \ \text{(from Appendix B1.)} \\ \Rightarrow z_1 &= -0.84 \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 80}{14} \\ \Rightarrow -0.84 &= \frac{X_1 - 80}{14} \\ \Rightarrow X_1 - 80 &= -11.76 \\ \Rightarrow X_1 &= -11.76 + 80 \\ \Rightarrow X_1 &= 68.24 \end{aligned}$$

80 percent of the observations will lie below 68.24.

Result

$$68.24$$

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Mean of the operating cost of airplane is $\mu = \$2100$ per hour and the standard deviation is $\sigma = \$250$.

Let X be the random variable.

Now,

$$\begin{aligned} z &= \frac{X-\mu}{\sigma} \\ \Rightarrow z &= \frac{X-2100}{250} \\ \text{and } z_1 &= \frac{X_1-2100}{250} \end{aligned}$$

3 percent of the observations lie below X_1.

$$\begin{aligned} \Rightarrow P(z < -z_1) &= 0.03 \\ \Rightarrow 0.5 - P(-z_1 < z < 0) &= 0.03 \\ \Rightarrow P(-z_1 < z < 0) &= 0.5 - 0.03 \\ \Rightarrow P(-z_1 < z < 0) &= 0.47 \\ \Rightarrow -z_1 &= 1.88 \ \ (\text{from Appendix B1.}) \\ \Rightarrow z_1 &= -1.88 \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1-50}{4} \\ \Rightarrow -1.88 &= \frac{X_1-2100}{250} \\ \Rightarrow X_1 - 2100 &= -470 \\ \Rightarrow X_1 &= 2100 - 470 \\ \Rightarrow X_1 &= 1630 \end{aligned}$$

Operating cost for the lowest 3 percent of the airlines is $\$1630$.

Result

$$\$1630$$

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Mean of the SAT scores is $\mu = 1500$ and the standard deviation is $\sigma = 300$.

Let X be the random variable.

Now,

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ \Rightarrow z &= \frac{X - 1500}{300} \\ \text{and } z_1 &= \frac{X_1 - 1500}{300} \end{aligned}$$

10 percent of the observations lie above X_1.

$$\begin{aligned} \Rightarrow P(z > z_1) &= 0.10 \\ \Rightarrow 0.5 - P(0 < z < z_1) &= 0.10 \\ \Rightarrow P(0 < z < z_1) &= 0.5 - 0.1 \\ \Rightarrow P(0 < z < z_1) &= 0.4 \\ \Rightarrow z_1 &= 1.28 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 1500}{300} \\ \Rightarrow 1.28 &= \frac{X_1 - 1500}{300} \\ \Rightarrow X_1 - 1500 &= 384 \\ \Rightarrow X_1 &= 1500 + 384 \\ \Rightarrow X_1 &= 1884 \end{aligned}$$

Minimum score to qualify for the scholarship is 1884..

Result

$$1884$$

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Given:

$$\mu = \text{Mean} = 195$$
$$\sigma = \text{Standard deviation} = 8.5$$

(a) The cuttoff value of the top 1% has the property that $100\% - 1\% = 99\%$ of the data values lie below the cutoff value.

Let us determine the z-score that corresponds with a probability of $99\% - 50\% = 49\%$ or 0.49 in the normal probability table of the appendix. We note that the closest probability is 0.4901 which lies in the row 2.3 and in the column .03 of the normal probability table and thus the corresponding z-score is then $2.3 + .03 = 2.33$.

$$z = 2.33$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{x - 195}{8.5}$$

The two found expressions of the z-score then have to be equal:

$$\frac{x - 195}{8.5} = 2.33$$

Multiply each side by 8.5:

$$x - 195 = 2.33(8.5)$$

Add 195 to each side:

$$x = 195 + 2.33(8.5)$$

Evaluate:

$$x = 4.544$$

Thus the 20th percentile is then 4.544 minutes.

Step 2

(b) Given:

$$\mu = \text{Mean} = 290$$
$$\sigma = \text{Standard deviation} = 8.5$$

The cutoff value of the bottom 1% has the property that 1% of the data values lie below the cutoff value.

Let us determine the z-score that corresponds with a probability of $100\% - 1\% - 50\% = 49\%$ or 0.49 in the normal probability table of the appendix (the z-value corresponding to the bottom 1% is then the negative of this z-value). We note that the closest probability is 0.4901 which lies in the row 2.3 and in the column .03 of the normal probability table and thus the corresponding z-score is then $2.3 + .03 = 2.33$.

$$z = -2.33$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{x - 290}{8.5}$$

The two found expressions of the z-score then have to be equal:

$$\frac{x - 290}{8.5} = -2.33$$

Multiply each side by 8.5:

$$x - 290 = -2.33(8.5)$$

Add 290 to each side:

$$x = 290 - 2.33(8.5)$$

Evaluate:

$$x = 270.195$$

Thus the bottom 1% listened for 270.195 hours or less.

Result

(a) At least 214.805 hours

(b) At most 270.195 hours

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Mean annual cost of private university is $\mu = \$26889$ and the standard deviation is $\sigma = \$4500$.

Let X be the annual cost.

Now,

$$\begin{aligned} z &= \frac{X-\mu}{\sigma} \\ \Rightarrow z &= \frac{X-50}{4} \\ \text{and } z_1 &= \frac{X_1-50}{4} \end{aligned}$$

95 percent of the students pay less than X_1.

$$\begin{aligned} \Rightarrow P(z < z_1) &= 0.95 \\ \Rightarrow 0.5 + P(0 < z < z_1) &= 0.95 \\ \Rightarrow P(0 < z < z_1) &= 0.95 - 0.5 \\ \Rightarrow P(0 < z < z_1) &= 0.45 \\ \Rightarrow z_1 &= 1.64 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 26889}{4500} \\ \Rightarrow 1.64 &= \frac{X_1 - 26889}{4500} \\ \Rightarrow X_1 - 26889 &= 7380 \\ \Rightarrow X_1 &= 26889 + 7380 \\ \Rightarrow X_1 &= 34269 \end{aligned}$$

95 percent of the students pay less than $\$34269$.

Result

$$34269$$

PROBLEM STATEMENT

Chapter 7, Page 241

Step 1

Given:

$$\mu = \text{Mean} = 12$$
$$\sigma = \text{Standard deviation} = 18$$

We are interested in the cutoff value for the top $\frac{1}{20} = 0.05 = 5\%$, which has the property that $100\% - 5\% = 95\%$ of the data values are below it.

Let us determine the z-score that corresponds with a probability of $95\% - 50\% = 45\%$ or 0.45 in the normal probability table of the appendix. We note that 0.45 lies exactly in the middle between the probabilities 0.4495 and 0.4505, which correspond to the z-scores 1.64 and 1.65. Thus we estimate the z-score of 0.45 as the z-score exactly in the middle between 1.64 and 1.65.

$$z = 1.645$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{x - 12}{18}$$

The two found expressions of the z-score then have to be equal:

$$\frac{x - 12}{0.9} = 1.645$$

Multiply each side by 18:

$$x - 12 = 1.645(18)$$

Add 12 to each side:

$$x = 12 + 1.645(18)$$

Evaluate:

$$x = 41.61$$

Thus the hurdle rate should be 41.61%.

Note: You obtain the hurdle rate 41.7% when using the z-score $z = 1.65$ instead.

🎯 Result

41.61%

PROBLEM STATEMENT

Chapter 7, Page 242

Step 1

Mean number of pages a cartridge will print is $\mu = 12200$ and the standard deviation is $\sigma = 820$.

Let X be the number of pages.

Now,

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ \Rightarrow z &= \frac{X - 12200}{820} \\ \text{and } \ z_1 &= \frac{X_1 - 12200}{820} \end{aligned}$$

Manufacturer is correct 99 percent of the time if number of pages is less than X_1.

$$\begin{aligned} \Rightarrow P(z < z_1) &= 0.99 \\ \Rightarrow 0.5 + P(0 < z < z_1) &= 0.99 \\ \Rightarrow P(0 < z < z_1) &= 0.99 - 0.5 \\ \Rightarrow P(0 < z < z_1) &= 0.49 \\ \Rightarrow z_1 &= 2.33 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 12200}{820} \\ \Rightarrow 2.33 &= \frac{X_1 - 12200}{820} \\ \Rightarrow X_1 - 12200 &= 1910.6 \\ \Rightarrow X_1 &= 12200 + 1910.6 \\ \Rightarrow X_1 &= 14110.6 \end{aligned}$$

99 percent of the time he will be correct of the number of pages is less than around 14110.

Result

$$14110$$

PROBLEM STATEMENT

Chapter 7, Page 245

Step 1

The binomial distribution has $n = 50$ and $\pi = 0.25$.

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 50 \times 0.25 \\ &= 12.5\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{50 \times 0.25 \times (1-0.25)} \\ &= \sqrt{9.375} \\ &= 3.062\end{aligned}$$

Step 2

$\textbf{b. Probability that X is 15 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 15 - 0.5 = 14.5$.

$$\textbf{z value with $X' = 14.5$}$$

$$\begin{aligned} z &= \frac{X' - \mu}{\sigma} \\ &= \frac{14.5 - 12.5}{3.062} \\ &= \frac{2}{3.062} \\ &= 0.653 \end{aligned}$$

Now,

$$\begin{aligned} P(12.5 < X' < 14.5) &= P(12.5 - \mu < X' - \mu < 14.5 - \mu) \\ &= P\left(\frac{12.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{14.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{12.5 - 12.5}{3.062} < z < \frac{14.5 - 12.5}{3.062}\right) \\ &= P(0 < z < 0.653) \\ &= 0.2422 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 14.5) &= 0.5 - P(12.5 < X' < 14.5) \\ &= 0.5 - 0.2422 \\ &= 0.2578 \end{aligned}$$

Step 3

$\textbf{c. Probability that X is 10 or less}$

For the probability X or more occurs, use the area below $X + 0.5$ which is $X' = 10 + 0.5 = 10.5$.

$$\textbf{z value with $X' = 10.5$}$$

$$\begin{aligned} z &= \frac{X' - \mu}{\sigma} \\ &= \frac{10.5 - 12.5}{3.062} \\ &= \frac{-2}{3.062} \\ &= -0.653 \end{aligned}$$

Now,

$$\begin{aligned} P(10.5 < X' < 12.5) &= P(10.5 - \mu < X' - \mu < 12.5 - \mu) \\ &= P\left(\frac{10.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{12.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{10.5 - 12.5}{3.062} < z < \frac{12.5 - 12.5}{3.062}\right) \\ &= P(-0.653 < z < 0) \\ &= P(0 < z < 0.653) \\ &= 0.2242 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' < 10.5) &= 0.5 - P(10.5 < X' < 12.5) \\ &= 0.5 - 0.2422 \\ &= 0.2578 \end{aligned}$$

Result

a. $\mu = 12.5$ and $\sigma = 3.062$

b. 0.2578

c. 0.2578

PROBLEM STATEMENT

Chapter 7, Page 245

Step 1

The binomial distribution has $n = 40$ and $\pi = 0.55$.

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 40 \times 0.55 \\ &= 22\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{40 \times 0.55 \times (1-0.55)} \\ &= \sqrt{9.9} \\ &= 3.146\end{aligned}$$

Step 2

$\textbf{b. Probability that X is 25 or greater}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 25 - 0.5 = 24.5$.

$$\textbf{z value with } X' = \textbf{24.5}$$

$$\begin{aligned} z &= \frac{X' - \mu}{\sigma} \\ &= \frac{24.5 - 22}{3.146} \\ &= \frac{2.5}{3.146} \\ &= 0.79 \end{aligned}$$

Now,

$$\begin{aligned} P(22 < X' < 24.5) &= P(22 - \mu < X' - \mu < 24.5 - \mu) \\ &= P\left(\frac{22 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{24.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{22 - 22}{3.146} < z < \frac{24.5 - 22}{3.146}\right) \\ &= P(0 < z < 0.79) \\ &= 0.2852 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 24.5) &= 0.5 - P(22 < X' < 24.5) \\ &= 0.5 - 0.2852 \\ &= 0.2148 \end{aligned}$$

Step 3

$\textbf{c. Probability that X is 15 or less}$

For the probability X or more occurs, use the area below $X + 0.5$ which is $X' = 15 + 0.5 = 15.5$.

$$\textbf{z value with $X' = 15.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{15.5 - 22}{3.146} \\ &= \frac{-6.5}{3.146} \\ &= -2.06 \end{aligned}$$

Now,

$$\begin{aligned} P(15.5 < X' < 22) &= P(15.5 - \mu < X' - \mu < 22 - \mu) \\ &= P\left(\frac{15.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{22 - \mu}{\sigma}\right) \\ &= P\left(\frac{15.5 - 22}{3.146} < z < \frac{22 - 22}{3.146}\right) \\ &= P(-2.06 < z < 0) \\ &= P(0 < z < 2.06) \\ &= 0.4803 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' < 15.5) &= 0.5 - P(15.5 < X' < 22) \\ &= 0.5 - 0.4803 \\ &= 0.0197 \end{aligned}$$

Step 4

$\textbf{c. Probability that X is 15 and 25 inclusive}$

$$P(15 < X < 25) = P(22 < X < 25) + P(15 < X < 22)$$

i. $P(22 < X < 25)$

For the probability X or less occurs, use the area below $X + 0.5$ which is $X' = 15 - 0.5 = 14.5$.

$$\textbf{z value with } X' = \textbf{25.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{25.5 - 22}{3.146} \\ &= \frac{3.5}{3.146} \\ &= 1.11 \end{aligned}$$

Now,

$$\begin{aligned} P(22 < X' < 25.5) &= P(22 - \mu < X' - \mu < 25.5 - \mu) \\ &= P\left(\frac{22 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{25.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{22 - 22}{3.146} < z < \frac{25.5 - 22}{3.146}\right) \\ &= P(0 < z < 1.11) \\ &= 0.3665 \ \text{ from the appendix B1} \end{aligned}$$

Step 5

ii. $P(15 < X < 22)$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 15 - 0.5 = 14.5$.

$$\textbf{z value with } X' = \textbf{14.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{14.5 - 22}{3.146} \\ &= \frac{-7.5}{3.146} \\ &= -2.38 \end{aligned}$$

Now,

$$\begin{aligned} P(14.5 < X' < 22) &= P(14.5 - \mu < X' - \mu < 22 - \mu) \\ &= P\left(\frac{14.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{22 - \mu}{\sigma}\right) \\ &= P\left(\frac{14.5 - 22}{3.146} < z < \frac{22 - 22}{3.146}\right) \\ &= P(-2.38 < z < 0) \\ &= P(0 < z < 2.38) \\ &= 0.4913 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(15 < X < 25) &= 0.3665 + 0.4913 \\ &= 0.8578 \end{aligned}$$

Result

a. $\mu = 22$ and $\sigma = 3.146$

b. 0.2148

c. 0.0197

d. 0.8578

PROBLEM STATEMENT

Chapter 7, Page 245

Step 1

The probability that an error was made on the tax returns is $\pi = 7\%$ or 0.07.

The number of tax returns $n = 80$.

Let X be the number of tax returns with errors.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 80 \times 0.07 \\ &= 5.6\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{80 \times 0.07 \times (1-0.07)} \\ &= \sqrt{5.208} \\ &= 2.28\end{aligned}$$

Step 2

$\textbf{a. Probability that X is more than 6}$

For the probability that more than X occurs, use the area above $X + 0.5$ which is $X' = 6 + 0.5 = 6.5$.

$$\textbf{z value with $X' = 6.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{6.5 - 5.6}{2.28} \\ &= \frac{0.9}{2.28} \\ &= 0.395 \end{aligned}$$

Now,

$$\begin{aligned} P(5.6 < X' < 6.5) &= P(5.6 - \mu < X' - \mu < 6.5 - \mu) \\ &= P\left(\frac{5.6 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{6.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{5.6 - 5.6}{2.28} < z < \frac{6.5 - 5.6}{2.28}\right) \\ &= P(0 < z < 0.395) \\ &= 0.1517 \ \text{ from the appendix B1} \end{aligned}$$

$$\begin{aligned} P(X' > 6.5) &= 0.5 - P(5.6 < X' < 6.5) \\ &= 0.5 - 0.1517 \\ &= 0.3483 \end{aligned}$$

Step 3

$\textbf{b. Probability that X is 6 or more}$

For the probability that X or more occurs, use the area above $X - 0.5$ which is $X' = 6 - 0.5 = 5.5$.

$$\textbf{z value with $X' = 5.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{5.5 - 5.6}{2.28} \\ &= \frac{-0.1}{2.28} \\ &= -0.04 \end{aligned}$$

Now,

$$\begin{aligned} P(5.5 < X' < 5.6) &= P(5.5 - \mu < X' - \mu < 5.6 - \mu) \\ &= P\left(\frac{5.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{5.6 - \mu}{\sigma}\right) \\ &= P\left(\frac{5.5 - 5.6}{2.28} < z < \frac{5.6 - 5.6}{2.28}\right) \\ &= P(-0.04 < z < 0) \\ &= P(0 < z < 0.04) \\ &= 0.0160 \ \text{ from the appendix B1} \end{aligned}$$

$$\begin{aligned} P(X' > 5.5) &= 0.5 + P(5.5 < X' < 6.5) \\ &= 0.5 + 0.0160 \\ &= 0.5160 \end{aligned}$$

Step 4

c. Probability that X is exactly 6

$$\begin{aligned} P(X = 6) &= P(X \text{ is 6 or more}) - P(X \text{ is more than 6} \\ &= 0.5160 - 0.3483 \\ &= 0.1677 \end{aligned}$$

🎯 Result

a. 0.3483

b. 0.5160

c. 0.1677

PROBLEM STATEMENT

Chapter 7, Page 246

Step 1

Probability that the mufflers were not installed in more than 30 minutes is $\pi = 20\%$ or 0.2.

Number of installed mufflers is $n = 50$.

Let X be the number of mufflers that took more than 30 minutes in installation.

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 50 \times 0.20 \\ &= 10\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{50 \times 0.20 \times (1-0.20)} \\ &= \sqrt{8} \\ &= 2.83\end{aligned}$$

Step 2

$\textbf{b. Probability that X is less than 8}$

For the probability that less than X occurs, use the area below $X - 0.5$ which is $X' = 8 - 0.5 = 7.5$.

$$\textbf{z value with $X' = 7.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{7.5 - 10}{2.83} \\ &= \frac{-2.5}{2.83} \\ &= -0.88 \end{aligned}$$

Now,

$$\begin{aligned} P(7.5 < X' < 10) &= P(7.5 - \mu < X' - \mu < 10 - \mu) \\ &= P\left(\frac{7.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{10 - \mu}{\sigma}\right) \\ &= P\left(\frac{7.5 - 10}{2.83} < z < \frac{10 - 10}{2.83}\right) \\ &= P(-0.88 < z < 0) \\ &= P(0 < z < 0.88) \\ &= 0.3106 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 7.5) &= 0.5 - P(7.5 < X' < 10) \\ &= 0.5 - 0.3106 \\ &= 0.1894 \end{aligned}$$

Step 3

$\textbf{c. Probability that X is 8 or less}$

For the probability that less than X occurs, use the area below $X + 0.5$ which is $X' = 8 + 0.5 = 8.5$.

z value with $X' = 8.5$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{8.5 - 10}{2.83} \\ &= \frac{-1.5}{2.83} \\ &= -0.53 \end{aligned}$$

Now,

$$\begin{aligned} P(8.5 < X' < 10) &= P(8.5 - \mu < X' - \mu < 10 - \mu) \\ &= P\left(\frac{8.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{10 - \mu}{\sigma}\right) \\ &= P\left(\frac{8.5 - 10}{2.83} < z < \frac{10 - 10}{2.83}\right) \\ &= P(-0.53 < z < 0) \\ &= P(0 < z < 0.53) \\ &= 0.2019 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 8.5) &= 0.5 - P(8.5 < X' < 10) \\ &= 0.5 - 0.2019 \\ &= 0.2981 \end{aligned}$$

Step 4

d. Probability that X is 8

$$\begin{aligned} P(X = 8) &= P(X \text{ is less than or equal to } 8) - P(X \text{ is less than } 8) \\ &= 0.2981 - 0.1894 \\ &= 0.1087 \end{aligned}$$

🎯 Result

a. 10

b. 0.1894

c. 0.2981

d. 0.1087

PROBLEM STATEMENT

Chapter 7, Page 246

Step 1

Probability that new members are significantly overweight is $\pi = 30\%$ or 0.3.

Number of new memebrs $n = 500$.

Let X be the new members

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 500 \times 0.3 \\ &= 150\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{500 \times 0.3 \times (1-0.3)} \\ &= \sqrt{105} \\ &= 10.25\end{aligned}$$

In this case, the normal probability distribution is a good approximation of the binomial distribution because $n\pi \geq 5$ and $n(1-\pi) \geq 5$.

Step 2

$\textbf{b. Probability that X is 175 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 175 - 0.5 = 174.5$.

$$\textbf{z value with } X' = \mathbf{174.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{174.5 - 150}{10.25} \\ &= \frac{24.5}{10.25} \\ &= 2.4 \end{aligned}$$

Now,

$$\begin{aligned} P(150 < X' < 174.5) &= P(150 - \mu < X' - \mu < 174.5 - \mu) \\ &= P\left(\frac{150 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{174.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{150 - 150}{10.25} < z < \frac{174.5 - 150}{10.25}\right) \\ &= P(0 < z < 2.4) \\ &= 0.4918 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 174.5) &= 0.5 - P(150 < X' < 174.5) \\ &= 0.5 - 0.4918 \\ &= 0.0082 \end{aligned}$$

Step 3

$\textbf{c. Probability that X is 140 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 140 - 0.5 = 139.5$.

$$\textbf{z value with } X' = \textbf{139.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{139.5 - 150}{10.25} \\ &= \frac{-10.5}{10.25} \\ &= -1.02 \end{aligned}$$

Now,

$$\begin{aligned} P(139.5 < X' < 150) &= P(139.5 - \mu < X' - \mu < 150 - \mu) \\ &= P\left(\frac{139.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{150 - \mu}{\sigma}\right) \\ &= P\left(\frac{139.5 - 150}{10.25} < z < \frac{150 - 150}{10.25}\right) \\ &= P(-1.02 < z < 0) \\ &= P(0 < z < 1.02) \\ &= 0.3461 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 139.5) &= 0.5 + P(139.5 < X' < 150) \\ &= 0.5 + 0.3461 \\ &= 0.8461 \end{aligned}$$

Result

a. $\mu = 150$ and $\sigma = 10.25$

b. 0.0082

c. 0.8461

PROBLEM STATEMENT

Chapter 7, Page 246

Step 1

Probability that those who are sent an invitation to respond that they will attend is $\pi = \frac{2}{3} = 0.67$.

Number of invitations is $n = 197$.

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 197 \times 0.67 \\ &= 131.99\end{aligned}$$

b. Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{197 \times 0.67 \times (1-0.67)} \\ &= \sqrt{43.56} \\ &= 6.6\end{aligned}$$

Step 2

$\textbf{c. Probability that X is 140 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 140 - 0.5 = 139.5$.

$$\textbf{z value with $X' = 139.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{139.5 - 131.99}{6.6} \\ &= \frac{7.51}{6.6} \\ &= 1.14 \end{aligned}$$

Now,

$$\begin{aligned} P(131.99 < X' < 139.5) &= P(131.99 - \mu < X' - \mu < 139.5 - \mu) \\ &= P\left(\frac{131.99 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{139.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{131.99 - 131.99}{6.6} < z < \frac{139.5 - 131.99}{6.6}\right) \\ &= P(0 < z < 1.14) \\ &= 0.3729 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 139.5) &= 0.5 - P(131.99 < X' < 139.5) \\ &= 0.5 - 0.3729 \\ &= 0.1271 \end{aligned}$$

Step 3

$\textbf{ Probability that X is more than 140}$

For the probability that more than X occurs, use the area above $X + 0.5$ which is $X' = 140 + 0.5 = 140.5$.

$$\textbf{z value with } X' = \textbf{140.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{140.5 - 131.99}{6.6} \\ &= \frac{8.51}{6.6} \\ &= 1.29 \end{aligned}$$

Now,

$$\begin{aligned} P(131.99 < X' < 140.5) &= P(131.99 - \mu < X' - \mu < 140.5 - \mu) \\ &= P\left(\frac{131.99 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{140.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{131.99 - 131.99}{6.6} < z < \frac{140.5 - 131.99}{6.6}\right) \\ &= P(0 < z < 1.29) \\ &= 0.4015 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 140.5) &= 0.5 - P(131.99 < X' < 140.5) \\ &= 0.5 - 0.4015 \\ &= 0.0985 \end{aligned}$$

d. Probability that X is 140

$$\begin{aligned} P(X = 140) &= P(X \text{ is 140 or more}) - P(X \text{ is more than 140}) \\ &= 0.1271 - 0.0985 \\ &= 0.0286 \end{aligned}$$

🎯 Result

a. 131.99

b. 6.6

c. 0.1271

d. 0.0286

PROBLEM STATEMENT

Chapter 7, Page 250

Step 1

Waiting time follows an exponential distribution with mean $\mu = 60$ seconds. This implies $\lambda = \frac{1}{\mu} = \frac{1}{60}$

Let X be the waiting time.

a. X is less than 30 seconds

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 30) &= 1 - e^{-\lambda \times 30} \\ &= 1 - e^{-\frac{1}{60} \times 30} \\ &= 1 - e^{-\frac{1}{2}} \\ &= 1 - 0.6065 \\ &= 0.3935 \end{aligned}$$

b. X is more than 120 seconds

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(X > 120) &= 1 - (1 - e^{-\lambda \times 120}) \\ &= e^{-\frac{1}{60} \times 120} \\ &= e^{-2} \\ &= 0.135 \end{aligned}$$

Step 2

c. X is between 45 and 75 seconds

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(45 < X < 75) &= 1 - e^{-\lambda \times 75} - (1 - e^{-\lambda \times 45}) \\ &= 1 - e^{-\frac{1}{60} \times 75} - 1 + e^{-\frac{1}{60} \times 45} \\ &= -e^{-\frac{5}{4}} + e^{-\frac{3}{4}} \\ &= -0.286 + 0.472 \\ &= 0.186 \end{aligned}$$

d. 50 percent patrons waiting for less than X_1 seconds

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.5 &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.5 &= 1 - e^{-\frac{1}{60} \times X_1} \\ \Rightarrow e^{-\frac{X_1}{60}} &= 0.5 \end{aligned}$$

Taking log on both sides

$$\begin{aligned} \Rightarrow -\frac{X_1}{60} \ln e &= \ln 0.5 \\ \Rightarrow -\frac{X_1}{60} &= -0.693 \\ \Rightarrow X_1 &= 41.58 \end{aligned}$$

So, median is 41.58 seconds.

🎯 Result

a. 0.3935

b. 0.135

c. 0.186

d. 41.58 seconds

PROBLEM STATEMENT

Chapter 7, Page 250

Step 1

The mean lifetime of TV is $\mu = 100000$ hours. This implies $\lambda = \frac{1}{\mu} = \frac{1}{100000}$

Let X be the failing time.

a. X is less than 10000 hours

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 10000) &= 1 - e^{-\lambda \times 10000} \\ &= 1 - e^{-\frac{1}{100000} \times 10000} \\ &= 1 - e^{-\frac{1}{10}} \\ &= 1 - 0.905 \\ &= 0.095 \end{aligned}$$

b. X is more than 120000 hours

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(X > 120000) &= 1 - (1 - e^{-\lambda \times 120000}) \\ &= e^{-\frac{1}{100000} \times 120000} \\ &= e^{-\frac{6}{5}} \\ &= 0.3012 \end{aligned}$$

Step 2

c. X is between 60000 and 100000 hours

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(60000 < X < 100000) &= 1 - e^{-\lambda \times 100000} - (1 - e^{-\lambda \times 60000}) \\ &= 1 - e^{-\frac{1}{100000} \times 100000} - 1 + e^{-\frac{1}{100000} \times 60000} \\ &= -e^{-1} + e^{-\frac{3}{5}} \\ &= -0.3679 + 0.472 \\ &= 0.186 \end{aligned}$$

d. 10 percent of TV sets lasts more than than X_1 seconds

$$\begin{gathered} P(X > X_1) = 1 - P(X < X_1) \\ \Rightarrow P(X > X_1) = 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow 0.1 = e^{-\lambda X_1} \\ \Rightarrow 0.1 = e^{-\frac{1}{100000} \times X_1} \\ \Rightarrow e^{-\frac{X_1}{100000}} = 0.1 \\ \text{Taking log on both sides} \\ \Rightarrow -\frac{X_1}{100000} \ln e = \ln 0.1 \\ \Rightarrow -\frac{X_1}{100000} = -2.302 \\ \Rightarrow X_1 = 230200 \end{gathered}$$

So, 10 percent of the TV sets could last more than 230200 seconds.

Result

a. 0.095

b. 0.3012

c. 0.186

d. 230200

PROBLEM STATEMENT

Chapter 7, Page 250

Step 1

The mean amount of time spent on computer for leisure by individuals aged 75 years and above is $\mu_1 = 18$ minutes. This implies $\lambda_1 = \frac{1}{\mu_1} = \frac{1}{18}$

The mean amount of time spent on computer for leisure by individuals aged 15 to 19 years is $\mu_2 = 60$ minutes. This implies $\lambda_2 = \frac{1}{\mu_2} = \frac{1}{60}$

Let X be the amount of time spent on computer for leisure.

$\textbf{a. X is less than 15 minutes }$

i. For individuals aged 75 years and above

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda_1 X_1} \\ \Rightarrow P(X < 15) &= 1 - e^{-\lambda_1 \times 15} \\ &= 1 - e^{-\frac{1}{18} \times 15} \\ &= 1 - e^{-\frac{5}{6}} \\ &= 1 - 0.4346 \\ &= 0.5654 \end{aligned}$$

ii. For individuals aged 15 to 19 years

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda_2 X_1} \\ \Rightarrow P(X < 15) &= 1 - e^{-\lambda_2 \times 15} \\ &= 1 - e^{-\frac{1}{60} \times 15} \\ &= 1 - e^{-\frac{1}{4}} \\ &= 1 - 0.7788 \\ &= 0.2212 \end{aligned}$$

Step 2

$\textbf{b. } X \text{ is more than 2 hours or 120 minutes }$

i. For individuals aged 75 years and above

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda_1 X_1}) \\ \Rightarrow P(X > 120) &= 1 - (1 - e^{-\lambda_1 \times 120}) \\ &= e^{-\frac{1}{18} \times 120} \\ &= e^{-\frac{20}{3}} \\ &= 0.0013 \end{aligned}$$

ii. For individuals aged 15 to 19 years

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda_2 X_1}) \\ \Rightarrow P(X > 120) &= 1 - (1 - e^{-\lambda_2 \times 120}) \\ &= e^{-\frac{1}{60} \times 120} \\ &= e^{-2} \\ &= 0.1353 \end{aligned}$$

Step 3

$\textbf{c. X is between 30 and 90 minutes }$

i. For individuals aged 75 years and above

$$
\begin{aligned}
P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\
&= 1 - e^{-\lambda_1 X_2} - (1 - e^{-\lambda_1 X_1}) \\
\Rightarrow P(30 < X < 90) &= 1 - e^{-\lambda_1 \times 90} - (1 - e^{-\lambda_1 \times 30}) \\
&= 1 - e^{-\frac{1}{18} \times 90} - 1 + e^{-\frac{1}{18} \times 30} \\
&= -e^{-5} + e^{-\frac{5}{3}} \\
&= -0.0067 + 0.1889 \\
&= 0.1822
\end{aligned}
$$

ii. For individuals aged 15 to 19 years

$$
\begin{aligned}
P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\
&= 1 - e^{-\lambda_2 X_2} - (1 - e^{-\lambda_2 X_1}) \\
\Rightarrow P(30 < X < 90) &= 1 - e^{-\lambda_2 \times 90} - (1 - e^{-\lambda_2 \times 30}) \\
&= 1 - e^{-\frac{1}{60} \times 90} - 1 + e^{-\frac{1}{60} \times 30} \\
&= -e^{-\frac{3}{2}} + e^{-\frac{1}{2}} \\
&= -0.2231 + 0.6065 \\
&= 0.3834
\end{aligned}
$$

Step 4

$\textbf{d. 80 percent spent more than than X_1 minutes }$

i. For individuals aged 75 years and above

$$
\begin{aligned}
P(X > X_1) &= 1 - P(X < X_1) \\
\Rightarrow P(X > X_1) &= 1 - (1 - e^{-\lambda_1 X_1}) \\
\Rightarrow 0.8 &= e^{-\lambda_1 X_1} \\
\Rightarrow 0.8 &= e^{-\frac{1}{18} \times X_1} \\
\Rightarrow e^{-\frac{X_1}{18}} &= 0.8 \\
&\text{Taking log on both sides} \\
\Rightarrow -\frac{X_1}{18} \ln e &= \ln 0.8 \\
\Rightarrow -\frac{X_1}{18} &= -0.2231 \\
\Rightarrow X_1 &= 4.0158
\end{aligned}
$$

So, 80 percent of the individuals aged 75 years and above spend more than 4.0158 minutes.

ii. For individuals aged 15 to 19 years

$$
\begin{aligned}
P(X > X_1) &= 1 - P(X < X_1) \\
\Rightarrow P(X > X_1) &= 1 - (1 - e^{-\lambda_2 X_1}) \\
\Rightarrow 0.8 &= e^{-\lambda_2 X_1} \\
\Rightarrow 0.8 &= e^{-\frac{1}{60} \times X_1} \\
\Rightarrow e^{-\frac{X_1}{60}} &= 0.8 \\
&\text{Taking log on both sides} \\
\Rightarrow -\frac{X_1}{60} \ln e &= \ln 0.8 \\
\Rightarrow -\frac{X_1}{60} &= -0.2231 \\
\Rightarrow X_1 &= 13.386
\end{aligned}
$$

So, 80 percent of the individuals aged 15 to 19 years spend more than 13.386 minutes.

🎯 Result

a. i. 0.5654 ii. 0.2212

b. i. 0.0013 ii. 0.1353

b. i. 0.1822 ii. 0.3834

b. i. 4.0158 ii. 13.386

PROBLEM STATEMENT

Chapter 7, Page 250

Step 1

The mean cost per item is $\mu = \$3.50$. This implies $\lambda = \frac{1}{\mu} = \frac{1}{3.50}$

Let X be the cost per item.

a. X is less than \$1

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 1) &= 1 - e^{-\lambda \times 1} \\ &= 1 - e^{-\frac{1}{3.50} \times 1} \\ &= 1 - e^{-\frac{1}{3.50}} \\ &= 1 - 0.0302 \\ &= 0.9698 \end{aligned}$$

b. X is more than \$4

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(X > 4) &= 1 - (1 - e^{-\lambda \times 4}) \\ &= e^{-\frac{1}{3.50} \times 4} \\ &= e^{-\frac{8}{7}} \\ &= 0.3189 \end{aligned}$$

Step 2

c. X is between \$2 and \$3

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(2 < X < 3) &= 1 - e^{-\lambda \times 3} - (1 - e^{-\lambda \times 2}) \\ &= 1 - e^{-\frac{1}{3.50} \times 3} - 1 + e^{-\frac{1}{3.50} \times 2} \\ &= -e^{-\frac{6}{7}} + e^{-\frac{4}{7}} \\ &= -0.4244 + 0.5647 \\ &= 0.1403 \end{aligned}$$

d. 60 percent of the supermarket items cost more than X_1

$$\begin{aligned} P(X > X_1) &= e^{-\lambda X_1} \\ \Rightarrow 0.6 &= e^{-\lambda X_1} \\ \Rightarrow 0.6 &= e^{-\frac{1}{3.50} \times X_1} \\ \Rightarrow e^{-\frac{X_1}{3.50}} &= 0.6 \end{aligned}$$

Taking log on both sides

$$\begin{aligned} \Rightarrow -\frac{X_1}{3.50} \ln e &= \ln 0.6 \\ \Rightarrow -\frac{X_1}{3.50} &= -0.5108 \\ \Rightarrow X_1 &= 1.7878 \end{aligned}$$

60 percent of the supermarket items cost more than \$1.7878

Result

a. 0.9698

b. 0.3189

c. 0.1403

d. \$1.7878

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

The amount of Coca Cola follows the uniform distribution between 11.96 ounces and 12.05 ounces.

$\Rightarrow a = 11.96$ and $b = 12.05$

a. Mean amount per can μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{11.96+12.05}{2} \\ &= \frac{24.01}{2} \\ &= 12.005\end{aligned}$$

b. Standard deviation of the amount per can σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(12.05-11.96)^2}{12}} \\ &= \sqrt{\frac{0.0081}{12}} \\ &= \sqrt{0.000675} \\ &= 0.026\end{aligned}$$

Step 2

c. Probability of selecting a can of cola and finding it has less than 12 ounces

If the amount in the can is less than 12 ounces, it means that the amount per can is between 11.96 and 12 ounce. So, the base is $12 - 11.96 = 0.04$.

$$\begin{aligned} P(11.96 < \text{amount} < 12) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 0.04 \\ &= \frac{1}{12.05 - 11.96} \times 0.04 \\ &= \frac{1}{0.09} \times 0.04 \\ &= 0.44 \end{aligned}$$

d. Probability of selecting a can of cola and finding it has more than 11.98 ounces

$$P(\text{amount} > 11.98) = 1 - P(\text{amount} < 11.98)$$

If the amount in the can is less than 11.98 ounces, it means that the amount per can is between 11.96 and 11.98 ounce. So, the base is $11.98 - 11.96 = 0.02$.

$$\begin{aligned} P(\text{amount} > 11.98) &= 1 - P(11.96 < \text{amount} < 11.98) \\ &= 1 - \text{height} \times \text{base} \\ &= 1 - \frac{1}{b-a} \times 0.02 \\ &= 1 - \frac{1}{12.05 - 11.96} \times 0.02 \\ &= 1 - \frac{1}{0.09} \times 0.02 \\ &= 0.22 \end{aligned}$$

Step 3

e. Probability of selecting a can of cola and finding it has more than 11 ounces

$$P(\text{amount} > 11) = 1 - P(\text{amount} < 11)$$

The amount in the can cannot be less than 11 because the amount of can is between 11.96 and 12.05 ounces. Amount in the can cannot be less than 11.96 ounces.

$\Rightarrow P(\text{amount} < 11) = 0$

So, $P(\text{amount} > 11) = 1$

🎯 Result

a. 12.005

b. 0.026

c. 0.44

d. 0.22

e. 1

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

The amount of toothpaste in tube follows the uniform distribution between 0 ounces and 4.2 ounces since the minimum amount that remains in the tube will be 0 ounce.

$\Rightarrow a = 0$ and $b = 4.2$

a. Expected amount in the tube μ

$$\begin{aligned}\mu &= \frac{a+b}{2} \\ &= \frac{0+4.2}{2} \\ &= \frac{4.2}{2} \\ &= 2.1\end{aligned}$$

b. Standard deviation of the amount in the tube σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(4.2-0)^2}{12}} \\ &= \sqrt{\frac{17.64}{12}} \\ &= \sqrt{1.47} \\ &= 1.212\end{aligned}$$

Step 2

c. Probability of remaining toothpaste in the tube is less than 3 ounces

If the amount in the can is less than 3 ounces, it means that the amount of remaining toothpaste is between 0 and 3 ounce. So, the base is $3 - 0 = 3$.

$$\begin{aligned} P(0 < \text{amount} < 3) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 3 \\ &= \frac{1}{4.2-0} \times 3 \\ &= \frac{1}{4.2} \times 3 \\ &= 0.714 \end{aligned}$$

d. Probability of remaining toothpaste in the tube is less than 1.5 ounces

$$P(\text{amount} > 1.5) = 1 - P(\text{amount} < 1.5)$$

If the amount in the can is less than 1.5 ounces, it means that the amount of remaining toothpaste is between 0 and 1.5 ounce. So, the base is $1.5 - 0 = 1.5$.

$$\begin{aligned} P(\text{amount} > 1.5) &= 1 - P(0 < \text{amount} < 1.5) \\ &= 1 - \text{height} \times \text{base} \\ &= 1 - \frac{1}{b-a} \times 1.5 \\ &= 1 - \frac{1}{4.2-0} \times 1.5 \\ &= 1 - \frac{1}{4.2} \times 1.5 \\ &= 1 - 0.357 \\ &= 0.643 \end{aligned}$$

Result

a. 2.1

b. 1.212

c. 0.714

d. 0.643

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

The time required for the credit application process follows the uniform distribution between 4 minutes and 10 minutes.

$\Rightarrow a = 4$ and $b = 10$

a. Mean time for the application process μ

$$\begin{aligned}\mu &= \frac{a+b}{2}\\ &= \frac{4+10}{2}\\ &= \frac{14}{2}\\ &= 7\end{aligned}$$

b. Standard deviation for the application process σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}}\\ &= \sqrt{\frac{(10-4)^2}{12}}\\ &= \sqrt{\frac{36}{12}}\\ &= \sqrt{3}\\ &= 1.73\end{aligned}$$

Step 2

c. Probability that a particular application will take less than 6 minutes

If the time for application is less than 6 minutes, it means that the timer for application is between 4 and 6 minutes. So, the base is $6 - 4 = 2$.

$$\begin{aligned} P(4 < \text{time} < 6) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 2 \\ &= \frac{1}{10-4} \times 2 \\ &= \frac{1}{6} \times 2 \\ &= 0.33 \end{aligned}$$

d. Probability that a particular application will take more than 5 minutes

$$P(\text{time} > 5) = 1 - P(\text{time} < 5)$$

If the time for application is less than 5 minutes, it means that the timer for application is between 4 and 5 minutes. So, the base is $5 - 4 = 1$.

$$\begin{aligned} P(\text{time} > 5) &= 1 - P(4 < (\text{time} < 5) \\ &= 1 - \text{height} \times \text{base} \\ &= 1 - \frac{1}{b-a} \times 1 \\ &= \frac{1}{10-4} \times 1 \\ &= \frac{1}{6} \times 1 \\ &= 0.167 \end{aligned}$$

🎯 Result

a. 7

b. 1.73

c. 0.33

d. 0.167

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

The time patrons spend waiting follows the uniform distribution between 0 minutes and 3.5 minutes.

$\Rightarrow a = 0$ and $b = 3.5$

$\textbf{a. Area of the distribution } A\}$

For the uniform distribution, the height of the rectangle is $\frac{1}{b-a}$ and the base is $b - a$. Hence,

$$\begin{aligned} A &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times b - a \\ &= 1 \end{aligned}$$

b. Mean time spent on waiting μ

$$\begin{aligned} \mu &= \frac{a+b}{2} \\ &= \frac{0+3.5}{2} \\ &= \frac{3.5}{2} \\ &= 1.75 \end{aligned}$$

Step 2

c. Standard deviation of the waiting time σ

$$\begin{aligned}\sigma &= \sqrt{\frac{(b-a)^2}{12}} \\ &= \sqrt{\frac{(3.5-0)^2}{12}} \\ &= \sqrt{\frac{12.25}{12}} \\ &= \sqrt{1.021} \\ &= 1.01\end{aligned}$$

d. Probability of the patrons waiting for less than a minute

If the waiting time is less than 1 minute, it means that the waiting time is between 0 and 1 minute. So, the base is $1-0=1$.

$$\begin{aligned}P(0 < \text{amount} < 1) &= \text{height} \times \text{base} \\ &= \frac{1}{b-a} \times 1 \\ &= \frac{1}{3.5-0} \times 1 \\ &= \frac{1}{3.5} \times 1 \\ &= 0.286\end{aligned}$$

Step 3

e. Probability of the patrons waiting for more than 2 minutes

$$P(\text{waiting time} > 2) = 1 - P(\text{waiting time} < 2)$$

If the waiting time is less than 2 minutes, it means that the waiting time is between 0 and 2 minutes. So, the base is $2 - 0 = 2$.

$$\begin{aligned} P(\text{waiting time} > 2) &= 1 - P(0 < \text{waiting time} < 2) \\ &= 1 - \text{height} \times \text{base} \\ &= 1 - \frac{1}{b-a} \times 2 \\ &= 1 - \frac{1}{3.5-0} \times 2 \\ &= 1 - \frac{1}{3.5} \times 2 \\ &= 1 - 0.571 \\ &= 0.421 \end{aligned}$$

🎯 Result

a. 1

b. 1.75

c. 1.01

d. 0.286

e. 0.421

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

Net sales has normal distribution with mean $\mu_1 = \$180$ million and standard deviation is $\sigma_1 = \$25$ million.

Number of employees has normal distribution with mean $\mu_2 = 1500$ and standard deviation is $\sigma_2 = 120$.

Let X_1 be the net sales and X_2 be the number of employees.

a. z_1 value with $X_1 = 170$

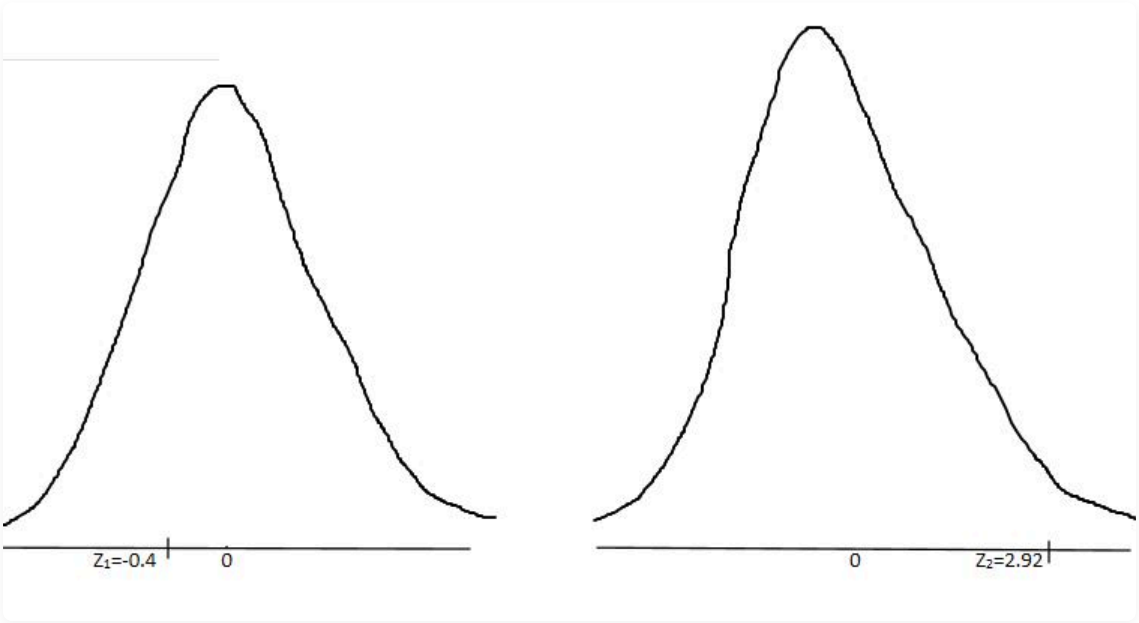
$$\begin{aligned} z_1 &= \frac{X_1 - \mu_1}{\sigma_1} \\ &= \frac{170 - 180}{25} \\ &= \frac{-10}{25} \\ &= -0.4 \end{aligned}$$

z_2 value with $X_2 = 1850$

$$\begin{aligned} z_2 &= \frac{X_2 - \mu_2}{\sigma_2} \\ &= \frac{1850 - 1500}{120} \\ &= \frac{350}{120} \\ &= 2.92 \end{aligned}$$

Step 2

There will be two bell shaped curves- one for the net sales and the other for the number of employees.



Step 3

c. The net sales of the Clarion is less than the average of the other fabricators' net sales and the number of employees of Clarion is more than the average of the other fabricators' employees.

🎯 Result

a. $z_1 = -0.4$ and $z_2 = 2.92$

PROBLEM STATEMENT

Chapter 7, Page 252

Step 1

Mean time taken by garage is $\mu = 32$ hours and standard deviation is $\sigma = 2$ hours.

Let X be the time taken by garage to erect the Red Barn model.

a. z value with $X = 29$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{29 - 32}{2} \\ &= \frac{-3}{2} \\ &= -1.5 \end{aligned}$$

z value with $X = 34$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{34 - 32}{2} \\ &= \frac{2}{2} \\ &= 1 \end{aligned}$$

X is between 32 and 34

$$\begin{aligned} P(32 < X < 34) &= P(32 - \mu < X - \mu < 34 - \mu) \\ &= P\left(\frac{32 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{34 - \mu}{\sigma}\right) \\ &= P\left(\frac{32 - 32}{2} < z < \frac{34 - 32}{2}\right) \\ &= P(0 < z < 1) \\ &= 0.3413 \text{ from the appendix B1} \end{aligned}$$

34.13% of the garages take between 32 and 34 hours.

Step 2

b. X is between 29 and 34

$$\begin{aligned} P(29 < X < 34) &= P(29 - \mu < X - \mu < 34 - \mu) \\ &= P\left(\frac{29 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{34 - \mu}{\sigma}\right) \\ &= P\left(\frac{29 - 32}{2} < z < \frac{34 - 32}{2}\right) \\ &= P(-1.5 < z < 1) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.5 < z < 0)$, $P(0 < z < 1.25)$ and add the two probabilities.

$$\begin{aligned} P(29 < X < 34) &= P(-1.5 < z < 0) + P(0 < z < 1) \\ &= P(0 < z < 1.5) + P(0 < z < 1) \\ &= 0.4332 + 0.3413 \ \ (\text{from Appendix B1.}) \\ &= 0.7745 \end{aligned}$$

77.45% of the garages take between 29 and 34 hours.

Step 3

$\textbf{c. X is less than 28.7 }$

$$\begin{aligned} P(X < 28.7) &= P(X - \mu < 28.7 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{28.7 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{28.7 - 32}{2}\right) \\ &= P\left(z < \frac{3.3}{2}\right) \\ &= P(z < 1.65) \\ &= 0.5 + P(0 < z < 1.65) \\ &= 0.5 + 0.4505 \ \ (\text{from Appendix B1.}) \\ &= 0.9505 \end{aligned}$$

95.05% of the garages take less than 28.7 hours.

Step 4

$$z = \frac{X - \mu}{\sigma}$$
$$\Rightarrow z = \frac{X - 32}{2}$$
$$\text{and } z_1 = \frac{X_1 - 32}{2}$$

5 percent of the garages are more than X_1.

$$\Rightarrow P(z > z_1) = 0.05$$
$$\Rightarrow 0.5 - P(0 < z < z_1) = 0.05$$
$$\Rightarrow P(0 < z < z_1) = 0.5 - 0.05$$
$$\Rightarrow P(0 < z < z_1) = 0.45$$
$$\Rightarrow z_1 = 1.64 \text{ (from Appendix B1.)}$$

Now,

$$z_1 = \frac{X_1 - 32}{2}$$
$$\Rightarrow 1.64 = \frac{X_1 - 32}{2}$$
$$\Rightarrow X_1 - 32 = 3.28$$
$$\Rightarrow X_1 = 32 + 3.28$$
$$\Rightarrow X_1 = 35.28$$

5% of the garages take 35.28 or more hours.

Result

a. 34.13% of the garages take between 32 and 34 hours.

b. 77.45% of the garages take between 29 and 34 hours.

c. 95.05% of the garages take less than 28.7 hours.

d. 35.28 or more hours.

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean food expenditure of the family of four is $\mu = \$490$ and the standard deviation is $\sigma = \$90$

Let X be the food expenditure of the family.

$\textbf{a. } X \text{ is between } 30 \text{ and } 490 \text{ }$

$$\begin{aligned} P(30 < X < 490) &= P(30 - \mu < X - \mu < 490 - \mu) \\ &= P\left(\frac{30-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{490-\mu}{\sigma}\right) \\ &= P\left(\frac{30-490}{90} < z < \frac{490-490}{90}\right) \\ &= P\left(\frac{-460}{90} < z < 0\right) \\ &= P(-5.11 < z < 0) \\ &= P(0 < z < 5.11) \\ &= 0.5 \ \ (\text{from Appendix B1.}) \end{aligned}$$

50% of the families spend between $\$30$ and $\$450$ on food.

$\textbf{b. } X \text{ is less than } 430 \text{ }$

$$\begin{aligned} P(X < 430) &= P(X - \mu < 430 - \mu) \\ &= P(\frac{X-\mu}{\sigma} < \frac{430-\mu}{\sigma}) \\ &= P(z < \frac{430-490}{90} \\ &= P(z < \frac{-60}{90}) \\ &= P(z < -0.67) \\ &= P(z > 0.67) \\ &= 0.5 - P(0 < z < 0.67) \\ &= 0.5 - 0.2486 \ \ (\text{from Appendix B1.}) \\ &= 0.2514 \end{aligned}$$

25.14% of the families spend less than $\$430$ on food.

Step 2

$\textbf{c. } X \text{ is between } 430 \text{ and } 600 \text{ }$

$$\begin{aligned} P(430 < X < 600) &= P(430 - \mu < X - \mu < 600 - \mu) \\ &= P\left(\frac{430 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{600 - \mu}{\sigma}\right) \\ &= P\left(\frac{430 - 490}{90} < z < \frac{600 - 490}{90}\right) \\ &= P\left(\frac{-60}{90} < z < \frac{110}{90}\right) \\ &= P(-0.67 < z < 1.22) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.67 < z < 0)$, $P(0 < z < 1.22)$ and add the two probabilities.

$$\begin{aligned} P(430 < X < 600) &= P(-0.67 < z < 0) + P(0 < z < 1.22) \\ &= P(0 < z < 0.67) + P(0 < z < 1.22) \\ &= 0.2486 + 0.3888 \ \ (\text{from Appendix B1.}) \\ &= 0.6374 \end{aligned}$$

63.74% of the families spend between $\$430$ and $\$600$ on food.

Step 3

$\textbf{d. } X \text{ is between } 500 \text{ and } 600 \text{ }$

$$\begin{aligned} P(500 < X < 600) &= P(500 - \mu < X - \mu < 600 - \mu) \\ &= P\left(\frac{500 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{600 - \mu}{\sigma}\right) \\ &= P\left(\frac{500 - 490}{90} < z < \frac{600 - 490}{90}\right) \\ &= P\left(\frac{10}{90} < z < \frac{110}{90}\right) \\ &= P(0.11 < z < 1.22) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0 < z < 0.11)$, $P(0 < z < 1.22)$ and subtract the two probabilities.

$$\begin{aligned} P(500 < X < 600) &= P(0 < z < 1.22) - P(0 < z < 0.11) \\ &= 0.3888 - 0.0438 \ \ (\text{from Appendix B1.}) \\ &= 0.345 \end{aligned}$$

34.5% of the families spend between $\$500$ and $\$600$ on food.

🎯 Result

a. 50%

b. 25.14%

c. 63.74%

d. 34.5%

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean length of time per call is $\mu = 4.2$ minutes and the standard deviation is $\sigma = 0.60$ minutes

Let X be the length of long distance phone calls.

$\textbf{a. } X \text{ is between } 4.2 \text{ and } 5 \}$

$$\begin{aligned} P(4.2 < X < 5) &= P(4.2 - \mu < X - \mu < 5 - \mu) \\ &= P\left(\frac{4.2 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{5 - \mu}{\sigma}\right) \\ &= P\left(\frac{4.2 - 4.2}{0.60} < z < \frac{5 - 4.2}{0.60}\right) \\ &= P\left(\frac{0}{4} < z < \frac{0.8}{0.6}\right) \\ &= P(0 < z < 1.33) \\ &= 0.4082 \ \ (\text{from Appendix B1.}) \end{aligned}$$

$\textbf{b. } X \text{ is greater than } 55 \}$

$$\begin{aligned} P(X > 5) &= P(X - \mu > 5 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{5 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{5 - 4.2}{0.6}\right) \\ &= P\left(z > \frac{0.8}{0.6}\right) \\ &= P(z > 1.33) \\ &= 0.5 - P(0 < z < 1.33) \\ &= 0.5 - 0.4082 \ \ (\text{from Appendix B1.}) \\ &= 0.0918 \end{aligned}$$

Step 2

$\textbf{c. X is between 5 and 6 }$

$$\begin{aligned} P(5<X<6) &= P(5-\mu<X-\mu<6-\mu) \\ &= P\left(\frac{5-\mu}{\sigma}<\frac{X-\mu}{\sigma}<\frac{6-\mu}{\sigma}\right) \\ &= P\left(\frac{5-4.2}{0.6}<z<\frac{6-4.2}{0.6}\right) \\ &= P\left(\frac{0.8}{0.6}<z<\frac{1.8}{0.6}\right) \\ &= P(1.33<z<3) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0<z<1.33)$, $P(0<z<3)$ and subtract the two probabilities.

$$\begin{aligned} P(5<X<6) &= P(0<z<3)-P(0<z<1.33) \\ &= 0.4987-0.4082 \ \ (\text{from Appendix B1.}) \\ &= 0.0905 \end{aligned}$$

Step 3

$\textbf{d. X is between 4 and 6 }$

$$\begin{aligned} P(4<X<6) &= P(4-\mu<X-\mu<6-\mu) \\ &= P\left(\frac{4-\mu}{\sigma}<\frac{X-\mu}{\sigma}<\frac{6-\mu}{\sigma}\right) \\ &= P\left(\frac{4-4.2}{0.6}<z<\frac{6-4.2}{0.6}\right) \\ &= P\left(\frac{-0.2}{0.6}<z<\frac{1.8}{0.6}\right) \\ &= P(-0.33<z<3) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.33<z<0)$, $P(0<z<3)$ and add the two probabilities.

$$\begin{aligned} P(4<X<6) &= P(-0.33<z<0)+P(0<z<3) \\ &= P(0<z<0.33)+P(0<z<3) \\ &= 0.1293+0.4987 \ \ (\text{from Appendix B1.}) \\ &= 0.628 \end{aligned}$$

Step 4

$$z = \frac{X - \mu}{\sigma}$$
$$\Rightarrow z = \frac{X - 4.2}{0.6}$$
$$\text{and } z_1 = \frac{X_1 - 4.2}{0.6}$$

e. 4 percent of the observations are above X_1

$$\Rightarrow P(z > z_1) = 0.04$$
$$\Rightarrow 0.5 - P(0 < z < z_1) = 0.04$$
$$\Rightarrow P(0 < z < z_1) = 0.5 - 0.04$$
$$\Rightarrow P(0 < z < z_1) = 0.46$$
$$\Rightarrow z_1 = 1.75 \text{ (from Appendix B1.)}$$

Now,

$$z_1 = \frac{X_1 - 4.2}{0.6}$$
$$\Rightarrow 1.75 = \frac{X_1 - 4.2}{0.6}$$
$$\Rightarrow X_1 - 4.2 = 1.05$$
$$\Rightarrow X_1 = 4.2 + 1.05$$
$$\Rightarrow X_1 = 5.25$$

The length of the longest 4 percent of the calls is 5.25 minutes.

Result

a. 0.4082

b. 0.0918

c. 0.0905

d. 0.628

e. 5.25 minutes

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean annual cost of insurance is $\mu = \$1280$ and the standard deviation is $\sigma = \$420$

Let X be the cost of insurance.

$\textbf{a. } X \text{ is greater than } 1500 \}$

$$\begin{aligned} P(X > 1500) &= P(X - \mu > 1500 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{1500 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{1500 - 1280}{420}\right) \\ &= P\left(z > \frac{220}{420}\right) \\ &= P(z > 0.52) \\ &= 0.5 - P(0 < z < 0.52) \\ &= 0.5 - 0.1985 \ \ (\text{from Appendix B1.}) \\ &= 0.3015 \end{aligned}$$

Step 2

$\textbf{b. } X \text{ is between } 1500 \text{ and } 2000 \text{ }$

$$\begin{aligned} P(1500 < X < 2000) &= P(1500 - \mu < X - \mu < 2000 - \mu) \\ &= P\left(\frac{1500 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{2000 - \mu}{\sigma}\right) \\ &= P\left(\frac{1500 - 1280}{420} < z < \frac{2000 - 1280}{420}\right) \\ &= P\left(\frac{220}{420} < z < \frac{720}{420}\right) \\ &= P(0.52 < z < 1.71) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0 < z < 0.52)$, $P(0 < z < 1.71)$ and subtract the two probabilities.

$$\begin{aligned} P(1500 < X < 2000) &= P(0 < z < 1.71) - P(0 < z < 0.52) \\ &= 0.4564 - 0.1985 \ \ (\text{from Appendix B1.}) \\ &= 0.2579 \end{aligned}$$

Step 3

$\textbf{c. } X \text{ is less than } 0 \text{ }$

$$\begin{aligned} P(X < 0) &= P(X - \mu < 0 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{0 - 1280}{420}\right) \\ &= P\left(z < \frac{-1280}{420}\right) \\ &= P(z < -3.05) \\ &= P(z > 3.05) \\ &= 0.5 - P(0 < z < 3.05) \\ &= 0.5 - 0.4989 \ \ (\text{from Appendix B1.}) \\ &= 0.0011 \end{aligned}$$

Step 4

$$z = \frac{X - \mu}{\sigma}$$
$$\Rightarrow z = \frac{X - 1280}{420}$$
$$\text{and } z_1 = \frac{X_1 - 1280}{420}$$

d. 10 percent of the employees incurred more than X_1

$$\Rightarrow P(z > z_1) = 0.10$$
$$\Rightarrow 0.5 - P(0 < z < z_1) = 0.10$$
$$\Rightarrow P(0 < z < z_1) = 0.5 - 0.10$$
$$\Rightarrow P(0 < z < z_1) = 0.4$$
$$\Rightarrow z_1 = 1.28 \text{ (from Appendix B1.)}$$

Now,

$$z_1 = \frac{X_1 - 1280}{420}$$
$$\Rightarrow 1.28 = \frac{X_1 - 1280}{420}$$
$$\Rightarrow X_1 - 1280 = 537.6$$
$$\Rightarrow X_1 = 1280 + 537.6$$
$$\Rightarrow X_1 = 1817.6$$

The cost for the 10 percent of employees who incurred the highest dental expense is $\$1817.6$.

Result

a. 0.3015

b. 0.2579

c. 0.0011

d. $\$1817.6$

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

The mean annual commissions is $\mu = \$40000$ and the standard deviation is $\sigma = \$5000$

Let X be the commissions earned by sales respresentative.

$\textbf{a. } X \text{ is greater than } 42000 \}$

$$\begin{aligned} P(X > 42000) &= P(X - \mu > 42000 - \mu) \\ &= P\left(\frac{X-\mu}{\sigma} > \frac{42000-\mu}{\sigma}\right) \\ &= P\left(z > \frac{42000-40000}{5000}\right) \\ &= P\left(z > \frac{2000}{5000}\right) \\ &= P(z > 0.4) \\ &= 0.5 - P(0 < z < 0.4) \\ &= 0.5 - 0.1554 \ \ (\text{from Appendix B1.}) \\ &= 0.3446 \end{aligned}$$

34.46% of the sales representatives earn more than $\$42000$.

Step 2

$\textbf{b. X is between 32000 and 42000 }$

$$\begin{aligned} P(32000 < X < 42000) &= P(32000 - \mu < X - \mu < 42000 - \mu) \\ &= P\left(\frac{32000 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{42000 - \mu}{\sigma}\right) \\ &= P\left(\frac{32000 - 40000}{5000} < z < \frac{42000 - 40000}{5000}\right) \\ &= P\left(\frac{-8000}{5000} < z < \frac{2000}{5000}\right) \\ &= P(-1.6 < z < 0.4) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.6 < z < 0)$, $P(0 < z < 0.4)$ and add the two probabilities.

$$\begin{aligned} P(32000 < X < 42000) &= P(-1.6 < z < 0) + P(0 < z < 0.4) \\ &= P(0 < z < 1.6) + P(0 < z < 0.4) \\ &= 0.4452 + 0.1554 \ \ (\text{from Appendix B1.}) \\ &= 0.6106 \end{aligned}$$

61.06% of the sales representatives earn between $\$32000$ and $\$42000$.

Step 3

$\textbf{c. } X \text{ is between } 32000 \text{ and } 35000 \text{ }$

$$\begin{aligned} P(32000 < X < 35000) &= P(32000 - \mu < X - \mu < 35000 - \mu) \\ &= P\left(\frac{32000 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{42000 - \mu}{\sigma}\right) \\ &= P\left(\frac{32000 - 40000}{5000} < z < \frac{35000 - 40000}{5000}\right) \\ &= P\left(\frac{-8000}{5000} < z < \frac{-5000}{5000}\right) \\ &= P(-1.6 < z < -1) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(-1.6 < z < 0)$, $P(-1 < z < 0)$ and subtract the two probabilities.

$$\begin{aligned} P(32000 < X < 42000) &= P(-1.6 < z < 0) - P(-1 < z < 0) \\ &= P(0 < z < 1.6) - P(0 < z < 1) \\ &= 0.4452 - 0.3413 \ \ (\text{from Appendix B1.}) \\ &= 0.1039 \end{aligned}$$

10.39% of the sales representatives earn between $\$32000$ and $\$35000$.

Step 4

$$z = \frac{X - \mu}{\sigma}$$
$$\Rightarrow z = \frac{X - 40000}{5000}$$
$$\text{and } z_1 = \frac{X_1 - 40000}{5000}$$

d. 20 percent of the representatives get bonus if they earn more than X_1

$$\Rightarrow P(z > z_1) = 0.20$$
$$\Rightarrow 0.5 - P(0 < z < z_1) = 0.20$$
$$\Rightarrow P(0 < z < z_1) = 0.5 - 0.20$$
$$\Rightarrow P(0 < z < z_1) = 0.3$$
$$\Rightarrow z_1 = 0.84 \text{ (from Appendix B1.)}$$

Now,

$$z_1 = \frac{X_1 - 40000}{5000}$$
$$\Rightarrow 0.84 = \frac{X_1 - 40000}{5000}$$
$$\Rightarrow X_1 - 40000 = 4200$$
$$\Rightarrow X_1 = 40000 + 4200$$
$$\Rightarrow X_1 = 44200$$

The cutoff point between those who earn a bonus and those who do not earn is $\$44200$.

Result

a. 34.46%

b. 61.06%

c. 10.39%

d. $\$44200$

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean hours spent by women on TV per week is $\mu_1 = 34$ hours and the standard deviation is $\sigma_1 = 4.5$ hours.

Let X_1 be the time spent by women in hours.

Mean hours spent by women on TV per week is $\mu_2 = 29$ hours and the standard deviation is $\sigma_2 = 5.1$ hours.

Let X_2 be the time spent by women in hours.

$\textbf{a. X_1 is less than 40 }$

$$\begin{aligned} P(X_1 < 40) &= P(X_1 - \mu_1 < 40 - \mu_1) \\ &= P\left(\frac{X_1 - \mu_1}{\sigma_1} < \frac{40 - \mu_1}{\sigma_1}\right) \\ &= P\left(z_1 < \frac{40 - 34}{4.5}\right) \\ &= P\left(z_1 < \frac{6}{4.5}\right) \\ &= P(z_1 < 1.33) \\ &= 0.5 + P(0 < z_1 < 1.33) \\ &= 0.5 + 0.4082 \ \ (\text{from Appendix B1.}) \\ &= 0.9082 \end{aligned}$$

90.82% of the women watch TV for less than 40 hours per week.

Step 2

$\textbf{b. X_2 is greater than 25 }$

$$
\begin{aligned}
P(X_2 > 25) &= P(X_2 - \mu_2 > 25 - \mu_2) \\
&= P\left(\frac{X_2 - \mu_2}{\sigma_2} > \frac{25 - \mu_2}{\sigma_2}\right) \\
&= P\left(z_2 > \frac{25 - 29}{5.1}\right) \\
&= P\left(z_2 > \frac{-4}{5.1}\right) \\
&= P(z_2 > -0.78) \\
&= P(z_2 < 0.78) \\
&= 0.5 + P(0 < z_2 < 0.78) \\
&= 0.5 + 0.2823 \ \ (\text{from Appendix B1.}) \\
&= 0.7823
\end{aligned}
$$

78.23% of the women watch TV for less than 25 hours per week.

Step 3

$$
\begin{aligned}
z_1 &= \frac{X_1 - \mu_1}{\sigma_1} \\
\Rightarrow z_1 &= \frac{X_1 - 34}{4.5} \\
\text{and } \ z_1^* &= \frac{X_1^* - 34}{4.5} \\
\text{Similarly } \ z_2^* &= \frac{X_2^* - 29}{5.1}
\end{aligned}
$$

c. 1 percent of women watch the TV for more than X_1^*

$$
\begin{aligned}
\Rightarrow P(z_1 > z_1^*) &= 0.01 \\
\Rightarrow 0.5 - P(0 < z < z_1^*) &= 0.01 \\
\Rightarrow P(0 < z < z_1^*) &= 0.5 - 0.01 \\
\Rightarrow P(0 < z < z_1^*) &= 0.49 \\
\Rightarrow z_1^* &= 2.33 \ \ (\text{from Appendix B1.})
\end{aligned}
$$

Now,

$$
\begin{aligned}
z_1^* &= \frac{X_1^* - 34}{4.5} \\
\Rightarrow 2.33 &= \frac{X_1^* - 34}{4.5} \\
\Rightarrow X_1^* - 34 &= 10.485 \\
\Rightarrow X_1^* &= 34 + 10.485 \\
\Rightarrow X_1^* &= 44.485
\end{aligned}
$$

One percent of women who watch the most TV per week watch for $\$44.485$ hours.

Step 4

1 percent of men watch the TV for more than X_2^*

$$
\begin{aligned}
&\Rightarrow P(z_2 > z_2^*) = 0.01 \\
\Rightarrow 0.5 - P(0 < z < z_2^*) &= 0.01 \\
\Rightarrow P(0 < z < z_2^*) &= 0.5 - 0.01 \\
\Rightarrow P(0 < z < z_2^*) &= 0.49 \\
\Rightarrow z_2^* &= 2.33 \ \ (\text{from Appendix B1.})
\end{aligned}
$$

Now,

$$
\begin{aligned}
z_2^* &= \frac{X_2^* - 29}{5.1} \\
\Rightarrow 2.33 &= \frac{X_2^* - 29}{5.1} \\
\Rightarrow X_2^* - 29 &= 11.883 \\
\Rightarrow X_2^* &= 29 + 11.883 \\
\Rightarrow X_2^* &= 40.883
\end{aligned}
$$

One percent of men who watch the most TV per week watch for $\$40.883$ hours.

🎯 Result

a. 90.82%

b. 78.23%

c. Women=44.485 hours, Men=40.883 hours

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean amount spent per year on reading and entertainment is $\mu = \$1994$ and the standard deviation is $\sigma = \$450$

Let X be the amount spent on reading and entertainment.

$\textbf{a. } X \text{ is greater than } 2500 \}$

$$\begin{aligned} P(X > 2500) &= P(X - \mu > 2500 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{2500 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{2500 - 1994}{450}\right) \\ &= P\left(z > \frac{506}{450}\right) \\ &= P(z > 1.12) \\ &= 0.5 - P(0 < z < 1.12) \\ &= 0.5 - 0.3686 \ \ (\text{from Appendix B1.}) \\ &= 0.1314 \end{aligned}$$

13.14% of the adults spend more than $\$2500$ per year.

Step 2

$\textbf{b. } X \text{ is between } 2500 \text{ and } 3000 \text{ }$

$$\begin{aligned} P(2500 < X < 3000) &= P(2500 - \mu < X - \mu < 3000 - \mu) \\ &= P\left(\frac{2500 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{3000 - \mu}{\sigma}\right) \\ &= P\left(\frac{2500 - 1994}{450} < z < \frac{3000 - 1994}{450}\right) \\ &= P\left(\frac{506}{450} < z < \frac{1006}{450}\right) \\ &= P(1.12 < z < 2.23) \end{aligned}$$

The two points are on the same side of the mean, so calculate $P(0 < z < 1.12)$, $P(0 < z < 2.23)$ and subtract the two probabilities.

$$\begin{aligned} P(2500 < X < 3000) &= P(0 < z < 2.23) - P(0 < z < 1.12) \\ &= 0.4871 - 0.3686 \ \ (\text{from Appendix B1.}) \\ &= 0.1185 \end{aligned}$$

11.85% of the adults spend between $\$2500$ and $\$3000$ per year.

Step 3

$\textbf{c. } X \text{ is less than } 1000 \text{ }$

$$\begin{aligned} P(X < 1000) &= P(X - \mu < 1000 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{1000 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{1000 - 1994}{450}\right) \\ &= P\left(z < \frac{-994}{450}\right) \\ &= P(z < -2.21) \\ &= P(z > 2.21) \\ &= 0.5 - P(0 < z < 2.21) \\ &= 0.5 - 0.4864 \ \ (\text{from Appendix B1.}) \\ &= 0.0136 \end{aligned}$$

1.36% of the adults spend less than $\$1000$ per year.

🎯 Result

a. 13.14%

b. 11.85%

c. 1.36%

PROBLEM STATEMENT

Chapter 7, Page 253

Step 1

Mean of the weekly production is $\mu = 4000$ units and the standard deviation is $\sigma = 60$ units.

Let X be the production per week.

Now,

$$z = \frac{X - \mu}{\sigma}$$
$$\Rightarrow z = \frac{X - 4000}{60}$$
$$\text{and } z_1 = \frac{X_1 - 4000}{60}$$

5 percent of the production is more than X_1

$$\Rightarrow P(z > z_1) = 0.05$$
$$\Rightarrow 0.5 - P(0 < z < z_1) = 0.05$$
$$\Rightarrow P(0 < z < z_1) = 0.5 - 0.05$$
$$\Rightarrow P(0 < z < z_1) = 0.45$$
$$\Rightarrow z_1 = 1.64 \text{ (from Appendix B1.)}$$

Now,

$$z_1 = \frac{X_1 - 4000}{60}$$
$$\Rightarrow 1.64 = \frac{X_1 - 4000}{60}$$
$$\Rightarrow X_1 - 4000 = 98.4$$
$$\Rightarrow X_1 = 4000 + 98.4$$
$$\Rightarrow X_1 = 4098.4$$

which is around 4100.

Bonus will be paid if the production is more than around 4100 units.

🎯 Result

4100

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Mean of the number of miles traveled is $\mu = 60000$ miles and the standard deviation is $\sigma = 2000$ miles

Let X be the number of miles traveled during the year by F-750.

$\textbf{a. } X \text{ is greater than } 65200 \}$

$$\begin{aligned} P(X > 65200) &= P(X - \mu > 65200 - \mu) \\ &= P\left(\frac{X-\mu}{\sigma} > \frac{65200-\mu}{\sigma}\right) \\ &= P\left(z > \frac{65200-60000}{2000}\right) \\ &= P\left(z > \frac{5200}{2000}\right) \\ &= P(z > 2.6) \\ &= 0.5 - P(0 < z < 2.6) \\ &= 0.5 - 0.4953 \ \ (\text{from Appendix B1.}) \\ &= 0.0047 \end{aligned}$$

0.47% of the Fords logged 65200 miles or more.

Step 2

$\textbf{b. X is between 57060 and 58280 }$

$$\begin{aligned} P(57060 < X < 58280) &= P(57060 - \mu < X - \mu < 58280 - \mu) \\ &= P\left(\frac{57060 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{58280 - \mu}{\sigma}\right) \\ &= P\left(\frac{57060 - 60000}{2000} < z < \frac{58280 - 60000}{2000}\right) \\ &= P\left(\frac{-2940}{2000} < z < \frac{-1720}{2000}\right) \\ &= P(-1.47 < z < -0.86) \end{aligned}$$

The two points are on the same sides of the mean, so calculate $P(-1.5 < z < 0)$, $P(-0.86 < z < 0)$ and subtract the two probabilities.

$$\begin{aligned} P(57060 < X < 58280) &= P(-1.5 < z < 0) - P(-0.86 < z < 0) \\ &= P(0 < z < 1.5) - P(0 < z < 0.86) \\ &= 0.4332 - 0.3051 \ \ (\text{from Appendix B1.}) \\ &= 0.1281 \end{aligned}$$

12.81% of the Fords logged more than 57060 miles but less than 58280 miles.

Step 3

$\textbf{c. X is less than 62000 }$

$$\begin{aligned} P(X < 62000) &= P(X - \mu < 62000 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{62000 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{62000 - 60000}{2000}\right) \\ &= P\left(z < \frac{200}{2000}\right) \\ &= P(z < 1) \\ &= 0.5 + P(0 < z < 1) \\ &= 0.5 + 0.3413 \ \ (\text{from Appendix B1.}) \\ &= 0.8413 \end{aligned}$$

84.13% of the Fords logged 62000 miles or less.

Step 4

$\textbf{d. X is greater than 70000 }$

$$
\begin{aligned}
P(X > 70000) &= P(X - \mu > 70000 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} > \frac{70000 - \mu}{\sigma}\right) \\
&= P\left(z > \frac{70000 - 60000}{2000}\right) \\
&= P\left(z > \frac{10000}{2000}\right) \\
&= P(z > 5) \\
&= 0.5 - P(0 < z < 5) \\
&= 0.5 - 0.5 \\
&\text{after } z = 3, \text{ probability of any } z \text{ value} \\
&\text{greater than 3 approaches 0} \\
&= 0
\end{aligned}
$$

Hence, none of the Fords were driven more than 70000 miles.

Result

a. 0.47%

b. 12.81%

c. 84.13%

d. 0

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Mean number of customer returns is $\mu = 10.3$ per day and the standard deviation is $\sigma = 2.25$ per day.

Let X be the number of customers returning items.

$\textbf{a. } X \text{ is less than } 8 \}$

$$\begin{aligned} P(X < 8) &= P(X - \mu < 8 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{8 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{8 - 10.3}{2.25}\right) \\ &= P\left(z < \frac{-2.3}{2.25}\right) \\ &= P(z < -1.022) \\ &= P(z > 1.022) \\ &= 0.5 - P(0 < z < 1.022) \\ &= 0.5 - 0.3461 \ \ (\text{from Appendix B1.}) \\ &= 0.1539 \end{aligned}$$

There are 8 or fewer customers returning items in 15.39% of the days.

Step 2

$\textbf{b. } X \text{ is between } 12 \text{ and } 14 \text{ }$

$$\begin{aligned} P(12 < X < 14) &= P(12 - \mu < X - \mu < 14 - \mu) \\ &= P\left(\frac{12 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{14 - \mu}{\sigma}\right) \\ &= P\left(\frac{12 - 10.3}{2.25} < z < \frac{14 - 10.3}{2.25}\right) \\ &= P\left(\frac{1.7}{2.25} < z < \frac{3.7}{2.25}\right) \\ &= P(0.76 < z < 1.64) \end{aligned}$$

The two points are on the same sides of the mean, so calculate $P(0 < z < 0.76)$, $P(0 < z < 1.64)$ and subtract the two probabilities.

$$\begin{aligned} P(0.76 < X < 1.64) &= P(0 < z < 1.64) - P(0 < z < 0.76) \\ &= 0.4495 - 0.2764 \ \ (\text{from Appendix B1.}) \\ &= 0.1731 \end{aligned}$$

In 17.31% of the days, there are 12 to 14 customer returning items.

Step 3

$\textbf{c. } X \text{ is less than } 0 \text{ }$

$$\begin{aligned} P(X < 0) &= P(X - \mu < 0 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{0 - 10.3}{2.25}\right) \\ &= P\left(z < \frac{-10.3}{2.25}\right) \\ &= P(z < -4.58) \\ &= P(z > 4.58) \\ &= 0.5 - P(0 < z < 4.58) \\ &= 0.5 - 0.5 \\ &\text{after } z = 3, \text{probability of any } z \text{ value} \\ &\text{greater than 3 approaches 0} \\ &= 0 \end{aligned}$$

Hence, there can't be any chance of a day with no returns.

🎯 **Result**

a. 15.39%

b. 17.31%

c. No chance

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Probability that employees steel from their companies is $\pi = 20\%$ or 0.20.

Total number of employees is $n = 50$.

Let X be the employees who steal.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 50 \times 0.20 \\ &= 10\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{50 \times 0.20 \times (1-0.20)} \\ &= \sqrt{8} \\ &= 2.83\end{aligned}$$

Step 2

$\textbf{a. Probability that X is less than 5}$

For the probability X or more occurs, use the area below $X - 0.5$ which is $X' = 5 - 0.5 = 4.5$.

$$\textbf{z value with $X' = 4.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{4.5 - 10}{2.83} \\ &= \frac{-5.5}{2.83} \\ &= -1.94 \end{aligned}$$

Now,

$$\begin{aligned} P(4.5 < X' < 10) &= P(4.5 - \mu < X' - \mu < 10 - \mu) \\ &= P\left(\frac{4.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{10 - \mu}{\sigma}\right) \\ &= P\left(\frac{4.5 - 10}{2.83} < z < \frac{10 - 10}{2.83}\right) \\ &= P(-1.94 < z < 0) \\ &= P(0 < z < 1.94) \\ &= 0.4738 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' < 4.5) &= 0.5 - P(4.5 < X' < 10) \\ &= 0.5 - 0.4738 \\ &= 0.0262 \end{aligned}$$

Step 3

$\textbf{b. Probability that X is more than 5}$

For the probability that more than X occurs, use the area above $X + 0.5$ which is $X' = 5 + 0.5 = 5.5$.

$$\textbf{z value with $X' = 5.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{5.5 - 10}{2.83} \\ &= \frac{-4.5}{2.83} \\ &= -1.59 \end{aligned}$$

Now,

$$\begin{aligned} P(5.5 < X' < 10) &= P(5.5 - \mu < X' - \mu < 10 - \mu) \\ &= P\left(\frac{4.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{10 - \mu}{\sigma}\right) \\ &= P\left(\frac{5.5 - 10}{2.83} < z < \frac{10 - 10}{2.83}\right) \\ &= P(-1.59 < z < 0) \\ &= P(0 < z < 1.59) \\ &= 0.4441 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 5.5) &= 0.5 + P(5.5 < X' < 10) \\ &= 0.5 + 0.4441 \\ &= 0.9441 \end{aligned}$$

Step 4

Probability that X is 5 or more

$$\begin{aligned} P(X \text{ is 5 or more}) &= 1 - P(X \text{ is less than 5}) \\ &= 1 - 0.0262 \\ &= 0.9738 \end{aligned}$$

c. Probability that X is 5

$$\begin{aligned} P(X = 5) &= P(X \text{ is 5 or more}) - P(X \text{ is more than 5}) \\ &= 0.9738 - 0.9441 \\ &= 0.0297 \end{aligned}$$

Step 5

$\textbf{c. Probability that X is more than 5 but less than 15}$

$$P(5 < X < 15) = P(5 < X < 10) + P(10 < X < 15)$$

i. $P(5 < X < 10)$

For the probability more than X occurs, use the area above $X + 0.5$ which is $X' = 5 + 0.5 = 5.5$.

Now,

$$P(5.5 < X' < 10) = 0.4441 \text{ from part b.}$$

ii. $P(10 < X < 15)$

For the probability less than X occurs, use the area below $X - 0.5$ which is $X' = 15 - 0.5 = 14.5$.

Now,

$$\begin{aligned} P(10 < X' < 14.5) &= P(10 - \mu < X' - \mu < 14.5 - \mu) \\ &= P\left(\frac{10 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{14.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{10 - 10}{2.83} < z < \frac{14.5 - 10}{2.83}\right) \\ &= P(0 < z < 1.49) \\ &= 0.4441 \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(5 < X < 15) &= P(5 < X < 10) + P(10 < X < 15) \\ &= 0.4441 + 0.4441 \\ &= 0.8882 \end{aligned}$$

Result

a. 0.0262

b. 0.9941

c. 0.0297

d. 0.8882

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Probability that men over the age of 18 consider nutrition a top priority is $\pi = 64\%$ or 0.64.

Total men in the sample $n = 60$

Let X be the men who consider nutrition important.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 60 \times 0.64 \\ &= 38.4\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{60 \times 0.64 \times (1-0.64)} \\ &= \sqrt{13.824} \\ &= 3.72\end{aligned}$$

Step 2

$\textbf{a. Probability that X is 32 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 32 - 0.5 = 31.5$.

$$\textbf{z value with $X' = 31.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{31.5 - 38.4}{3.72} \\ &= \frac{-6.9}{3.72} \\ &= -1.85 \end{aligned}$$

Now,

$$\begin{aligned} P(31.5 < X' < 38.4) &= P(31.5 - \mu < X' - \mu < 38.4 - \mu) \\ &= P\left(\frac{31.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{38.4 - \mu}{\sigma}\right) \\ &= P\left(\frac{31.5 - 38.4}{3.72} < z < \frac{38.4 - 38.4}{3.72}\right) \\ &= P(-1.85 < z < 0) \\ &= P(0 < z < 1.85) \\ &= 0.4678 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 31.5) &= 0.5 + P(31.5 < X' < 38.4) \\ &= 0.5 + 0.4678 \\ &= 0.9678 \end{aligned}$$

Step 3

$\textbf{b. Probability that X is 44 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 44 - 0.5 = 43.5$.

$$\textbf{z value with } X' = \textbf{43.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{43.5 - 38.4}{3.72} \\ &= \frac{5.1}{3.72} \\ &= 1.37 \end{aligned}$$

Now,

$$\begin{aligned} P(38.4 < X' < 43.5) &= P(38.4 - \mu < X' - \mu < 43.5 - \mu) \\ &= P\left(\frac{38.4 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{43.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{38.4 - 38.4}{3.72} < z < \frac{43.5 - 38.4}{3.72}\right) \\ &= P(0 < z < 1.37) \\ &= 0.4147 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 43.5) &= 0.5 - P(38.4 < X' < 43.5) \\ &= 0.5 - 0.4147 \\ &= 0.0853 \end{aligned}$$

Step 4

$\textbf{c. Probability that X is more than 32 but less than 43}$

$$P(32 < X < 43) = P(32 < X < 38.4) + P(38.4 < X < 43)$$

i. $P(32 < X < 38.4)$

For the probability more than X occurs, use the area above $X + 0.5$ which is $X' = 32 + 0.5 = 32.5$.

Now,

$$\begin{aligned} P(32.5 < X' < 38.4) &= P(32.5 - \mu < X' - \mu < 38.4 - \mu) \\ &= P\left(\frac{32.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{38.4 - \mu}{\sigma}\right) \\ &= P\left(\frac{32.5 - 38.4}{3.72} < z < \frac{38.4 - 38.4}{3.72}\right) \\ &= P(-1.59 < z < 0) \\ &= P(0 < z < 1.59) \\ &= 0.4441 \ \text{ from the appendix B1} \end{aligned}$$

ii. $P(38.4 < X < 43)$

For the probability less than X occurs, use the area below $X - 0.5$ which is $X' = 43 - 0.5 = 42.5$.

Now,

$$\begin{aligned} P(38 < X' < 42.5) &= P(38 - \mu < X' - \mu < 42.5 - \mu) \\ &= P\left(\frac{38.4 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{42.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{38.4 - 38.4}{3.72} < z < \frac{42.5 - 38.4}{3.72}\right) \\ &= P(0 < z < 1.10) \\ &= 0.3643 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(32 < X < 43) &= P(32 < X < 38.4) + P(38.4 < X < 43) \\ &= 0.4441 + 0.3643 \\ &= 0.8084 \end{aligned}$$

Step 5

$\textbf{ Probability that X is more than 44}$

For the probability that more than X occurs, use the area above $X + 0.5$ which is $X' = 44 + 0.5 = 44.5$.

$$\textbf{z value with } X' = \textbf{44.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{44.5 - 38.4}{3.72} \\ &= \frac{6.1}{3.72} \\ &= 1.64 \end{aligned}$$

Now,

$$\begin{aligned} P(38.4 < X' < 44.5) &= P(38.4 - \mu < X' - \mu < 44.5 - \mu) \\ &= P\left(\frac{38.4 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{44.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{38.4 - 38.4}{3.72} < z < \frac{44.5 - 38.4}{3.72}\right) \\ &= P(0 < z < 1.64) \\ &= 0.4495 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 44.5) &= 0.5 - P(38.4 < X' < 44.5) \\ &= 0.5 - 0.4495 \\ &= 0.0505 \end{aligned}$$

d. Probability that X is 44

$$\begin{aligned} P(X = 44) &= P(X \text{ is } 44 \text{ or more}) - P(X \text{ is more than } 44) \\ &= 0.0853 - 0.0505 \\ &= 0.0348 \end{aligned}$$

🎯 Result

a. 0.9678

b. 0.0853

c. 0.8084

d. 0.0348

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Probability that the students taking the quantitative methods portion of the CPA examination fail is $\pi = 10\%$ or 0.10.

Number of students taking the exam $n = 60$.

Let X be the number of students who failed in the exam.

a. Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 60 \times 0.10 \\ &= 6\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{60 \times 0.10 \times (1-0.10)} \\ &= \sqrt{5.4} \\ &= 2.32\end{aligned}$$

Step 2

$\textbf{c. Probability that X is 2 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 2 - 0.5 = 1.5$.

$$\textbf{z value with $X' = 1.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{1.5 - 6}{2.32} \\ &= \frac{-4.5}{2.32} \\ &= -1.99 \end{aligned}$$

Now,

$$\begin{aligned} P(1.5 < X' < 6) &= P(2 - \mu < X' - \mu < 6 - \mu) \\ &= P\left(\frac{2 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{6 - \mu}{\sigma}\right) \\ &= P\left(\frac{2 - 6}{2.32} < z < \frac{6 - 6}{2.32}\right) \\ &= P(-1.99 < z < 0) \\ &= P(0 < z < 1.99) \\ &= 0.4767 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 1.5) &= 0.5 + P(1.5 < X' < 6) \\ &= 0.5 + 0.4767 \\ &= 0.9767 \end{aligned}$$

Step 3

$\textbf{Probability that X is more than 2}$

For the probability X or more occurs, use the area above $X + 0.5$ which is $X' = 2 + 0.5 = 2.5$.

$$\textbf{z value with $X' = 2.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{2.5 - 6}{2.32} \\ &= \frac{-3.5}{2.32} \\ &= -1.51 \end{aligned}$$

Now,

$$\begin{aligned} P(2.5 < X' < 6) &= P(2.5 - \mu < X' - \mu < 6 - \mu) \\ &= P\left(\frac{2.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{6 - \mu}{\sigma}\right) \\ &= P\left(\frac{2.5 - 6}{2.32} < z < \frac{6 - 6}{2.32}\right) \\ &= P(-1.51 < z < 0) \\ &= P(0 < z < 1.51) \\ &= 0.4345 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 2.5) &= 0.5 + P(2.5 < X' < 6) \\ &= 0.5 + 0.4345 \\ &= 0.9345 \end{aligned}$$

b. Probability that X is 2

$$\begin{aligned} P(X = 2) &= P(X \text{ is 2 or more}) - P(X \text{ is more than 2}) \\ &= 0.9767 - 0.9345 \\ &= 0.0422 \end{aligned}$$

🎯 **Result**

a. $\mu = 6$ and $\sigma = 2.32$

b. 0.0422

c. 0.9767

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Probability that high-speed chases involving automobiles result in a minor or major accident is $\pi = 40\%$ or 0.40.

Total number of high-speed chases is $n = 50$.

Let X be the number of high-speed chases resulting in accident.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 50 \times 0.40 \\ &= 20\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{50 \times 0.40 \times (1-0.40)} \\ &= \sqrt{12} \\ &= 3.46\end{aligned}$$

Step 2

$\textbf{Probability that X is 25 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 25 - 0.5 = 24.5$.

$$\textbf{z value with } X' = \textbf{24.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{24.5 - 20}{3.46} \\ &= \frac{4.5}{3.46} \\ &= 1.30 \end{aligned}$$

Now,

$$\begin{aligned} P(20 < X' < 24.5) &= P(20 - \mu < X' - \mu < 24.5 - \mu) \\ &= P\left(\frac{20 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{24.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{20 - 20}{3.46} < z < \frac{24.5 - 20}{3.46}\right) \\ &= P(0 < z < 1.30) \\ &= 0.4032 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 24.5) &= 0.5 - P(20 < X' < 24.5) \\ &= 0.5 - 0.4032 \\ &= 0.0968 \end{aligned}$$

Result

$$0.0968$$

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Probability that the rooms of the ship are occupied is $\pi = 80\%$ or 0.80.

Total number of rooms are $n = 800$.

Let X be the number of rooms that are occupied.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 800 \times 0.80 \\ &= 640\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{800 \times 0.80 \times (1-0.80)} \\ &= \sqrt{128} \\ &= 11.31\end{aligned}$$

Step 2

$\textbf{Probability that X is 665 or more}$

For the probability X or more occurs, use the area above $X - 0.5$ which is $X' = 665 - 0.5 = 664.5$.

$$\textbf{z value with } X' = \textbf{664.5}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{644.5 - 640}{11.31} \\ &= \frac{4.5}{11.31} \\ &= 0.40 \end{aligned}$$

Now,

$$\begin{aligned} P(640 < X' < 664.5) &= P(640 - \mu < X' - \mu < 644.5 - \mu) \\ &= P\left(\frac{640 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{644.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{640 - 640}{11.31} < z < \frac{644.5 - 640}{11.31}\right) \\ &= P(0 < z < 0.40) \\ &= 0.1554 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 644.5) &= 0.5 - P(640 < X' < 664.5) \\ &= 0.5 - 0.1554 \\ &= 0.3446 \end{aligned}$$

Result

$$0.3446$$

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

(a) Given:

$$\begin{aligned} \mu &= \text{Mean} = \text{unknown} \\ \sigma &= \text{Standard deviation} = 5 \\ P(X \leq 45) &= 95\% = 0.95 \end{aligned}$$

Let us determine the z-score that corresponds with a probability of $95\% - 50\% = 45\%$ or 0.45 in the normal probability table of the appendix. We note that 0.45 lies exactly in the middle between the probabilities 0.4495 and 0.4505, which correspond to the z-scores 1.64 and 1.65. Thus we estimate the z-score of 0.45 as the z-score exactly in the middle between 1.64 and 1.65.

$$z = 1.645$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{45 - \mu}{5}$$

The two found expressions of the z-score then have to be equal:

$$\frac{45 - \mu}{5} = 1.645$$

Multiply each side by 5:

$$45 - \mu = 1.645(5)$$

Subtract 45 from each side:

$$-\mu = -45 + 1.645(5)$$

Multiply each side by -1:

$$\mu = 45 - 1.645(5)$$

Evaluate:

$$x = 36.775$$

Thus the mean is 36.775 minutes.

Step 2

(b) Given:

$$\begin{aligned} \mu &= \text{Mean} = \text{unknown} \\ \sigma &= \text{Standard deviation} = 10 \\ P(X \leq 45) &= 95\% = 0.95 \end{aligned}$$

Let us determine the z-score that corresponds with a probability of $95\% - 50\% = 45\%$ or 0.45 in the normal probability table of the appendix. We note that 0.45 lies exactly in the middle between the probabilities 0.4495 and 0.4505, which correspond to the z-scores 1.64 and 1.65. Thus we estimate the z-score of 0.45 as the z-score exactly in the middle between 1.64 and 1.65.

$$z = 1.645$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{45 - \mu}{10}$$

The two found expressions of the z-score then have to be equal:

$$\frac{45 - \mu}{10} = 1.645$$

Multiply each side by 10:

$$45 - \mu = 1.645(10)$$

Subtract 45 from each side:

$$-\mu = -45 + 1.645(10)$$

Multiply each side by -1:

$$\mu = 45 - 1.645(10)$$

Evaluate:

$$x = 28.55$$

Thus the mean is 28.55 minutes.

Step 3

(c) By part (b), we know that the mean is 28.55 minutes.

$$\mu = \text{Mean} = 28.55$$
$$\sigma = \text{Standard deviation} = 10$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{30 - 28.55}{10} \approx 0.15$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 0.15)$ is given in the row starting with 0.1 and in the column starting with .05 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X < 30) &= P(Z < 0.15) \\ &= P(X < 0) + P(0 < Z < 0.15) \\ &= 0.5000 + 0.0596 \\ &= 0.5596 \\ &= 55.96\% \end{aligned}$$

Result

(a) 36.775 minutes

(b) 28.55 minutes

(c) $0.5596 = 55.96\%$

PROBLEM STATEMENT

Chapter 7, Page 254

Step 1

Mean funds dispensed at the ATM is $\mu = \$4200$ per day and the standard deviation is $\sigma = \$720$ per day.

Let X be the amount dispensed.

a. X is less than 2500

$$\begin{aligned} P(X < 2500) &= P(X - \mu < 2500 - \mu) \\ &= P\left(\frac{X-\mu}{\sigma} < \frac{2500-\mu}{\sigma}\right) \\ &= P\left(z < \frac{2500 - 4200}{720}\right) \\ &= P\left(z < \frac{-1700}{720}\right) \\ &= P(z < -2.36) \\ &= P(z > 2.36) \\ &= 0.5 - P(0 < z < 2.36) \\ &= 0.5 - 0.4909 \ \ (\text{from Appendix B1.}) \\ &= 0.0091 \end{aligned}$$

0.91% of the days the bank will be notified because the amount dispensed is less than 2500.

Step 2

b. X is less than 6000

$$\begin{aligned}
P(X > 6000) &= P(X - \mu > 6000 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} > \frac{6000 - \mu}{\sigma}\right) \\
&= P\left(z > \frac{6000 - 4200}{720}\right) \\
&= P\left(z > \frac{1800}{720}\right) \\
&= P(z > 2.5) \\
&= 0.5 - P(0 < z < 2.5) \\
&= 0.5 - 0.4938 \ \ (\text{from Appendix B1.}) \\
&= 0.0062
\end{aligned}$$

0.62% of the days the bank will be notified because the amount dispensed is more than 6000.

Step 3

$\textbf{c. } X \text{ is between } 2500 \text{ and } 6000 \}$

The bank will not be notified about the funds when funds dispensed are between 2500 and 6000.

$$\begin{aligned}
P(2500 < X < 6000) &= P(2500 - \mu < X - \mu < 6000 - \mu) \\
&= P\left(\frac{2500 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{6000 - \mu}{\sigma}\right) \\
&= P\left(\frac{2500 - 4200}{720} < z < \frac{6000 - 4200}{720}\right) \\
&= P\left(\frac{-1700}{720} < z < \frac{1800}{720}\right) \\
&= P(-2.36 < z < 2.5)
\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-2.36 < z < 0)$, $P(0 < z < 2.5)$ and add the two probabilities.

$$\begin{aligned}
P(2500 < X < 6000) &= P(-2.36 < z < 0) + P(0 < z < 2.5) \\
&= 0.4909 + 0.4938 \ \ (\text{from Appendix B1.}) \\
&= 0.9847
\end{aligned}$$

98.47% of the time the bank will not be notified regarding the amount of funds dispersed.

🎯 Result

a. 0.91%

b. 0.62%

c. 98.47%

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

The weights of canned hams has mean $\mu = 9.20$ pounds and the standard deviation is $\sigma = 0.25$ pounds.

Let X be the label weight.

$\textbf{a. } X$ is less than 9 pounds }$

$$\begin{aligned} P(X < 9) &= P(X - \mu < 9 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{9 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{9 - 9.20}{0.25}\right) \\ &= P\left(z < \frac{-0.2}{0.25}\right) \\ &= P(z < -0.8) \\ &= P(z > 0.8) \\ &= 0.5 - P(0 < z < 0.8) \\ &= 0.5 - 0.2881 \ \ (\text{from Appendix B1.}) \\ &= 0.2119 \end{aligned}$$

21.19% of the hams actually weigh less than the amount claimed on the label.

Step 2

b.

Mean weight is increased to $\mu' = 9.25$ pounds and the standard deviation σ is same

$$\begin{aligned}
P(X < 9) &= P(X - \mu' < 9 - \mu') \\
&= P\left(\frac{X - \mu'}{\sigma} < \frac{9 - \mu'}{\sigma}\right) \\
&= P\left(z < \frac{9 - 9.25}{0.25}\right) \\
&= P\left(z < \frac{-0.25}{0.25}\right) \\
&= P(z < -1) \\
&= P(z > 1) \\
&= 0.5 - P(0 < z < 1) \\
&= 0.5 - 0.3413 \ \text{(from Appendix B1.)} \\
&= 0.1587
\end{aligned}$$

15.87% of the hams actually weigh less than the amount claimed on the label.

Mean weight is same at $\mu = 9.20$ pounds and the standard deviation is decreased to $\sigma' =$

$$\begin{aligned}
P(X < 9) &= P(X - \mu < 9 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma'} < \frac{9 - \mu}{\sigma'}\right) \\
&= P\left(z < \frac{9 - 9.20}{0.15}\right) \\
&= P\left(z < \frac{-0.20}{0.15}\right) \\
&= P(z < -1.33) \\
&= P(z > 1.33) \\
&= 0.5 - P(0 < z < 1.33) \\
&= 0.5 - 0.4082 \ \text{(from Appendix B1.)} \\
&= 0.0918
\end{aligned}$$

9.18% of the hams actually weigh less than the amount claimed on the label.

Step 3

With increase in the mean keeping standard deviation same, the proportion of hams weighing less than amount claimed on the label falls from 21.19% to 15.87%. But with the decrease in the standard deviation keeping mean weight constant, the proportion of hams weighing less than amount claimed on the label falls from 21.19% to 9.18%.

Decrease in the proportion of hams is more in the case of decrease in standard deviation. So, mean weight should be kept constant and standard deviation should be reduced.

Result

a. 21.19%

b. Mean weight should be kept constant and the standard deviation should be reduced.

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

(a) Given:

$$\mu = \text{Mean} = 43.9$$
$$\sigma = \text{Standard deviation} = 0.9$$
$$P(X < 40) = \frac{1}{3} \approx 0.3333$$

Let us determine the z-score that corresponds with a probability of $1 - 0.3333 - 0.5 = 0.1667$ in the normal probability table of the appendix (the z-score corresponding to $P(X < 40) = 0.3333$ will be the negative of this z-score). We note that the closest probability is 0.1664 which lies in the row 0.4 and in the column .03 of the normal probability table and thus the corresponding z-score is then $0.4 + .03 = 0.43$.

$$z = -0.43$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{40 - 43.9}{\sigma}$$

The two found expressions of the z-score then have to be equal:

$$\frac{40 - 43.9}{\sigma} = -0.43$$

Multiply each side by σ:

$$40 - 43.9 = -0.43\sigma$$

Divide each side by -0.43:

$$\frac{40 - 43.9}{-0.43} = \sigma$$

Evaluate:

$$x = 9.0698$$

Thus the standard deviation is 9.0698 hours.

Step 2

(b) Given:

$$\begin{aligned} \mu &= \text{Mean} = 43.9 \\ \sigma &= \text{Standard deviation} = 0.9 \\ P(X > 49) &= 20\% = 0.20 \end{aligned}$$

Let us determine the z-score that corresponds with a probability of $1 - 0.2 - 0.5 = 0.3$ in the normal probability table of the appendix. We note that the closest probability is 0.2995 which lies in the row 0.8 and in the column .04 of the normal probability table and thus the corresponding z-score is then $0.8 + .04 = 0.84$.

$$z = 0.84$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{49 - 43.9}{\sigma}$$

The two found expressions of the z-score then have to be equal:

$$\frac{49 - 43.9}{\sigma} = 0.84$$

Multiply each side by σ:

$$49 - 43.9 = 0.84\sigma$$

Divide each side by 0.84:

$$\frac{49 - 43.9}{0.84} = \sigma$$

Evaluate:

$$x = 6.0714$$

Thus the standard deviation is 6.0714 hours.

Result

(a) 9.0698 hours

(b) 6.0714 hours

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

The mean miles driven on four-year leases is $mu = 52000$ miles and the standard deviation is $\sigma = 5000$ miles.

Let X be the miles driven on four-year leases.

$\textbf{a. } X$ is greater than 60000 }$

The lease will yield a penalty if the miles driven is greater than 60000.

$$\begin{aligned} P(X > 60000) &= P(X - \mu > 60000 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{60000 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{60000 - 52000}{5000}\right) \\ &= P\left(z > \frac{8000}{5000}\right) \\ &= P(z > 1.6) \\ &= 0.5 - P(0 < z < 1.6) \\ &= 0.5 - 0.4452 \ \ (\text{from Appendix B1.}) \\ &= 0.0548 \end{aligned}$$

5.48% of the leases will yield a penalty.

Step 2

b.

Now,

$$\begin{aligned} z &= \frac{X-\mu}{\sigma} \\ \Rightarrow z &= \frac{X-52000}{5000} \\ \text{and } z_1 &= \frac{X_1-52000}{5000} \end{aligned}$$

25 percent of the leases lie above X_1.

$$\begin{aligned} \Rightarrow P(z > z_1) &= 0.25 \\ \Rightarrow 0.5 - P(0 < z < z_1) &= 0.25 \\ \Rightarrow P(0 < z < z_1) &= 0.5 - 0.25 \\ \Rightarrow P(0 < z < z_1) &= 0.25 \\ \Rightarrow z_1 &= 0.67 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1-52000}{5000} \\ \Rightarrow 0.67 &= \frac{X_1-52000}{5000} \\ \Rightarrow X_1 - 52000 &= 3350 \\ \Rightarrow X_1 &= 52000 + 3350 \\ \Rightarrow X_1 &= 55350 \end{aligned}$$

The new upper limit should be 55350 miles.

Step 3

$\textbf{c. X is less than 45000 }$

A car is of low mileage if the miles driven is less than 45000.

$$\begin{aligned}
P(X < 45000) &= P(X - \mu < 45000 - \mu) \\
&= P\left(\frac{X - \mu}{\sigma} < \frac{45000 - \mu}{\sigma}\right) \\
&= P\left(z < \frac{45000 - 52000}{5000}\right) \\
&= P\left(z < \frac{-7000}{5000}\right) \\
&= P(z < -1.4) \\
&= P(z > 1.4) \\
&= 0.5 - P(0 < z < 1.4) \\
&= 0.5 - 0.4192 \ \ (\text{from Appendix B1.}) \\
&= 0.0808
\end{aligned}$$

8.08% of the cars are of low mileage.

Result

a. 5.48%

b. 55350 miles

c. 8.08%

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

The mean price of shares was $\mu = \$42$ and the standard deviation is $\sigma = \$2.25$.

Let X be the price of share.

a. X is greater than 45

$$\begin{aligned} P(X > 45) &= P(X - \mu > 45 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} > \frac{45 - \mu}{\sigma}\right) \\ &= P\left(z > \frac{45 - 42}{2.25}\right) \\ &= P\left(z > \frac{3}{2.25}\right) \\ &= P(z > 1.33) \\ &= 0.5 - P(0 < z < 1.33) \\ &= 0.5 - 0.4082 \ \ (\text{from Appendix B1.}) \\ &= 0.0918 \end{aligned}$$

9.18% of the time was the price greater than 45. Number of days would be $240 \times 9.18\% = 22.032$ days which is approximately 22 days.

Step 2

b. X is between 38 and 40

$$\begin{aligned} P(38 < X < 40) &= P(38 - \mu < X - \mu < 40 - \mu) \\ &= P\left(\frac{38 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{40 - \mu}{\sigma}\right) \\ &= P\left(\frac{38 - 42}{2.25} < z < \frac{40 - 42}{2.25}\right) \\ &= P\left(\frac{-4}{2.25} < z < \frac{-2}{2.25}\right) \\ &= P(-1.78 < z < 0.89) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.78 < z < 0)$, $P(0 < z < 0.89)$ and add the two probabilities.

$$\begin{aligned} P(38 < X < 40) &= P(-1.78 < z < 0) + P(0 < z < 0.89) \\ &= P(0 < z < 1.78) + P(0 < z < 0.89) \\ &= 0.4625 + 0.3133 \ \ (\text{from Appendix B1.}) \\ &= 0.7758 \end{aligned}$$

77.58% of the days was the price between $\$38$ and $\$40$.

Step 3

c. X is greater than X_1 on the highest 15 percent of days

$$\begin{aligned} z &= \frac{X-\mu}{\sigma} \\ \Rightarrow z &= \frac{X-42}{2.25} \\ \text{and } z_1 &= \frac{X_1-42}{2.25} \end{aligned}$$

15 percent of the observations lie above X_1.

$$\begin{aligned} \Rightarrow P(z > z_1) &= 0.15 \\ \Rightarrow 0.5 - P(0 < z < z_1) &= 0.15 \\ \Rightarrow P(0 < z < z_1) &= 0.5 - 0.15 \\ \Rightarrow P(0 < z < z_1) &= 0.35 \\ \Rightarrow z_1 &= 1.04 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1-42}{2.25} \\ \Rightarrow 1.04 &= \frac{X_1-42}{2.25} \\ \Rightarrow X_1 - 42 &= 2.34 \\ \Rightarrow X_1 &= 42 + 2.34 \\ \Rightarrow X_1 &= 44.34 \end{aligned}$$

Stock's price on the highest 15 percent of the days was $\$44.34$.

Result

a. 9.18% and 22 days

b. 77.58%

c. $\$44.34$

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

Given:

$$P(X > 470,000) = 40\% = 0.40$$
$$P(X > 500,000) = 10\% = 0.10$$

Let us determine the z-score that corresponds with a probability of $1 - 0.40 - 0.50 = 0.10$ in the normal probability table of the appendix. We note that the closest probability is 0.0987 which lies in the row 0.2 and in the column .05 of the normal probability table and thus the corresponding z-score is then $0.2 + .05 = 0.25$.

$$z = 0.25$$

Let us determine the z-score that corresponds with a probability of $1 - 0.10 - 0.50 = 0.40$ in the normal probability table of the appendix. We note that the closest probability is 0.3997 which lies in the row 1.2 and in the column .08 of the normal probability table and thus the corresponding z-score is then $1.2 + .08 = 1.28$.

$$z = 1.28$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma}$$

The two found expressions of the z-score then have to be equal:

$$\frac{470,000 - \mu}{\sigma} = 0.25$$
$$\frac{500,000 - \mu}{\sigma} = 1.28$$

Step 2

Determine standard deviation

Multiply each side of each equation by σ:

$$470,000 - \mu = 0.25\sigma$$
$$500,000 - \mu = 1.28\sigma$$

Subtract the second equation from the first equation:

$$(470,000 - \mu) - (500,000 - \mu) = 0.25\sigma - 1.28\sigma$$

Combine like terms:

$$-30,000 = -1.03\sigma$$

Divide each side by -1.03:

$$\frac{-30,000}{-1.03} = \sigma$$

Evaluate:

$$\sigma \approx 29,126$$

Thus the standard deviation is about 29,126.

Step 3

Determine mean

Next, we use the equation $470,000 - \mu = 0.25\sigma$ with $\sigma = 29,126$:

$$470,000 - \mu = 0.25(29,126) \approx 7282$$

Subtract 470,000 from each side of the equation:

$$-\mu = -462,718$$

Multiply each side of the previous equation by -1:

$$\mu = 462,718$$

Thus the mean is 462,718.

🎯 Result

Mean 462,718

Standard deviation 29,126

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

Mean number of months until repairs are needed is $\mu = 36.84$ and the standard deviation is $\sigma = 3.34$ months.

Let X be the number of months until repairs are needed.

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ \Rightarrow z &= \frac{X - 36.84}{3.34} \\ \text{and } \ z_1 &= \frac{X_1 - 36.84}{3.34} \end{aligned}$$

10 percent of the HDTVs need repairs if X lies above X_1.

$$\begin{aligned} \Rightarrow P(z > z_1) &= 0.10 \\ \Rightarrow 0.5 - P(0 < z < z_1) &= 0.10 \\ \Rightarrow P(0 < z < z_1) &= 0.5 - 0.1 \\ \Rightarrow P(0 < z < z_1) &= 0.4 \\ \Rightarrow z_1 &= 1.28 \ \ (\text{from Appendix B1.}) \end{aligned}$$

Now,

$$\begin{aligned} z_1 &= \frac{X_1 - 36.84}{3.34} \\ \Rightarrow 1.28 &= \frac{X_1 - 36.84}{3.34} \\ \Rightarrow X_1 - 36.84 &= 4.2752 \\ \Rightarrow X_1 &= 36.84 + 4.2752 \\ \Rightarrow X_1 &= 41.1152 \end{aligned}$$

Warranty limits should be set it 41 months so that only 10% of the HDTVs need repair.

Result

41 months

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

Probability that the machine reduces the calling to business phones is $\pi = 15\%$ or 0.15.

Sample of phone numbers is $n = 150$.

Le X be the number of business calls.

Mean of the distribution μ

$$\begin{aligned}\mu &= n\pi \\ &= 150 \times 0.15 \\ &= 22.5\end{aligned}$$

Standard deviation of the distribution σ

$$\begin{aligned}\sigma &= \sqrt{n\pi(1-\pi)} \\ &= \sqrt{150 \times 0.15 \times (1-0.15)} \\ &= \sqrt{19.125} \\ &= 4.37\end{aligned}$$

Step 2

$\textbf{Probability that X is more than 30}$

For the probability more than X occurs, use the area above $X + 0.5$ which is $X' = 30 + 0.5 = 30.5$.

$$\textbf{z value with $X' = 30.5$}$$

$$\begin{aligned} z &= \frac{X - \mu}{\sigma} \\ &= \frac{30.5 - 22.5}{4.37} \\ &= \frac{8}{4.37} \\ &= 1.83 \end{aligned}$$

Now,

$$\begin{aligned} P(22.5 < X' < 30.5) &= P(22.5 - \mu < X' - \mu < 30.5 - \mu) \\ &= P\left(\frac{22.5 - \mu}{\sigma} < \frac{X' - \mu}{\sigma} < \frac{30.5 - \mu}{\sigma}\right) \\ &= P\left(\frac{22.5 - 22.5}{4.37} < z < \frac{30.5 - 22.5}{4.37}\right) \\ &= P(0 < z < 1.83) \\ &= 0.4664 \ \text{ from the appendix B1} \end{aligned}$$

Therefore,

$$\begin{aligned} P(X' > 30.5) &= 0.5 - P(22.5 < X' < 30.5) \\ &= 0.5 - 0.4664 \\ &= 0.0336 \end{aligned}$$

Result

$$0.0336$$

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

A carbon monoxide detector in the Wheelock household activates once every 200 days on average. So mean is $\mu = 200$ days. This implies $\lambda = \frac{1}{\mu} = \frac{1}{200}$

Let X be the time taken by the detector to activate.

a. X is less than 60 days

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 60) &= 1 - e^{-\lambda \times 60} \\ &= 1 - e^{-\frac{1}{200} \times 60} \\ &= 1 - e^{-\frac{3}{10}} \\ &= e^{-0.3} \\ &= 1 - 0.7408 \\ &= 0.2592 \end{aligned}$$

Step 2

b. X is at least 400 days

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(X > 400) &= 1 - (1 - e^{-\lambda \times 400}) \\ &= e^{-\frac{1}{200} \times 400} \\ &= e^{-2} \\ &= 0.135 \end{aligned}$$

Step 3

c. X is between 150 and 250 days

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(150 < X < 250) &= 1 - e^{-\lambda \times 250} - (1 - e^{-\lambda \times 150}) \\ &= 1 - e^{-\frac{1}{200} \times 250} - 1 + e^{-\frac{1}{200} \times 150} \\ &= -e^{-\frac{5}{4}} + e^{-\frac{3}{4}} \\ &= -0.286 + 0.472 \\ &= 0.186 \end{aligned}$$

Step 4

d. 50 percent chance that alarm will activate in less than X_1 days

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.5 &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.5 &= 1 - e^{-\frac{1}{200} \times X_1} \\ \Rightarrow e^{-\frac{X_1}{200}} &= 0.5 \end{aligned}$$

Taking log on both sides

$$\begin{aligned} \Rightarrow -\frac{X_1}{200} \ln e &= \ln 0.5 \\ \Rightarrow -\frac{X_1}{200} &= -0.693 \\ \Rightarrow X_1 &= 138.6 \end{aligned}$$

So, median is 138.6 days.

Result

a. 0.2592

b. 0.135

c. 0.186

d. 138.6 days

PROBLEM STATEMENT

Chapter 7, Page 255

Step 1

Mean boot time is $\mu = 27$ seconds. This implies $\lambda = \frac{1}{\mu} = \frac{1}{27}$

Let X be the boot time.

a. X is less than 15 seconds

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 15) &= 1 - e^{-\lambda \times 15} \\ &= 1 - e^{-\frac{1}{27} \times 15} \\ &= 1 - e^{-\frac{5}{9}} \\ &= 1 - 0.5737 \\ &= 0.4263 \end{aligned}$$

Step 2

b. X is more than 60 seconds

$$\begin{aligned} P(X > X_1) &= 1 - P(X < X_1) \\ &= 1 - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(X > 60) &= 1 - (1 - e^{-\lambda \times 60}) \\ &= e^{-\frac{1}{27} \times 60} \\ &= e^{-\frac{20}{9}} \\ &= 0.1084 \end{aligned}$$

Step 3

c. X is between 30 and 45 seconds

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(30 < X < 45) &= 1 - e^{-\lambda \times 45} - (1 - e^{-\lambda \times 30}) \\ &= 1 - e^{-\frac{1}{27} \times 45} - 1 + e^{-\frac{1}{27} \times 30} \\ &= -e^{-\frac{5}{3}} + e^{-\frac{10}{9}} \\ &= -0.1889 + 0.3292 \\ &= 0.1403 \end{aligned}$$

Step 4

d. 10 percent of the boots occur below X_1 seconds

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.1 &= 1 - e^{-\lambda X_1} \\ \Rightarrow 0.1 &= 1 - e^{-\frac{1}{27} \times X_1} \\ \Rightarrow e^{-\frac{X_1}{27}} &= 0.9 \end{aligned}$$

Taking log on both sides

$$\begin{aligned} \Rightarrow -\frac{X_1}{27} \ln e &= \ln 0.9 \\ \Rightarrow -\frac{X_1}{27} &= -0.1054 \\ \Rightarrow X_1 &= 2.8458 \end{aligned}$$

10 percent of the boots occur below 2.8458 seconds

🎯 Result

a. 0.4263

b. 0.1084

c. 0.1403

d. 2.8458 seconds

PROBLEM STATEMENT

Chapter 7, Page 256

Step 1

Mean time between visits to an emergency room $\mu = 2.5$ years. This implies $\lambda = \frac{1}{\mu} = \frac{1}{2.5}$

Let X be the time between visits to an emergency room.

a. X is less than six months or 0.5 year

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 30) &= 1 - e^{-\lambda \times 30} \\ &= 1 - e^{-\frac{1}{60} \times 30} \\ &= 1 - e^{-\frac{1}{2}} \\ &= 1 - 0.6065 \\ &= 0.3935 \end{aligned}$$

39.35% of the population will visit ER within 6 months.

Step 2

b. Not visit the ER over the next six years

Let E be the event that people do not visit RE over the next six years.

$$\begin{aligned} P(E) &= 1 - P(\text{visits within next six years}) \\ \Rightarrow P(E) &= 1 - P(X < 6) \\ &= 1 - (1 - e^{-\lambda \times 6}) \\ &= e^{-\frac{1}{2.5} \times 6} \\ &= e^{-2.4} \\ &= 0.0907 \end{aligned}$$

9.07% of the population do not visit RE over the next 6 years.

Step 3

c. Next year but not this year

This implies X is between 1 and 2 year.

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(1 < X < 2) &= 1 - e^{-\lambda \times 2} - (1 - e^{-\lambda \times 1}) \\ &= 1 - e^{-\frac{1}{2.5} \times 2} - 1 + e^{-\frac{1}{2.5} \times 1} \\ &= -e^{-\frac{4}{5}} + e^{-\frac{2}{5}} \\ &= -0.4493 + 0.6703 \\ &= 0.221 \end{aligned}$$

22.1% of the population will visit next year but not this year.

Step 4

d. First quartile of the distribution i.e 25 percent of population will visit ER within X_1 y

$$P(X < X_1) = 1 - e^{-\lambda X_1}$$
$$\Rightarrow 0.25 = 1 - e^{-\lambda X_1}$$
$$\Rightarrow 0.25 = 1 - e^{-\frac{1}{2.5} \times X_1}$$
$$\Rightarrow e^{-\frac{X_1}{2.5}} = 0.75$$

Taking log on both sides

$$\Rightarrow -\frac{X_1}{2.5} \ln e = \ln 0.75$$
$$\Rightarrow -\frac{X_1}{2.5} = -0.2877$$
$$\Rightarrow X_1 = 0.72$$

First Quartile is 0.72 years.

Third quartile of the distribution i.e 75 percent of population will visit ER within X_2 ye

$$P(X < X_2) = 1 - e^{-\lambda X_2}$$
$$\Rightarrow 0.75 = 1 - e^{-\lambda X_2}$$
$$\Rightarrow 0.75 = 1 - e^{-\frac{1}{2.5} \times X_2}$$
$$\Rightarrow e^{-\frac{X_2}{2.5}} = 0.25$$

Taking log on both sides

$$\Rightarrow -\frac{X_2}{2.5} \ln e = \ln 0.25$$
$$\Rightarrow -\frac{X_2}{2.5} = -1.386$$
$$\Rightarrow X_2 = 3.465$$

Third Quartile is 3.465 years.

Result

a. 39.35%

b. 9.07%

c. 22.1%

d. First quartile= 0.72 years and Third quartile= 3.465 years.

PROBLEM STATEMENT

Chapter 7, Page 256

Step 1

The times between failures on a personal computer has a mean $\mu = 300,000$ hours. This implies $\lambda = \frac{1}{\mu} = \frac{1}{300000}$

Let X be the failure time.

a. X is less than 100000 hours

$$\begin{aligned} P(X < X_1) &= 1 - e^{-\lambda X_1} \\ \Rightarrow P(X < 100000) &= 1 - e^{-\lambda \times 100000} \\ &= 1 - e^{-\frac{1}{300000} \times 100000} \\ &= 1 - e^{-\frac{1}{3}} \\ &= 1 - 0.7165 \\ &= 0.2835 \end{aligned}$$

b. No failure in the next 500,000 hours

Let E be the event that no failures happen in the next 500000 hours.

$$\begin{aligned} \Rightarrow P(E) &= 1 - P(\text{failures happen in the next } 500000 \text{ hours}) \\ &= 1 - P(X < 500000) \\ &= 1 - (1 - e^{-\lambda \times 500000}) \\ &= e^{-\frac{1}{300000} \times 500000} \\ &= e^{-\frac{5}{3}} \\ &= 0.1889 \end{aligned}$$

Step 2

c. X is between 200000 and 350000 hours

$$\begin{aligned} P(X_1 < X < X_2) &= P(X < X_2) - P(X < X_1) \\ &= 1 - e^{-\lambda X_2} - (1 - e^{-\lambda X_1}) \\ \Rightarrow P(200000 < X < 350000) &= 1 - e^{-\lambda \times 350000} - (1 - e^{-\lambda \times 200000}) \\ &= 1 - e^{-\frac{1}{300000} \times 350000} - 1 + e^{-\frac{1}{300000} \times 200000} \\ &= -e^{-\frac{7}{6}} + e^{-\frac{2}{3}} \\ &= -0.3114 + 0.5134 \\ &= 0.202 \end{aligned}$$

Step 3

The mean of the times between failures is $\mu = 300,000$ hours.

The standard deviation of the times between failures is also λ i.e. $\sigma = 300,000$ hours.

Result

a. 0.2835

b. 0.1889

c. 0.202

d. $\mu = 300,000$ and $\sigma = 300,000$.

PROBLEM STATEMENT

Chapter 7, Page 256

Step 1

(a) Given:

$$\begin{aligned}\mu &= \text{Mean} = 221.10\\ \sigma &= \text{Standard deviation} = 47.11\\ x &= 280.0\end{aligned}$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x-\mu}{\sigma} = \frac{280.0 - 221.10}{47.11} \approx 1.25$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.25)$ is given in the row starting with 1.2 and in the column starting with .05 of the standard normal probability table in the appendix.

$$\begin{aligned}P(X > 280.0) &= P(Z > 1.25)\\ &= 1 - P(Z < 1.25)\\ &= 1 - P(Z < 0) - P(0 < Z < 1.25)\\ &= 1 - 0.5000 - 0.3944\\ &= 0.1056\\ &= 10.56\%\end{aligned}$$

Thus about 10.56% of the homes sell for more than $\$280.0$.

14 of the 105 homes sold for more than $\$280.0$ in the given data set, which corresponds with $\frac{14}{105} \approx 0.1333 = 13.33\%$.

Since 13.33% is quite a bit higher than 10.56%, the normal distribution does not yield a good approximation of the actual results in this case.

Step 2

(b) Given:

$$\mu = \text{Mean} = 14.629$$
$$\sigma = \text{Standard deviation} = 4.874$$
$$x = 18; 22$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{18 - 14.629}{4.874} \approx 0.69$$
$$z = \frac{x - \mu}{\sigma} = \frac{22 - 14.629}{4.874} \approx 1.51$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.51)$ is given in the row starting with 1.5 and in the column starting with .01 of the standard normal probability table in the appendix. $P(0 < Z < 0.69)$ is given in the row starting with 0.6 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(18 \leq X < 22) &= P(0.69 < Z < 1.51) \\ &= P(0 < z < 1.51) - P(0 < Z < 0.69) \\ &= 0.4345 - 0.2549 \\ &= 0.1796 \\ &= 17.96\% \end{aligned}$$

Thus about 17.96% of the homes are at least 18 miles but less than 22 miles from the center of the city.

24 of the 105 homes are at least 18 miles but less than 22 miles from the center of the city in the given data set, which corresponds with $\frac{24}{105} \approx 0.2286 = 22.86\%$.

Since 22.86% is quite a bit higher than 17.96%, the normal distribution does not yield a good approximation of the actual results in this case.

Result

(a) No

(b) No

PROBLEM STATEMENT

Chapter 7, Page 256

Step 1

(a) Given:

$$\mu = \text{Mean} = 2.448$$
$$\sigma = \text{Standard deviation} = 0.698$$
$$x = 3.5$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{3.5 - 2.448}{0.698} = 1.51$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.51)$ is given in the row starting with 1.5 and in the column starting with .01 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 3.5) &= P(Z > 1.51) \\ &= 1 - P(Z < 1.51) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.51) \\ &= 1 - 0.5000 - 0.4345 \\ &= 0.0655 \\ &= 6.55\% \end{aligned}$$

Thus about 6.55% of the teams are estimated to have an attendance of more than 3.5 million.

3 of the 30 homes in the data set had an attendance of more than 3.5 million, which corresponds with $\frac{3}{30} = \frac{1}{10} = 0.1 = 10\%$.

We note that the actual value 10% is quite a bit higher than the estimate of 6.55%.

Step 2

(b) Given:

$$\mu = \text{Mean} = 88.51$$
$$\sigma = \text{Standard deviation} = 33.90$$
$$x = 50$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{50 - 88.51}{33.90} = -1.14$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.14)$ is given in the row starting with 1.1 and in the column starting with .04 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 50) &= P(Z > -1.14) \\ &= P(Z < 1.14) \\ &= P(Z < 0) + P(0 < Z < 1.14) \\ &= 0.5000 + 0.3729 \\ &= 0.8729 \\ &= 87.29\% \end{aligned}$$

Thus about 87.29% of the teams are estimated to have a team salary of more than $\$50$ million.

27 of the 30 homes in the data set had a team salary of more than $\$50$ million, which corresponds with $\frac{27}{30} = \frac{9}{10} = 0.9 = 90\%$.

We note that the actual value 90% is quite a bit higher than the estimate of 87.29%.

Result

(a) $0.0655 = 6.55\%$

(b) $0.8729 = 87.29\%$

PROBLEM STATEMENT

Chapter 7, Page 256

Step 1

(a) Given:

$$\mu = \text{Mean} = 450.29$$
$$\sigma = \text{Standard deviation} = 53.69$$
$$x = 500$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{500 - 450.29}{53.69} = 0.93$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 0.93)$ is given in the row starting with 0.9 and in the column starting with .03 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 500) &= P(Z > 0.93) \\ &= 1 - P(Z < 0.93) \\ &= 1 - P(Z < 0) - P(0 < Z < 0.93) \\ &= 1 - 0.5000 - 0.3238 \\ &= 0.1762 \\ &= 17.62\% \end{aligned}$$

Thus about 17.62% of the buses are estimated to have a maintenance cost exceeding $\$500$.

We note that 14 of the 80 buses in the data set have a maintenance cost exceeding $\$500$, which corresponds to $\frac{14}{80} = 0.175 = 17.5\%$.

We note that the estimation 17.62% is very close to 17.5% and thus the normal approximation yields a good approximation in this case.

Step 2

(b) Given:

$$\mu = \text{Mean} = 830.11$$
$$\sigma = \text{Standard deviation} = 42.19$$
$$x = 900$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{900 - 830.11}{42.19} = 1.66$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.66)$ is given in the row starting with 1.6 and in the column starting with .06 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 900) &= P(Z > 1.66) \\ &= 1 - P(Z < 1.66) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.66) \\ &= 1 - 0.5000 - 0.4515 \\ &= 0.0485 \\ &= 4.85\% \end{aligned}$$

Thus about 4.85% of the buses are estimated to travel more than 900 miles.

We note that 2 of the 80 buses in the data set traveled more than 900 miles, which corresponds to $\frac{2}{80} = 0.025 = 2.5\%$.

We note that the estimation 4.85% is quite a bit higher than the actual percentage of 2.5%.

Result

(a) $0.1762 = 17.62\%$

(b) $0.0485 = 4.85\%$

Statistics

Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 6

PROBLEM STATEMENT

Chapter 7, Page 260

Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X = k) =_n C_k \cdot \pi^k \cdot (1 - \pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1 - \pi)^{n-k}$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 15$$
$$\pi = \text{Probability of success} = 80\% = 0.80$$

The number of successes among a fixed number of independent trials with a constant probability of success follows a **binomial distribution**.

(a) Evaluate the definition of binomial probability at $k = 15$:

$$\begin{aligned} P(X = 15) &=_{15} C_{15} \cdot 0.80^{15} \cdot (1 - 0.80)^{15-15} \\ &= \frac{15!}{15!(15-15)!} \cdot 0.80^{15} \cdot 0.20^0 \\ &= 1 \cdot 0.80^{15} \cdot 1 \\ &= 0.80^{15} \\ &\approx 0.0352 \\ &= 3.52\% \end{aligned}$$

Step 3

(b) Evaluate the definition of binomial probability at $k = 9, 10, 11, 12, 13, 14$:

$$
\begin{aligned}
P(X = 9) &=_{15} C_9 \cdot 0.80^9 \cdot (1 - 0.80)^{15-9} \\
&= \frac{15!}{9!(15-9)!} \cdot 0.80^9 \cdot 0.20^6 \\
&= 5005 \cdot 0.80^9 \cdot 0.20^6 \\
&\approx 0.0430 \\
P(X = 10) &=_{15} C_{10} \cdot 0.80^{10} \cdot (1 - 0.80)^{15-10} \\
&= \frac{15!}{10!(15-10)!} \cdot 0.80^{10} \cdot 0.20^5 \\
&= 3003 \cdot 0.80^{10} \cdot 0.20^5 \\
&\approx 0.1032 \\
P(X = 11) &=_{15} C_{11} \cdot 0.80^{11} \cdot (1 - 0.80)^{15-11} \\
&= \frac{15!}{11!(15-11)!} \cdot 0.80^{11} \cdot 0.20^4 \\
&= 1365 \cdot 0.80^{11} \cdot 0.20^4 \\
&\approx 0.1876 \\
P(X = 12) &=_{15} C_{12} \cdot 0.80^{12} \cdot (1 - 0.80)^{15-12} \\
&= \frac{15!}{12!(15-12)!} \cdot 0.80^{12} \cdot 0.20^3 \\
&= 455 \cdot 0.80^{12} \cdot 0.20^3 \\
&\approx 0.2501 \\
P(X = 13) &=_{15} C_{13} \cdot 0.80^{13} \cdot (1 - 0.80)^{15-13} \\
&= \frac{15!}{13!(15-13)!} \cdot 0.80^{13} \cdot 0.20^2 \\
&= 105 \cdot 0.80^{13} \cdot 0.20^2 \\
&\approx 0.2309 \\
P(X = 14) &=_{15} C_{14} \cdot 0.80^{14} \cdot (1 - 0.80)^{15-14} \\
&= \frac{15!}{14!(15-14)!} \cdot 0.80^{14} \cdot 0.20^1 \\
&= 15 \cdot 0.80^{14} \cdot 0.20^1 \\
&\approx 0.1319
\end{aligned}
$$

Step 4

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes in a single sample) and use the result of part (a):

$$\begin{aligned} P(X \geq 9) &= P(X = 9) + P(X = 10) + P(X = 11) + P(X = 12) \\ &\quad + P(X = 13) + P(X = 14) + P(X = 15) \\ &= 0.0430 + 0.1032 + 0.1876 + 0.2501 \\ &\quad + 0.2309 + 0.1319 + 0.0352 \\ &= 0.9819 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X < 9) &= 1 - P(X \geq 9) \\ &= 1 - 0.9819 \\ &= 0.0181 \\ &= 1.81\% \end{aligned}$$

Step 5

(c) Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes in a single sample) and use the result of part (a)-(b):

$$\begin{aligned} P(X \geq 12) &= P(X = 12) + P(X = 13) + P(X = 14) + P(X = 15) \\ &= 0.2501 + 0.2309 + 0.1319 + 0.0352 \\ &= 0.6481 \\ &= 64.81\% \end{aligned}$$

Result

(a) $0.0352 = 3.52\%$

(b) $0.0181 = 1.81\%$

(c) $0.6481 = 64.81\%$

PROBLEM STATEMENT

Chapter 7, Page 260

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 400$$
$$\pi = \text{Probability of success} = 0.005$$

(a) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^0 e^{-2}}{0!} \\ &= \frac{1e^{-2}}{1} \\ &= e^{-2} \\ &\approx 0.1353 \\ &= 13.53\% \end{aligned}$$

Step 3

(b) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Thus 2 of the 400 loans are expected to default, which implies that $400 - 2 = 398$ loans are not expected to default.

Step 4

(c) The mean is the product of the sample size and the probability of success:

$$\mu = n\pi = 400(0.005) = 2$$

Evaluate the probability of Poisson probability at $x = 1, 2$:

$$\begin{aligned} P(X = 1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^1 e^{-2}}{1!} \\ &= \frac{2e^{-2}}{1} \\ &= 2e^{-2} \\ &\approx 0.2707 \\ P(X = 2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{2^2 e^{-2}}{2!} \\ &= \frac{4e^{-2}}{2} \\ &= 2e^{-2} \\ &\approx 0.2707 \end{aligned}$$

Step 5

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same sample) and the results of part (a):

$$\begin{aligned} P(X < 3) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= 0.1353 + 0.2707 + 0.2707 \\ &= 0.6767 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) \\ &= 1 - 0.6767 \\ &= 0.3233 \\ &= 32.33\% \end{aligned}$$

🎯 Result

(a) $e^{-2} \approx 0.1353 = 13.53\%$

(b) 398 loans

(c) $0.3233 = 32.33\%$

PROBLEM STATEMENT

Chapter 7, Page 260

Step 1

Given:

$$\mu = \text{Mean} = 10000$$
$$\sigma = \text{Standard deviation} = 2000$$

(a) The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{13500 - 10000}{2000} = 1.75$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.75)$ is given in the row starting with 1.7 and in the column starting with .05 of the standard normal probability table in the appendix.

$$\begin{aligned} P(X > 13500) &= P(Z > 1.75) \\ &= 1 - P(Z < 1.75) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.75) \\ &= 1 - 0.5000 - 0.4599 \\ &= 0.0401 \\ &= 4.01\% \end{aligned}$$

Step 2

(b) The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{8000 - 10000}{2000} = -1$$
$$z = \frac{x - \mu}{\sigma} = \frac{11500 - 10000}{2000} = 0.75$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 0.75)$ is given in the row starting with 0.7 and in the column starting with .05 of the standard normal probability table in the appendix. $P(0 < Z < 1)$ is given in the row starting with 1 and in the column starting with .01 of the standard normal probability table in the appendix.

$$\begin{aligned}
&P(8000 < X < 11500) \\
&= P(-1 < Z < 0.75) \\
&= P(Z < 0.75) - P(Z < -1) \\
&= P(Z < 0.75) - P(Z > 1) \\
&= P(Z < 0.75) - (1 - P(Z < 1)) \\
&= P(Z < 0) + P(0 < Z < 0.75) - (1 - P(Z < 0) - P(0 < Z < 1)) \\
&= 0.5000 + 0.2734 - (1 - 0.5000 - 0.3413) \\
&= 0.7734 - 0.1587 \\
&= 0.6147 \\
&= 61.47\%
\end{aligned}$$

Step 3

(c) Let us determine the z-score that corresponds with a probability of $100\% - 10\% - 50\% = 40\%$ in the normal probability table of the appendix (the z-score corresponding with the bottom 10% is then the negative of this z-score). We note that the closest probability is 0.3997 which lies in the row 1.2 and in the column .08 of the normal probability table and thus the corresponding z-score is then $1.2 + .08 = 1.28$.

$$z = 1.28$$

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x - \mu}{\sigma} = \frac{x - 10000}{2000}$$

The two found expressions of the z-score then have to be equal:

$$\frac{x - 10000}{2000} = 1.28$$

Multiply each side by 2000:

$$x - 10000 = 1.28(2000)$$

Add 10000 to each side:

$$x = 10000 - 1.28(2000)$$

Evaluate:

$$x = 7440$$

Thus 10% of the games have an attendance of 7440 or less.

Result

(a) $0.0401 = 4.01\%$

(b) $0.6147 = 61.47\%$

(c) 7440 or less

PROBLEM STATEMENT

Chapter 7, Page 260

Step 1

Given:

x	0	40	300
$P(X = x)$	0.98	0.016	0.004

(a) The expected value (or mean) is the sum of the product of each possibility x with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x) \\ &= 0 \times 0.98 + 40 \times 0.016 + 300 \times 0.004 \\ &= 1.84\end{aligned}$$

Thus the expected amount of loss is 1.84 millions.

Step 2

(b) By part (a), we know that the expected loss is 1.84 million.

The only ways to lose less than 1.8 million is when we makes a loss of 0 million, which has a probability of 0.98.

$$P(X < 1.8) = P(X = 0) = 0.98 = 98\%$$

Step 3

(c) Daniel-James suffers a loss when he loses either 40 million or 300 million.

$$P(X < 1.8) = P(X = 0) = 0.98 = 98\%$$

Use the **Addition rule** for disjoint or mutually exclusive events: $P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$

$$\begin{aligned} P(\text{loss}) &= P(X = 40) + P(X = 300) \\ &= 0.016 + 0.004 \\ &= 0.020 \end{aligned}$$

Use the Definition of **Conditional probability**: $P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(\$300 \text{ million}|\text{loss}) &= \frac{P(\$300 \text{ million and loss})}{P(\text{loss})} \\ &= \frac{P(\$300 \text{ million})}{P(\text{loss})} \\ &= \frac{0.004}{0.020} \\ &= \frac{4}{20} \\ &= \frac{1}{5} \\ &= 0.2 \\ &= 20\% \end{aligned}$$

Step 4

(d) The premium of $\$2.0$ million exceeds the expected loss of $\$1.8$ million and thus the premium is fair.

However, this premium does not cover the risk, because if Daniel-James makes a loss, then this loss will far exceed $\$2.0$ million.

🎯 **Result**

(a) 1.84 millions

(b) $0.98 = 98\%$

(c) $\frac{1}{5} = 0.2 = 20\%$

(d) Fair, Not covered

PROBLEM STATEMENT

Chapter 7, Page 261

Step 1

Given:

x	0	1	2	3	4
Percent	40%	30%	15%	10%	5%
Probability	0.40	0.30	0.15	0.10	0.05

(a) The expected value (or mean) is the sum of the product of each possibility x with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x) \\ &= 0 \times 0.40 + 1 \times 0.30 + 2 \times 0.15 + 3 \times 0.10 + 4 \times 0.05 \\ &= 1.1\end{aligned}$$

The variance is the expected value of the squared deviation from the mean:

$$\begin{aligned}\sigma^2 &= \sum (x-\mu)^2 P(x) \\ &= (0-1.1)^2 \times 0.40 + (1-1.1)^2 \times 0.30 + (2-1.1)^2 \times 0.15 \\ &+ (3-1.1)^2 \times 0.10 + (4-1.1)^2 \times 0.05 \\ &= 1.39\end{aligned}$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{1.39} \approx 1.1790$$

Step 2

(b) By part (a), the expected number of school-age children per family in Whitehall Estates is 1.1 child, as the mean is 1.1.

$$\text{Number of families} = 500$$

The expected number of children is then the product of the expected number of children per family and the number of families:

$$\begin{aligned}&\text{Expected number of children}\\&= \text{Expected number of children per family}\\&\times \text{Number of families}\\&= 1.1 \times 500\\&= 500\end{aligned}$$

Step 3

(c) Since we are interested in the expected number of children among families with children, we exclude the case of 0 children.

However, we then also need to adjust the probabilities by dividing the current probability by $0.30 + 0.15 + 0.10 + 0.05 = 0.60$ (total probability of a family having a child.

x	1	2	3	4
Probability	$\frac{0.30}{0.60} = \frac{1}{2}$	$\frac{0.15}{0.60} = \frac{1}{4}$	$\frac{0.10}{0.60} = \frac{1}{6}$	$\frac{0.05}{0.60} = \frac{1}{12}$

The expected value (or mean) is the sum of the product of each possibility x with its probability $P(x)$:

$$\begin{aligned}\mu &= \sum xP(x)\\&= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{6} + 4 \times \frac{1}{12}\\&= \frac{11}{6}\\&\approx 1.8333\end{aligned}$$

🎯 **Result**

(a) Mean 1.1 and Standard deviation 1.1790

(b) 500 children

(c) $\frac{11}{6} \approx 1.8333$

PROBLEM STATEMENT

Chapter 7, Page 261

Step 1

Given:

	Democrats	Republicans	Total
House	236	199	435
Senate	48	52	100
Total	284	251	535

(a) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

251 of the 535 people are Republican, because 251 is mentioned in the row "Total" and in the column "Republicans."
The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Republican}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{251}{535} \approx 0.4692 = 46.92\%$$

Step 2

(b) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

435 of the 535 people are members of the House of Representatives, because 435 is mentioned in the row "House" and in the column "Total."

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{House}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{435}{535}$$

199 of the 535 people are republicans and members of the House of Representatives, because 199 is mentioned in the row "House" and in the column "Republicans."

$$P(\text{House and Republican}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{199}{535}$$

Use the Definition of **Conditional probability**: $P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(\text{Republican}|\text{House}) &= \frac{P(\text{House and Republican})}{P(\text{House})} \\ &= \frac{199/535}{435/535} \\ &= \frac{199}{435} \\ &\approx 0.4575 \\ &= 45.75\% \end{aligned}$$

Step 3

(c) The table describes 535 people in total, because 535 is mentioned in the bottom right corner of the given table.

435 of the 535 people are members of the House of Representatives, because 435 is mentioned in the row "House" and in the column "Total."

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{House}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{435}{535}$$

284 of the 535 people are Democrats and members of the House of Representatives, because 284 is mentioned in the row "Total" and in the column "Democrats."

$$P(\text{Democrat}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{284}{535}$$

236 of the 535 people are Democrats and members of the House of Representatives, because 236 is mentioned in the row "House" and in the column "Democrats."

$$P(\text{House and Democrat}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{236}{535}$$

Use the **General addition rule** for any two events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$\begin{aligned}
&P(\text{House and Democrat}) \\
&= P(\text{House}) + P(\text{Democrat}) - P(\text{House and Democrat}) \\
&= \frac{435}{535} + \frac{284}{535} - \frac{236}{535} \\
&= \frac{483}{535} \\
&\approx 0.9028 \\
&= 90.28\%
\end{aligned}$$

Result

(a) $\frac{251}{535} \approx 0.4692 = 46.92\%$

(b) $\frac{199}{435} \approx 0.4575 = 45.75\%$

(c) $\frac{483}{535} \approx 0.9028 = 90.28\%$

Statistics

Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 15

Chapter 7 **Exercise 1**

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

A probability is always between 0 and 1 (inclusive), or equivalently, between 0% and 100% (inclusive).

This then implies that a probability can **never** be greater than 1 or 100%.

Result

Never

Chapter 7 Exercise 2

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

An **experiment** deliberately imposes some treatment on individuals in order to observe their responses.

Moreover, an experiment observes and/or measures some activity.

Result

Experiment

Chapter 7 Exercise 3

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The collection of one or multiple outcomes to an experiment is called an **event**.

Result

Event

Chapter 7 Exercise 4

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The probability that two or more events will happen at the same time is called the **joint** probability.

Result

Joint

Chapter 7 Exercise 5

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The order in which events are counted is important in a **permutation**, while the order is not important in a **combination**.

Result

(a) Permutation

(b) Combination

Chapter 7 Exercise 6

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

Each probability distribution requires that the total probability of all possible outcomes is equal to 1.

A discrete probability distribution then also requires that the sum of the (probability of) possible outcomes is equal to 1.

Result

1

Chapter 7 Exercise 7

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The four conditions for a binomial setting are: binary (success/failure), independent trials, fixed number of trials and probability of success is the same for each trial.

We note that a binomial distribution requires that there are only two possible outcomes (a success or a failure) and thus "three or more outcomes" is not one of the requirements for a binomial distribution.

Result

Three or more outcomes

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The **normal probability distribution** is dependent on the value of the mean and the standard deviation.

However, this then implies that a different mean and/or a different standard deviation leads to a different normal distribution and thus there are many possible normal probability distributions.

Since there are infinitely many possible values for the mean and infinitely many possible values for the standard deviation, there are infinitely many possible normal distributions.

🎯 Result

Infinite

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

The **normal probability distribution** is dependent on the value of the mean and the standard deviation.

However, this then implies that a different mean and/or a different standard deviation leads to a different normal distribution and thus there are many possible normal probability distributions.

Since there are infinitely many possible values for the mean and infinitely many possible values for the standard deviation, there are infinitely many possible normal distributions.

However, a standard normal distribution is the normal distribution with mean 0 and standard deviation 1, which implies that there is exactly 1 standard normal distribution.

🎯 Result

1

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

Given:

$$P(-0.76 < Z < 0)$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 0.76)$ is given in the row starting with 0.7 and in the column starting with .06 of the standard normal probability table in the appendix.

Since the standard normal distribution is symmetric about 0, we then obtain the probability:

$$\begin{aligned} P(-0.76 < Z < 0) &= P(0 < Z < 0.76) \\ &= 0.2764 \\ &= 27.64\% \end{aligned}$$

🎯 Result

$$0.2764 = 27.64\%$$

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

Given:

$$P(Z < 1.67)$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 1.67)$ is given in the row starting with 1.6 and in the column starting with .07 of the standard normal probability table in the appendix.

$$\begin{aligned} P(Z < 1.67) &= P(Z < 0) + P(0 < Z < 1.67) \\ &= 0.5000 + 0.4525 \\ &= 0.9525 \\ &= 95.25\% \end{aligned}$$

🎯 Result

$$0.9525 = 95.25\%$$

Chapter 7 Exercise 12

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

Result

Independent

Chapter 7 Exercise 13

Page 263

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

Two events are **disjoint** or **mutually exclusive**, if the events cannot occur at the same time.

Result

Mutually exclusive

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

An asymptotic distribution is a probability distribution that is the limiting distribution of a sequence of distributions. Since normal distributions are the limiting distribution for many other distributions and thus normal distributions are asymptotic.

Normal distributions are considered to be a family of distributions, because you obtain a different normal distribution when you have a different mean and/or a different standard deviation.

A normal distribution is also symmetric about its mean and thus 50% of the observations in a normal distribution are greater than the mean.

The property that does not hold for a normal probability distribution is then "only two outcomes," which is true as a normal probability distribution has infinitely many possible outcomes.

Result

Only two outcomes

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

A normal probability distribution has a peak centered about the mean, while the normal probability distributions approaches zero on either side of the mean.

Moreover, this particular shape is called a bell-shape and thus the shape of a normal probability distribution is **bell-shaped**.

Result

Bell-shaped

Statistics

Chapter 7 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 7

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

DEFINITIONS

General multiplication rule:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

(a) 5 of the 20 tax returns are business.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{5}{20}$$

Once one of the business tax returns have been selected, 4 of the 19 remaining tax returns are business.

$$P(\text{Second business}|\text{First business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{4}{19}$$

Use the general multiplication rule:

$$\begin{aligned} P(\text{both business}) &= P(\text{First business}) \\ &\quad \times P(\text{Second business}|\text{First business}) \\ &= \frac{5}{20} \times \frac{4}{19} \\ &= \frac{1}{4} \times \frac{4}{19} \\ &= \frac{1 \times 4}{4 \times 19} \\ &= \frac{1}{19} \\ &\approx 0.0526 \\ &= 5.26\% \end{aligned}$$

Step 3

(b) 5 of the 20 tax returns are business, thus $20 - 5 = 15$ of the 20 tax returns are not business.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First not business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{15}{20}$$

Once one of the business tax returns have been selected, 4 of the 19 remaining tax returns are business.

$$P(\text{Second not business}|\text{First not business}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{14}{19}$$

Use the general multiplication rule:

$$\begin{aligned} P(\text{none business}) &= P(\text{First not business}) \\ &\quad \times P(\text{Second not business}|\text{First not business}) \\ &= \frac{15}{20} \times \frac{14}{19} \\ &= \frac{3}{4} \times \frac{14}{19} \\ &= \frac{3 \times 14}{4 \times 19} \\ &= \frac{3 \times 7}{2 \times 19} \\ &= \frac{21}{38} \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(\text{at least one business}) &= 1 - P(\text{none business}) \\ &= 1 - \frac{21}{38} \\ &= \frac{17}{38} \\ &\approx 0.4474 \\ &= 44.74\% \end{aligned}$$

Result

(a) $\frac{1}{19} \approx 0.0526 = 5.26\%$

(b) $\frac{17}{38} \approx 0.4474 = 44.74\%$

PROBLEM STATEMENT

Chapter 7, Page 263

Step 1

DEFINITIONS

Definition **binomial probability**:

$$P(X=k) =_n C_k \cdot \pi^k \cdot (1-\pi)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \pi^k \cdot (1-\pi)^{n-k}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

$$n = \text{Sample size} = 16$$
$$\pi = \text{Probability of success} = 15\% = 0.15$$

(a) Evaluate the definition of binomial probability at $k = 1$:

$$\begin{aligned} P(X=1) &=_{16} C_1 \cdot 0.15^1 \cdot (1-0.15)^{16-1} \\ &= \frac{16!}{1!(16-1)!} \cdot 0.15 \cdot 0.85^{15} \\ &= 16 \cdot 0.15 \cdot 0.85^{15} \\ &\approx 0.2097 \\ &= 20.97\% \end{aligned}$$

Step 3

(b) Evaluate the definition of binomial probability at $k = 0$:

$$\begin{aligned} P(X = 0) &= {}_{16}C_0 \cdot 0.15^0 \cdot (1 - 0.15)^{16-0} \\ &= \frac{16!}{0!(16-0)!} \cdot 1 \cdot 0.85^{16} \\ &= 1 \cdot 1 \cdot 0.85^{16} \\ &= 0.85^{16} \\ &\approx 0.0743 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - 0.0743 \\ &= 0.9257 \\ &= 92.57\% \end{aligned}$$

Result

(a) $0.2097 = 20.97\%$

(b) $0.9257 = 92.57\%$

PROBLEM STATEMENT

Chapter 7, Page 264

Step 1

Fundamental counting principle (multiplication rule): If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

There are 6 possible CPAs (Fred and five others), while there are 5 parking spots for the 6 CPAs.

First parking spot: 6 ways

Second parking spot: 5 ways $\text{\color{#4257b2}(different from CPA in previous parking spot)\color{default}

Third parking spot: 4 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}

Fourth parking spot: 3 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}

Fifth parking spot: 2 ways \color{#4257b2}(different from CPA in previous parking spots)\color{default}\

Use the fundamental counting principle:

$$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$$

Thus there are 720 different ways the cars can be arranged in the five parking spots.}$

Result

720 different ways

PROBLEM STATEMENT

Chapter 7, Page 264

Step 1

Let x be the number of exemptions per return and $P(x)$ be the percent of the people who availed exemptions.

a.The mean number of exemptions per return

Mean number of exemptions per return is μ.

$$\begin{aligned}\mu &= \Sigma x P(x)\\ &= 1 \times \frac{20}{100} + 2 \times \frac{50}{100} + 3 \times \frac{20}{100} + 4 \times \frac{10}{100}\\ &= 0.2 + 2 \times 0.5 + 3 \times 0.2 + 4 \times 0.10\\ &= 0.2 + 1 + 0.6 + 0.4\\ &= 2.2\end{aligned}$$

Step 2

b.The variance of the number of exemptions per return

Variance of the number of exemptions per return is σ^2.

$$\begin{aligned}\sigma^2 &= \Sigma x^2 P(x) - \mu^2\\ &= 1^2 \times \frac{20}{100} + 2^2 \times \frac{50}{100} + 3^2 \times \frac{20}{100} + 4^2 \times \frac{10}{100} - (2.2)^2\\ &= 1 \times 0.2 + 4 \times 0.5 + 9 \times 0.2 + 16 \times 0.1 - 4.84\\ &= 0.2 + 2 + 1.8 + 1.6 - 4.84\\ &= 5.6 - 4.84\\ &= 0.76\end{aligned}$$

Result

$$\mu = 2.2$$

$$\sigma^2 = 0.76$$

PROBLEM STATEMENT

Chapter 7, Page 264

Step 1

Mean amount of the refund is $\mu = \$1600$ and the standard deviation is $\sigma = \$850$

Let X be the refunds.

$\textbf{a. X is between 1600 and 2000 }$

$$\begin{aligned} P(1600 < X < 2000) &= P(1600 - \mu < X - \mu < 2000 - \mu) \\ &= P\left(\frac{1600 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{2000 - \mu}{\sigma}\right) \\ &= P\left(\frac{1600 - 1600}{850} < z < \frac{2000 - 1600}{850}\right) \\ &= P\left(\frac{0}{850} < z < \frac{400}{850}\right) \\ &= P(0 < z < 0.47) \\ &= 0.1808 \ \ (\text{from Appendix B1.}) \end{aligned}$$

18.08% of the refunds are between $\$1600$ and $\$2000$.

Step 2

$\textbf{b. } X \text{ is between } 900 \text{ and } 2000 \text{ }}$

$$\begin{aligned} P(900 < X < 2000) &= P(900 - \mu < X - \mu < 2000 - \mu) \\ &= P\left(\frac{900 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{2000 - \mu}{\sigma}\right) \\ &= P\left(\frac{900 - 1600}{850} < z < \frac{2000 - 1600}{850}\right) \\ &= P\left(\frac{-700}{850} < z < \frac{400}{850}\right) \\ &= P(-0.82 < z < 0.47) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.82 < z < 0)$, $P(0 < z < 0.47)$ and add the two probabilities.

$$\begin{aligned} P(900 < X < 2000) &= P(-0.82 < z < 0) + P(0 < z < 0.47) \\ &= P(0 < z < 0.82) + P(0 < z < 0.47) \\ &= 0.2939 + 0.1808 \ \ (\text{from Appendix B1.}) \\ &= 0.4747 \end{aligned}$$

47.47% of the refunds are between $\$900$ and $\$2000$.

Step 3

$\textbf{c. } X \text{ is less than } 0 \text{ }}$

$$\begin{aligned} P(X < 0) &= P(X - \mu < 0 - \mu) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\right) \\ &= P\left(z < \frac{0 - 1600}{850}\right) \\ &= P\left(z < \frac{-1600}{850}\right) \\ &= P(z < -1.88) \\ &= P(z > 1.88) \\ &= 0.5 - P(0 < z < 1.88) \\ &= 0.5 - 0.4699 \ \ (\text{from Appendix B1.}) \\ &= 0.0301 \end{aligned}$$

3.01% of the funds are less than 0.

🎯 **Result**

a. 18.08%

b. 47.47%

c. 3.01%

PROBLEM STATEMENT

Chapter 7, Page 264

Step 1

Given:

Refund	1	2	3 or more	Total
Yes	20	20	10	50
No	10	20	0	30
Total	30	40	10	80

(a) A table that contains counts for two different (qualitative) variables is called a **contingency table**.

Step 2

(b) The table describes 80 returns in total.

50 of the 80 returns result in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80} = \frac{5}{8} = 0.625 = 62.5\%$$

Step 3

(c) The table describes 80 returns in total.

50 of the 80 returns resulted in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80}$$

30 of the 80 returns had 1 dependent, because 30 is mentioned in the row "Total" and in the column "1" of the given table.

$$P(1\text{ dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{30}{80}$$

20 of the 80 returns resulted in a refund and had 1 dependent, because 20 is mentioned in the row "Yes" and in the column "1" of the given table.

$$P(\text{refund and 1 dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{20}{80}$$

Use the **General addition rule** for any two events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$\begin{aligned}
&P(\text{refund or 1 dependent}) \\
&= P(\text{refund}) + P(1\text{ dependent}) - P(\text{refund and 1 dependent}) \\
&= \frac{50}{80} + \frac{30}{80} - \frac{20}{80} \\
&= \frac{50 + 30 - 20}{80} \\
&= \frac{60}{80} \\
&= \frac{3}{4} \\
&= 0.75 \\
&= 75\%
\end{aligned}$$

Step 4

(d) The table describes 80 returns in total.

50 of the 80 returns resulted in a refund, because 50 is mentioned in the row "Yes" and in the column "Total" of the given table.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{refund}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{50}{80}$$

20 of the 80 returns resulted in a refund and had 1 dependent, because 20 is mentioned in the row "Yes" and in the column "1" of the given table.

$$P(\text{refund and 1 dependent}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{20}{80}$$

Use the Definition **Conditional probability**: $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$

$$\begin{aligned} P(1 \text{ dependent}|\text{refund}) &= \frac{P(\text{refund and 1 dependent})}{P(\text{refund})} \\ &= \frac{20/80}{50/80} \\ &= \frac{20}{50} \\ &= \frac{2}{5} \\ &= 0.4 \\ &= 40\% \end{aligned}$$

Step 5

(e) The table describes 80 returns in total.

10 of the 80 returns resulted in no refund and had 1 dependent, because 10 is mentioned in the row "No" and in the column "1" of the given table.

$$\begin{aligned} P(\text{no refund and 1 dependent}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{10}{80} \\ &= \frac{1}{8} \\ &= 0.125 \\ &= 12.5\% \end{aligned}$$

Result

(a) Contingency table

(b) $\frac{5}{8} = 0.625 = 62.5\%$

(c) $\frac{3}{4} = 0.75 = 75\%$

(d) $\frac{2}{5} = 0.4 = 40\%$

(e) $\frac{1}{8} = 0.125 = 12.5\%$

PROBLEM STATEMENT

Chapter 7, Page 264

Step 1

DEFINITIONS

Formula **Poisson probability**:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Multiplication rule for independent events:

$$P(A \cap B) = P(A \text{ and } B) = P(A) \times P(B)$$

Step 2

SOLUTION

$$\mu = \text{Mean} = 3$$

(a) Evaluate the probability of Poisson probability at $x = 0$:

$$\begin{aligned} P(X = 0) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^0 e^{-3}}{0!} \\ &= \frac{1 e^{-3}}{1} \\ &= e^{-3} \\ &\approx 0.0498 \\ &= 4.98\% \end{aligned}$$

Step 3

(b) Evaluate the probability of Poisson probability at $x = 3$:

$$\begin{aligned} P(X = 3) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^3 e^{-3}}{3!} \\ &= \frac{27e^{-3}}{6} \\ &= \frac{9}{2}e^{-3} \\ &\approx 0.2240 \\ &= 22.40\% \end{aligned}$$

Step 4

(c) Evaluate the probability of Poisson probability at $x = 1, 2, 4$:

$$\begin{aligned} P(X = 1) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^1 e^{-3}}{1!} \\ &= \frac{3e^{-3}}{1} \\ &= 3e^{-3} \\ &\approx 0.1494 \\ P(X = 2) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^2 e^{-3}}{2!} \\ &= \frac{9e^{-3}}{2} \\ &= \frac{9}{2}e^{-3} \\ &\approx 0.2240 \\ P(X = 4) &= \frac{\mu^k e^{-\mu}}{k!} \\ &= \frac{3^4 e^{-3}}{4!} \\ &= \frac{81e^{-3}}{24} \\ &= \frac{27}{8}e^{-3} \\ &\approx 0.1680 \end{aligned}$$

Step 5

Use the addition rule for mutually exclusive events (which is appropriate as it is not possible to obtain two different number of successes for the same month):

$$\begin{aligned} P(X<5) &= P(X=0)+P(X=1)+P(X=2) \\ &\quad +P(X=3)+P(X=4) \\ &= 0.0498+0.1494+0.2240 \\ &\quad +0.2240+0.1680 \\ &= 0.8152 \end{aligned}$$

Use the complement rule:

$$\begin{aligned} P(X\geq 5) &= 1-P(X<5) \\ &= 1-0.8152 \\ &= 0.1848 \\ &= 18.45\% \end{aligned}$$

Step 6

(d) Use the multiplication rule for independent events (assuming that the number of requests on a single day is not affected by the number of requests on the previous days) and the results of part (a):

$$\begin{aligned} P(X=0 \text{ for two days}) &= P(X=0)\times P(X=0) \\ &= 0.0498\times 0.0498 \\ &= 0.0498^2 \\ &\approx 0.0025 \\ &= 0.25\% \end{aligned}$$

Result

(a) $e^{-3}\approx 0.0498 = 4.98\%$

(b) $\frac{9}{2}e^{-3}\approx 0.2240 = 22.40\%$

(c) $0.1848 = 18.45\%$

(d) $0.0025 = 0.25\%$